Lecture 1 Review Handout

Statistics 367-1-4361

Lecture Overview

Lecture 1 introduces the central problem of statistics: we want to understand real-world processes, but we only observe measurements. Statistics provides methods for reasoning from imperfect data to explanations of the processes that generated them. It establishes the essential cycle **reality → data → models → reality** and the relationshpi between data, models, and measurement. The lecture concludes by introducing probability distributions and the parameters used to describe them.

Lecture Sections

1.1 — Why Statistics Exists

Statistics exists because we only observe **data produced by measurements**, not reality itself. Scientific reasoning therefore requires systematic methods for moving from imperfect observations back toward the underlying processes that generated them.

New Definitions and Terms

- **Statistics** — a set of methods for learning about the world using data

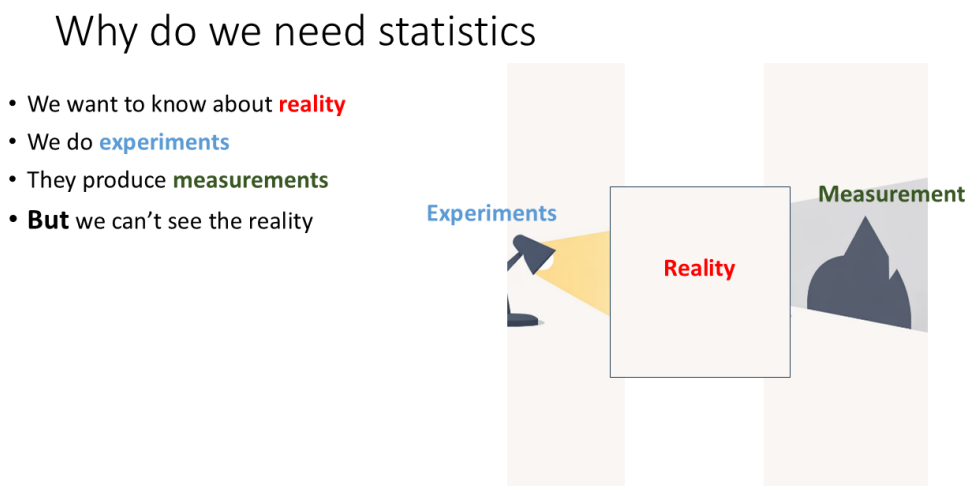


Figure 1: Slide 3 — Why do we need statistics

1.2 — What Is Statistics

Statistics provides tools for connecting **reality, data, and models**, allowing us to use observations to reason about the processes that generated them. In biomedical engineering, data come from many sources such as physiological signals, behavioral measurements, imaging data, and experiments, and statistical models help organize these observations into explanations of biological systems.

New Definitions and Terms

- **Model** — a formal representation of a process that could generate the observed data

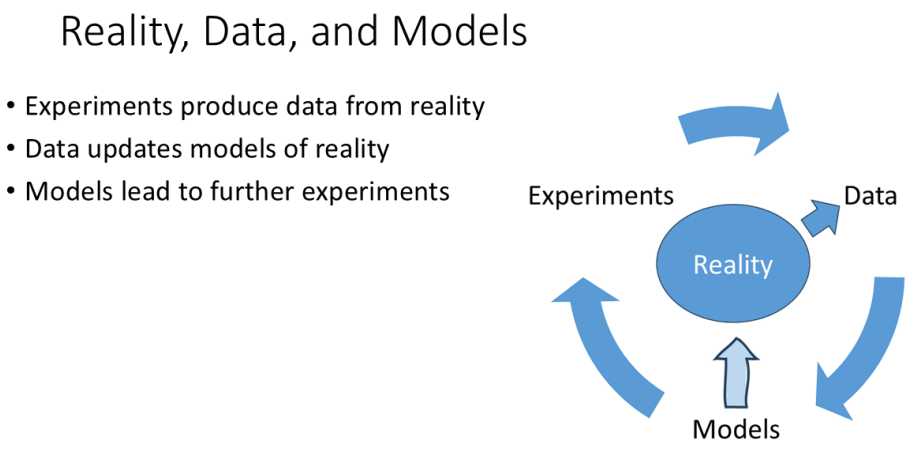


Figure 2: Slide 13 — Reality, Data, and Models

1.3 — What Is Data

Data are the recorded results of **measurements**, and understanding how measurements are produced is essential for interpreting data correctly. Measurements convert real-world phenomena into recorded values, and data may also include quantities derived from those measurements. Because data must ultimately be stored and analyzed, they must be finite and representable.

New Definitions and Terms

- **Measurement** — the process of converting a real-world phenomenon into recorded values

Important measurement properties:

- **Precision** — consistency of repeated measurements
- **Validity** — whether the measurement reflects the intended quantity
- **Reliability** — stability of measurements across time or conditions

1.4 — Imperfect Data

All real data are imperfect because measurements are affected by noise, bias, and experimental conditions. These imperfections limit the conclusions that can be drawn from data and require

careful interpretation. Researchers also have ethical responsibilities regarding how data are collected, analyzed, and communicated.

New Definitions and Terms

Types of data imperfection discussed in the lecture:

- **Sampling bias** — systematic distortion caused by which observations enter the dataset
- **Measurement error** — mismatch between the recorded value and the true quantity
- **Missing data** — absent observations that increase uncertainty and may introduce bias
- **Proxy measures** — indirect measurements used in place of the quantity of real interest
- **Data ethics** — principles governing responsible data collection, analysis, and interpretation

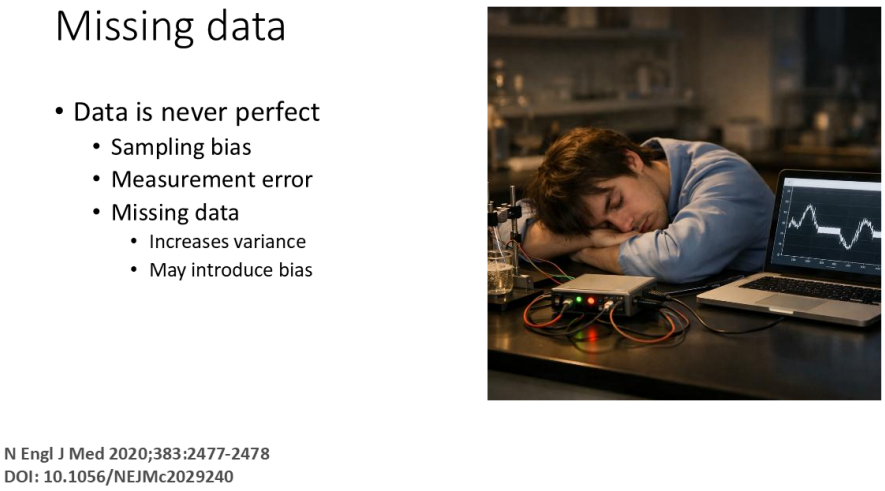


Figure 3: Slide 28 — Missing data

1.5 — Types of Data

Different kinds of measurements produce different **types of data**, and recognizing the structure of the data is an essential step in choosing an appropriate statistical model. Scientific datasets often include continuous measurements as well as categorical or ordered variables, and many analyses combine several types of data simultaneously.

New Definitions and Terms

- **Continuous data** — numerical measurements on a continuous scale
- **Categorical data** — observations grouped into discrete categories

- **Ordinal data** — categorical values with a meaningful ordering

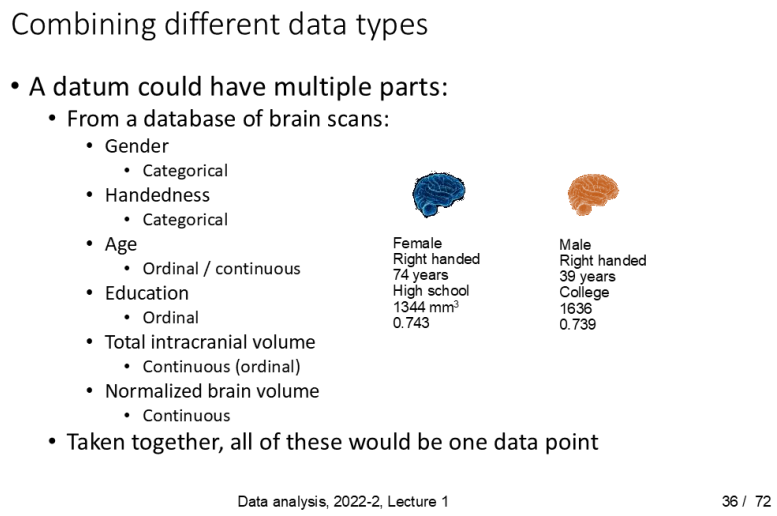


Figure 4: Slide 36 — Combining different data types

1.6 — Probability and Uncertainty

Because data are imperfect, our conclusions about the world are uncertain. Probability provides a mathematical framework for representing and reasoning about this uncertainty. In scientific contexts, uncertainty arises either from inherent randomness in the world or from incomplete knowledge about a system. Probability distributions provide a way to represent these uncertainties quantitatively.

New Definitions and Terms

- **Probability** — a numerical representation of the likelihood of events
- **Probability distribution** — a function describing probabilities of possible outcomes
- **Aleatory uncertainty** — variability inherent in the system or process being studied
- **Epistemic uncertainty** — uncertainty due to incomplete knowledge about the system

Formulas

$$P(A)$$

Probability assigned to event A.

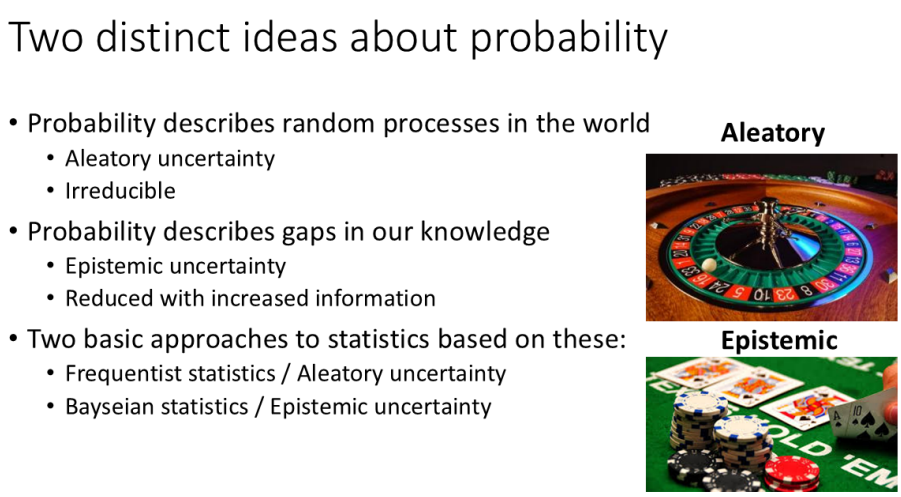


Figure 5: Slide 47 — Two distinct ideas about probability

1.7 — Probability Distributions

A **probability distribution** describes how probability is assigned across the possible values of a random variable. Instead of asking whether a single event occurs, we describe the probability associated with *each possible outcome*.

Probability distributions are the main way we represent uncertainty in statistics. Different scientific processes produce different characteristic distributions, which describe how values are likely to be distributed.

Two broad classes of distributions are commonly used:

- **Discrete distributions** — probabilities assigned to distinct outcomes (for example, the number of successes in repeated trials)
- **Continuous distributions** — probability density spread across a continuous range of values

Distributions describe the **shape of uncertainty** in a variable. In the next section we will see how **parameters** describe the specific instance of a distribution for a particular dataset.

Random variables

- A discrete random variable may have different specific values
  - Probability associated with each value
- A continuous random variable can have any value in a range
  - Probability associated with a range of values

x	p(x)
0	0.5
1	0.2
2	0.15
3	0.1
4	0.05

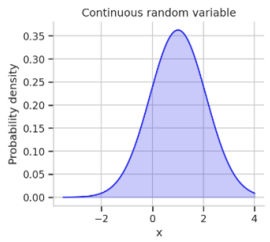


Figure 6: Slide 50 — Random variables

1.8 — Parameters and Describing Distributions

Probability distributions can be summarized using numerical parameters that describe key properties such as their center, spread, and shape. Measures such as the mean or median describe central tendency, standard deviation and interquartile range describe spread, while other quantities describe other aspects of the distribution's shape such as asymmetry or tail behavior.

New Definitions and Terms

Measures of central tendency:

- **Mean** — the average value of a distribution
- **Median** — the central value that divides the distribution in half
- **Mode** — the point with the largest probability

Shape descriptors:

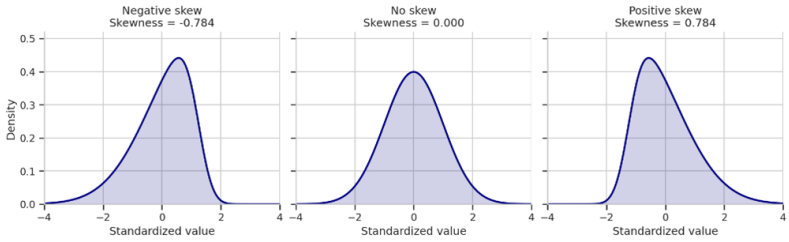
- **Skewness** — a measure of asymmetry in a distribution
- **Kurtosis** — a measure of tail weight and peak sharpness
- **Unimodal distribution** — a distribution with a single mode
- **Multimodal distribution** — a distribution with more than one mode

Formulas

Mean:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i$$

Measure of assymetry: Skewness

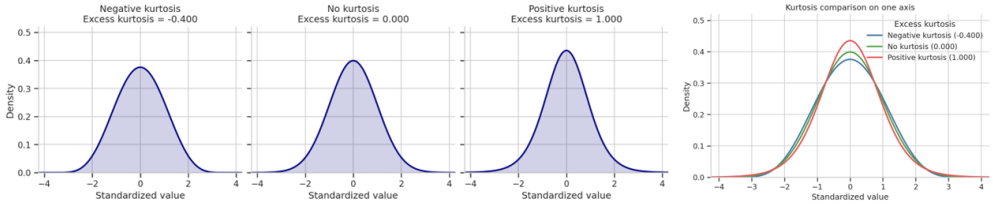


- For the normal distribution, we only need mean and variance
- For any other distribution, it will vary in either skewness or kurtosis
- Skewness measures the symmetry of the distribution

$$\text{Skewness} = \frac{1}{\sigma^3} E\left(\left(x-\mu\right)^3\right)$$

Figure 7: Slide 66 — Measure of assymetry: Skewness

Sharpness of peak: Kurtosis



- Kurtosis measures the size of the tails
- This also captures the pointiness of the peak

$$\text{Kurtosis} = \frac{1}{\sigma^4} E\left(\left(x-\mu\right)^4\right) - 3$$

Figure 8: Slide 67 — Sharpness of peak: Kurtosis

1.9 — About This Course

The course focuses on using probability and statistical models to analyze data and reason about uncertainty in biomedical engineering. Students will learn how to construct models, interpret data, and evaluate evidence using modern statistical tools.

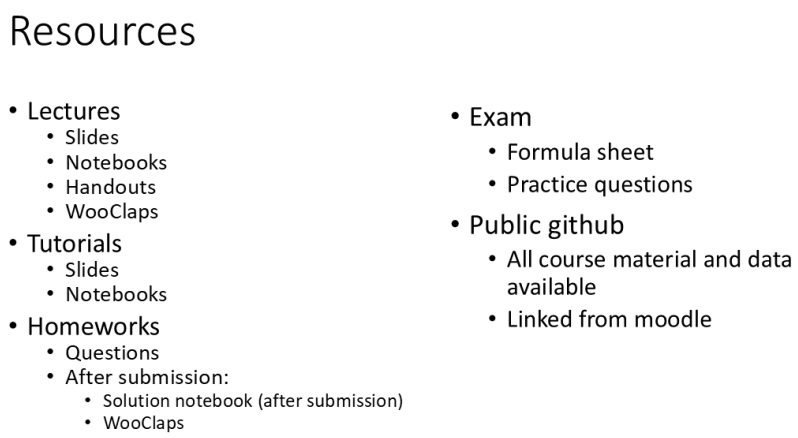


Figure 9: Slide 73 — Resources

Lecture 2 handout: Beta-binomial conjugacy

Analytic derivation moved out of the lecture slides

Statistics 367-1-4361

Table of contents

1 Why this handout exists 1

2 Model and notation 2

3 Step 1: Write Bayes' rule 2

4 Step 2: Write the prior and likelihood explicitly 3

5 Step 3: Multiply prior and likelihood 3

6 Step 4: Recognize the kernel of a Beta distribution 4

7 Step 5: Show the normalizing constant explicitly 5

8 Interpretation of the update 5

9 Posterior summaries 6

9.1 Posterior mean . . . 6

9.2 Posterior mode . . . 6

9.3 Posterior variance . . . 6

10 Sequential updating 6

11 What to remember 7

1 Why this handout exists

In the lecture, the main point of *conjugate priors* is conceptual:

- a Beta prior is a natural prior for an unknown probability,
- a binomial likelihood updates that prior analytically, and
- the posterior stays in the Beta family.

The algebra needed to prove that result is useful, but it is not the best use of slide time. This handout contains the derivation carefully so that the lecture slides can stay focused on updating, interpretation, and intuition.

2 Model and notation

We observe z heads in N coin tosses, and we want to learn about the unknown probability of heads θ.

We use the model

$$\theta \sim \text{Beta}(\alpha, \beta)$$

and

$$z \mid \theta \sim \text{Binomial}(N, \theta).$$

Here:

- $\theta \in [0, 1]$ is the probability of heads,
- $\alpha, \beta > 0$ are prior shape parameters,
- z is the number of heads,
- $N - z$ is the number of tails.

3 Step 1: Write Bayes' rule

Bayes' rule says

$$p(\theta \mid z, N) = \frac{p(z \mid \theta, N)\, p(\theta)}{p(z \mid N)}.$$

The denominator

$$p(z \mid N) = \int_0^1 p(z \mid \theta, N)\, p(\theta)\, d\theta$$

is the **evidence** (also called the marginal likelihood). It is the normalizing constant that makes the posterior integrate to 1.

For the derivation, it is easiest to start with the proportional form

$$p(\theta \mid z, N) \propto p(z \mid \theta, N)\, p(\theta).$$

4 Step 2: Write the prior and likelihood explicitly

The Beta prior density is

$$p(\theta) = \frac{1}{B(\alpha, \beta)}\, \theta^{\alpha-1}(1-\theta)^{\beta-1}, \qquad 0 \le \theta \le 1,$$

where

$$B(\alpha, \beta) = \int_0^1 \theta^{\alpha-1}(1-\theta)^{\beta-1}\, d\theta$$

is the Beta function.

The binomial likelihood is

$$p(z \mid \theta, N) = \binom{N}{z} \theta^z (1-\theta)^{N-z}.$$

5 Step 3: Multiply prior and likelihood

Now multiply them:

$$p(z \mid \theta, N)p(\theta) = \binom{N}{z} \theta^z (1-\theta)^{N-z} \cdot \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1}(1-\theta)^{\beta-1}.$$

Group the constants together and combine powers of θ and $(1-\theta)$:

$$p(z \mid \theta, N)p(\theta) = \frac{\binom{N}{z}}{B(\alpha, \beta)} \theta^{z+\alpha-1}(1-\theta)^{N-z+\beta-1}.$$

Since the factor

$$\frac{\binom{N}{z}}{B(\alpha, \beta)}$$

does not depend on θ, we can absorb it into the proportionality constant. Therefore

$$p(\theta \mid z, N) \propto \theta^{\alpha+z-1}(1-\theta)^{\beta+N-z-1}.$$

6 Step 4: Recognize the kernel of a Beta distribution

A Beta(a, b) density has the form

$$p(\theta) \propto \theta^{a-1}(1-\theta)^{b-1}.$$

Comparing this with

$$p(\theta \mid z, N) \propto \theta^{\alpha+z-1}(1-\theta)^{\beta+N-z-1},$$

we immediately identify

$$a = \alpha + z, \qquad b = \beta + N - z.$$

So the posterior must be

$$\theta \mid z, N \sim \text{Beta}(\alpha + z, \beta + N - z).$$

This is the conjugacy result.

7 Step 5: Show the normalizing constant explicitly

To make the posterior a proper density, we divide by the Beta-function normalizing constant for the updated parameters:

$$p(\theta \mid z, N) = \frac{1}{B(\alpha + z, \beta + N - z)} \theta^{\alpha+z-1}(1-\theta)^{\beta+N-z-1}.$$

Equivalently, the evidence is

$$p(z \mid N) = \int_0^1 p(z \mid \theta, N) p(\theta)\, d\theta = \frac{\binom{N}{z}}{B(\alpha, \beta)} \int_0^1 \theta^{\alpha+z-1}(1-\theta)^{\beta+N-z-1}\, d\theta.$$

The integral is exactly a Beta function:

$$\int_0^1 \theta^{\alpha+z-1}(1-\theta)^{\beta+N-z-1}\, d\theta = B(\alpha + z, \beta + N - z).$$

So the evidence is

$$p(z \mid N) = \frac{\binom{N}{z} B(\alpha + z, \beta + N - z)}{B(\alpha, \beta)}.$$

You do not need this expression to identify the posterior family, but it is the missing normalization step in the full derivation.

8 Interpretation of the update

The posterior update is

$$\mathrm{Beta}(\alpha, \beta) \xrightarrow{z \text{ heads in } N \text{ tosses}} \mathrm{Beta}(\alpha + z, \beta + N - z).$$

So the data update the parameters directly:

- add the number of heads to α,
- add the number of tails to β.

This is why conjugacy is so attractive pedagogically: the update rule is exact, compact, and easy to interpret.

9 Posterior summaries

Once we know the posterior is $\text{Beta}(\alpha + z, \beta + N - z)$, we can read off useful summaries.

9.1 Posterior mean

$$\mathbb{E}[\theta \mid z, N] = \frac{\alpha + z}{\alpha + \beta + N}.$$

9.2 Posterior mode

When $\alpha + z > 1$ and $\beta + N - z > 1$,

$$\text{mode}(\theta \mid z, N) = \frac{\alpha + z - 1}{\alpha + \beta + N - 2}.$$

9.3 Posterior variance

$$\text{Var}(\theta \mid z, N) = \frac{(\alpha + z)(\beta + N - z)}{(\alpha + \beta + N)^2(\alpha + \beta + N + 1)}.$$

These formulas make the next lecture section on posterior updating much easier to interpret.

10 Sequential updating

Suppose we observe the tosses one at a time. If the current posterior is

$$\theta \mid y_{1:n} \sim \text{Beta}(\alpha_n, \beta_n),$$

then the next toss updates as follows:

- if $y_{n+1} = 1$ (a head), then

$$\alpha_{n+1} = \alpha_n + 1, \qquad \beta_{n+1} = \beta_n;$$

- if $y_{n+1} = 0$ (a tail), then

$$\alpha_{n+1} = \alpha_n, \qquad \beta_{n+1} = \beta_n + 1.$$

This makes the repeated posterior-updating plots in the lecture completely natural.

11 What to remember

The main lesson is not the integral itself. The main lesson is the structure:

1. start with a prior family,
2. multiply by the likelihood,
3. recognize the kernel of a known distribution,
4. normalize.

For the Beta prior and binomial likelihood, this process produces another Beta distribution:

$$\boxed{\theta \mid z, N \sim \text{Beta}(\alpha + z, \beta + N - z).}$$

That is exactly why the Beta prior is called **conjugate** to the binomial likelihood.

Lecture 2 Review Handout

Statistics 367-1-4361

Lecture Overview

Lecture 2 introduces **statistical models** as explicit ideas about how data are generated, then develops the Bayesian update from **conditional probability** to **Bayes' rule**. The lecture uses the coin-flip example to define the **likelihood**, shows how a posterior can be computed on a **grid**, and then introduces the **prior** and the **Beta distribution** as a useful prior family for probabilities. It shows how Beta and Binomial models combine analytically through **conjugacy**, how posteriors evolve as data accumulate, and why prior choice matters. The lecture ends by introducing **Preliz** as a practical tool for prior elicitation and pointing ahead to sampling methods beyond grid calculations.

Lecture Sections

2.1 — What Is a Statistical Model

This section introduces a statistical model as an explicit account of how data could have been generated. The slides contrast a period when statistics focused mainly on **estimators** with a return to models as scientific representations that can be updated by evidence. The central idea is that a model contains **probability distributions** and **parameters**, and that data analysis is not only about producing estimates but about revising a generative explanation in light of new evidence.

New Definitions and Terms

- **Statistical model** — a formal probabilistic account of how observed data could be generated
- **Parameter** — a quantity that determines the specific form of a probability distribution
- **Estimator** — a rule for producing a numerical guess for a parameter from data

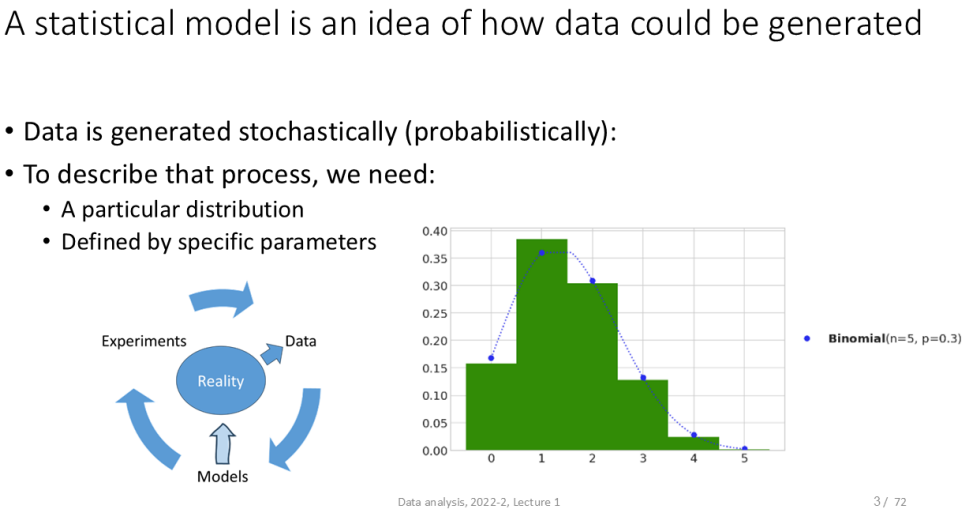


Figure 1: Slide 3 — A statistical model is an idea of how data could be generated

2.2 — Conditional Probability

This section reviews how probabilities change when we condition on additional information. The lecture distinguishes **joint**, **marginal**, and **conditional** probabilities, and emphasizes that conditioning has a direction: $p(r \mid c)$ and $p(c \mid r)$ are generally different. The section ends by connecting this idea to modeling, where we condition data on parameters and later condition parameters on data.

New Definitions and Terms

- **Joint probability** — the probability of two variables or events considered together
- **Marginal probability** — the probability of one variable after ignoring or summing over the other
- **Conditional probability** — the probability of one variable given information about another

Formulas

$$P(A \mid B) = \frac{P(A, B)}{P(B)}$$

Conditional probability is the joint probability, normalized by the probability of the condition.

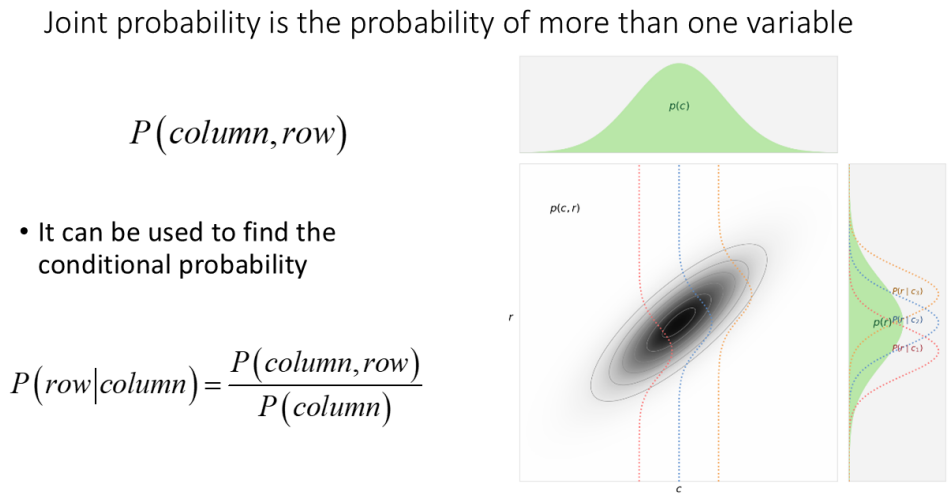


Figure 2: Slide 13 — Joint probability is the probability of more than one variable

2.3 — Bayes rule

This section turns conditional probability into the main updating rule of the course. Bayes' rule is introduced as the way to revise a model after observing data, and the lecture names the four parts that will be used repeatedly: **prior**, **likelihood**, **posterior**, and **evidence**. The slides also emphasize that Bayesian updating can feel counterintuitive at first, because we are learning about parameters indirectly through their consequences for the data.

New Definitions and Terms

- **Prior** — the probability distribution over parameter values before seeing the current data
- **Likelihood** — the probability of the observed data as a function of the parameter
- **Posterior** — the updated probability distribution over parameter values after observing the data
- **Evidence** — the normalizing constant in Bayes' rule; the overall probability of the observed data under the model

Formulas

$$p(\theta \mid y) = \frac{p(y \mid \theta)\, p(\theta)}{p(y)}$$

The posterior is proportional to likelihood times prior, with the evidence ensuring the result is a valid probability distribution.

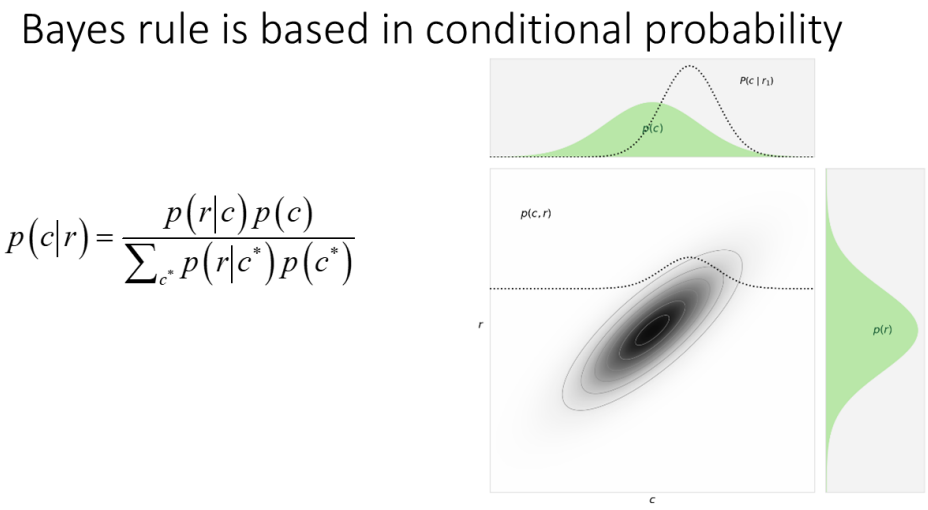


Figure 3: Slide 18 — Bayes rule is based in conditional probability

2.4 — Likelihood

This section introduces the coin-flip model as the main running example and defines the likelihood more concretely. The slides show both a graphical representation of the model and a mathematical one, then move from a single flip to the **Binomial** model for multiple flips. The key idea is that the likelihood tells us which parameter values make the observed data more or less plausible, and that multiplying the likelihood by the prior yields the posterior.

New Definitions and Terms

- **Binomial model** — a model for the number of successes in a fixed number of Bernoulli trials
- **Likelihood function** — the probability of the observed data viewed as a function of the unknown parameter

Formulas

$$z \mid \theta, N \sim \text{Binomial}(N, \theta)$$

The number of heads z in N flips is modeled with a Binomial distribution whose parameter is the unknown coin probability θ.

$$p(z \mid \theta, N) = \binom{N}{z} \theta^z (1-\theta)^{N-z}$$

This is the likelihood for the coin-flip example.

$$p(\theta \mid z, N) \propto p(z \mid \theta, N)\, p(\theta)$$

In grid calculation, the posterior is obtained point by point by multiplying prior and likelihood and then normalizing.

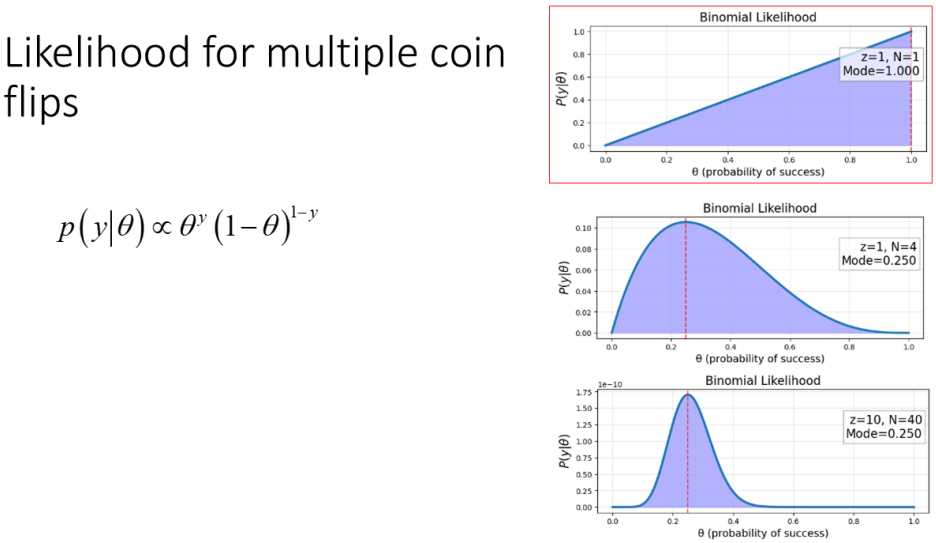


Figure 4: Slide 29 — Likelihood for multiple coin flips

The likelihood has a strong influence on the posterior.

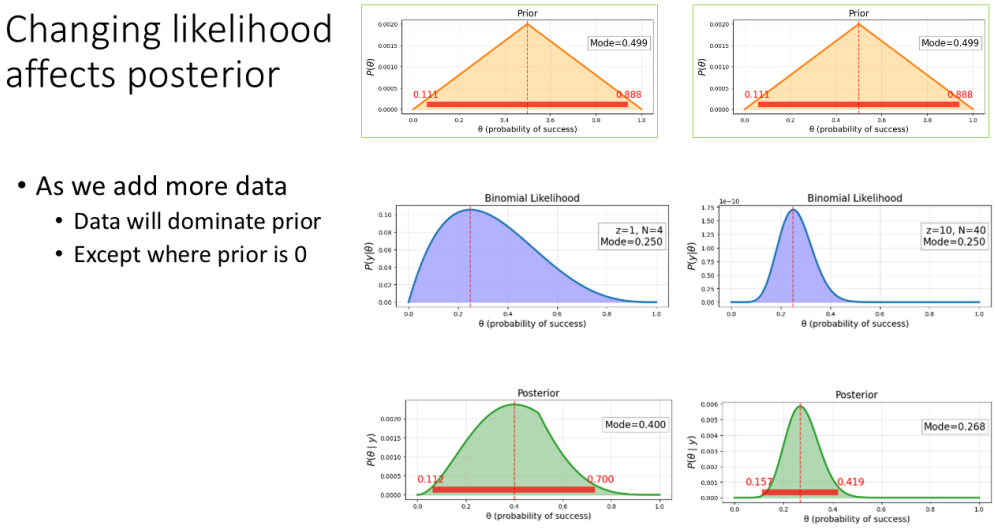


Figure 5: Slide 31 — Changing likelihood affects posterior

2.5 — Grid calculation

This section shows how to compute a posterior directly on a dense grid of parameter values. The lecture's strategy is simple: evaluate the **prior** and the **likelihood** at many fixed values of θ, multiply them point by point, and then normalize the resulting curve so that it becomes a proper probability distribution. The grid method is practical for one-parameter examples like the coin-flip model and gives a concrete way to see what Bayes' rule is doing computationally.

New Definitions and Terms

- **Grid calculation** — computing a posterior by evaluating it on a fixed grid of parameter values
- **Normalization** — rescaling a curve so that its total area is 1

Formulas

$$p(\theta \mid z, N) \propto p(z \mid \theta, N)\, p(\theta)$$

On a grid, we compute this product at each sampled value of θ.

$$p(\theta \mid z, N) = \frac{p(z \mid \theta, N)\, p(\theta)}{\int p(z \mid \theta, N)\, p(\theta)\, d\theta}$$

After taking the pointwise product, we normalize the curve so that it becomes a proper posterior density.

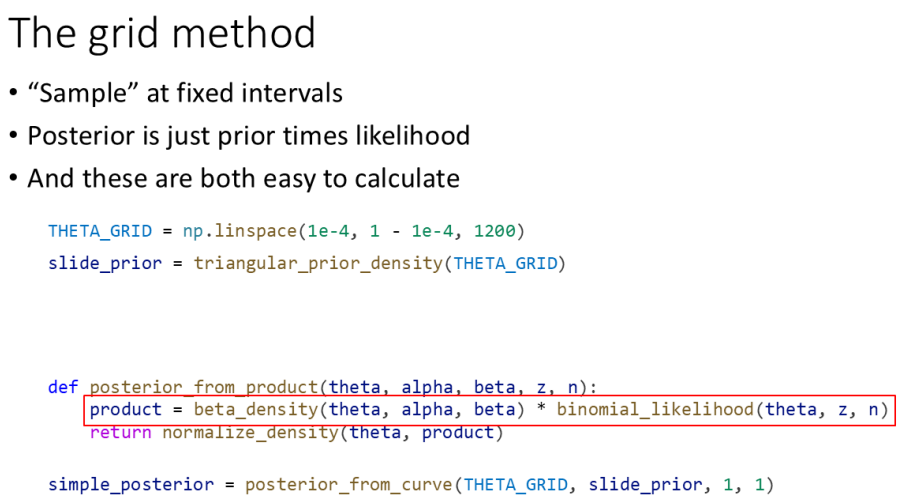


Figure 6: Slide 34 — The grid method

2.6 — Prior

This section focuses on the prior as a probability distribution over parameter values before the current data are observed. The slides stress that a prior must be a proper probability distribution and that it expresses what values are considered possible or plausible under the model. They also show that the posterior is shaped jointly by the prior and the likelihood, so prior assumptions matter most when the data are limited.

New Definitions and Terms

- **Prior distribution** — a probability distribution representing uncertainty about a parameter before seeing the current dataset
- **Parameter uncertainty** — uncertainty about the true value of a model parameter

Formulas

$$\int p(\theta)\, d\theta = 1$$

A prior must integrate to 1 because it is a probability distribution.



Figure 7: Slide 39 — Posterior is affected by both priors and likelihood

2.7 — Beta distribution

This section introduces the **Beta distribution** as a flexible prior family for probabilities. Because it is defined on $[0, 1]$, it matches the possible range of a coin-flip probability, and by changing its shape parameters it can represent uniform, biased, broad, or narrow prior beliefs. The lecture also introduces the **Beta function** as the normalizing constant and notes that the **Gamma function** extends the factorial to non-integer values.

New Definitions and Terms

- **Beta distribution** — a probability distribution on $[0, 1]$ commonly used as a prior for unknown probabilities
- **Shape parameters** — parameters such as α and β that determine the form of a distribution
- **Beta function** — the normalizing constant that makes the Beta density integrate to 1
- **Gamma function** — a generalization of the factorial function

Formulas

$$\theta \sim \text{Beta}(\alpha, \beta)$$

The Beta distribution is used to express prior uncertainty about the unknown probability θ.

$$p(\theta) = \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1}(1-\theta)^{\beta-1}$$

The Beta density is controlled by the shape parameters α and β.

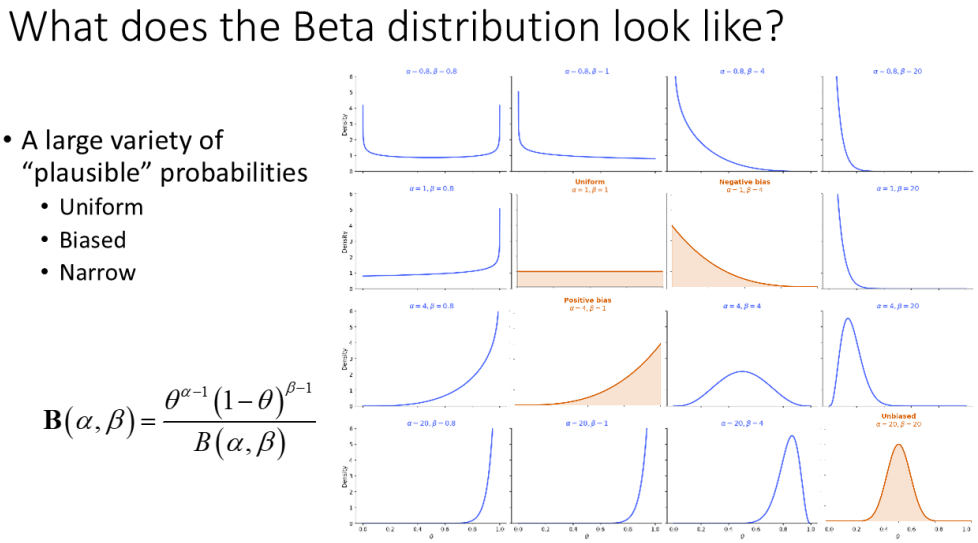


Figure 8: Slide 43 — What does the Beta distribution look like?

2.8 — Conjugate priors

This section shows that a Beta prior combined with a Binomial likelihood produces a Beta posterior. That analytic compatibility is what makes the Beta prior **conjugate** to the Binomial likelihood. The slides sketch the algebra, introduce the idea of simplifying Bayesian derivations with proportionality, and give the **pseudo-count** interpretation; the full algebraic derivation is available separately in the conjugate-prior derivation handout.

New Definitions and Terms

- **Conjugate prior** — a prior distribution that leads to a posterior in the same family after updating with the likelihood
- **Pseudo-count** — an interpretation of prior parameters as if they represented prior observations
- **Proportionality** — keeping only the terms that depend on the parameter and absorbing constants into the normalizing factor

Formulas

$$\theta \sim \text{Beta}(\alpha, \beta), \qquad z \mid \theta, N \sim \text{Binomial}(N, \theta)$$

This is the Beta-Binomial model discussed in the lecture.

$$\theta \mid z, N \sim \text{Beta}(\alpha + z,\ \beta + N - z)$$

Observed heads add to α and observed tails add to β, producing an analytic posterior update.

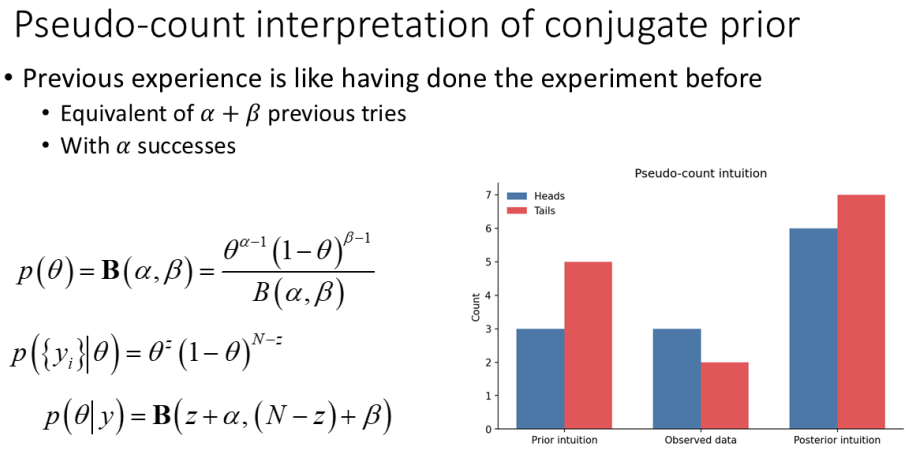


Figure 9: Slide 52 — Pseudo-count interpretation of conjugate prior

2.9 — Posterior distribution

This section applies the Beta-Binomial update repeatedly to the same coin-flip problem under several different priors. The main lesson is that the posterior remains a **distribution**, not just a point estimate, and that as more data arrive the posteriors become narrower and different priors tend to matter less. The lecture also emphasizes that for this model the final posterior depends on the total number of heads and tails, not on the order in which they arrived.

Formulas

$$\theta \mid z, N \sim \text{Beta}(\alpha + z,\ \beta + N - z)$$

Repeated updating keeps the same form; only the sufficient counts change as more data accumulate.

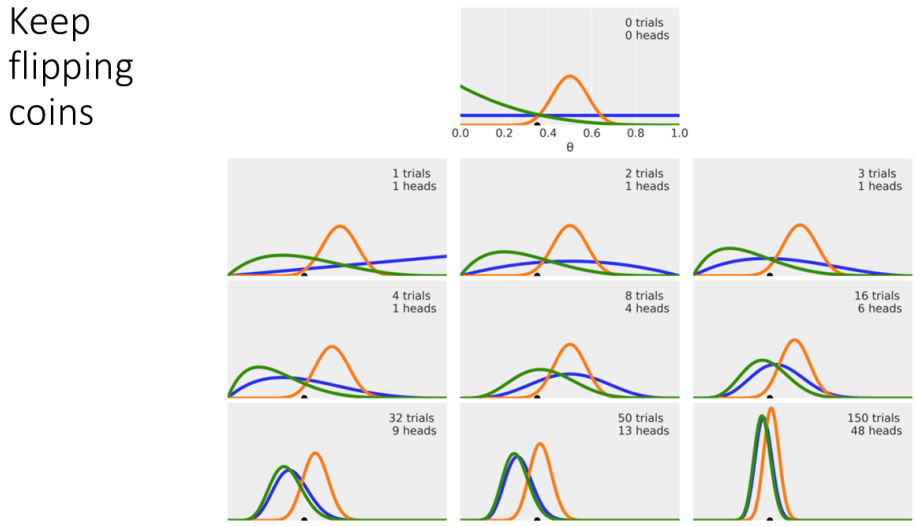


Figure 10: Slide 58 — Keep flipping coins

2.10 — Choosing priors

This section steps back from the coin example to ask how priors should be chosen in practice. The slides briefly review the history of so-called objective, maximum-entropy, vague, empirical, weakly informative, and automatic priors, then argue that prior choice should depend on the scientific problem and the strength of available background knowledge. The section ends with **prior elicitation**, where expert knowledge is translated into a prior distribution in an explicit, revisable way.

New Definitions and Terms

- **Weak prior** — a prior that places relatively broad constraints on plausible parameter values
- **Strong prior** — a prior that places substantial probability mass in a narrower set of values
- **Prior elicitation** — the process of translating domain knowledge into a prior distribution
- **Maximum entropy prior** — a prior chosen to satisfy stated constraints while adding as little extra structure as possible
- **Empirical prior** — a prior informed by external or previously collected data
- **Weakly informative prior** — a prior that regularizes an analysis without trying to dominate the data

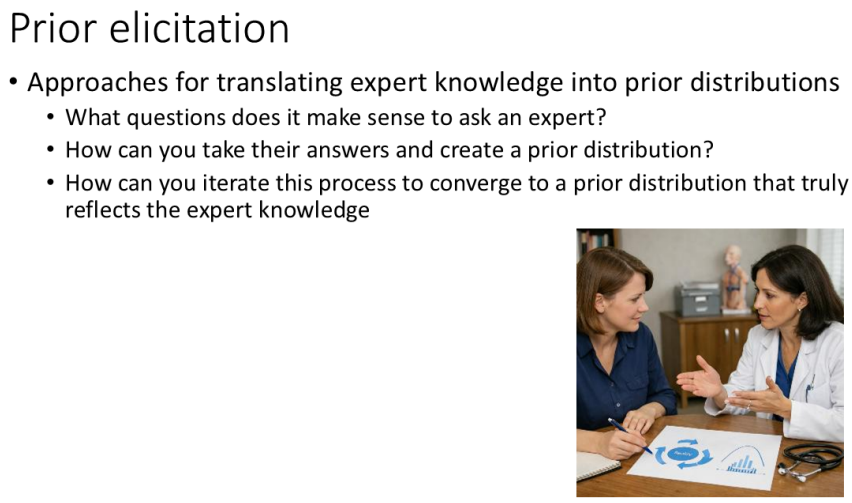


Figure 11: Slide 65 — Prior elicitation

2.11 — Preliz

This section introduces **Preliz** as a practical tool for exploring distributions and constructing priors from constraints. The lecture example asks for a Beta prior whose mass lies mostly between two bounds, and `maxent` is used to find a distribution that matches those constraints. Preliz is also presented as a broader reference for learning the shapes and uses of many probability distributions that will appear later in the course.

New Definitions and Terms

- **Preliz** — a Python package for exploring probability distributions and eliciting priors
- **maxent** — a Preliz function for finding a distribution that satisfies user-specified constraints

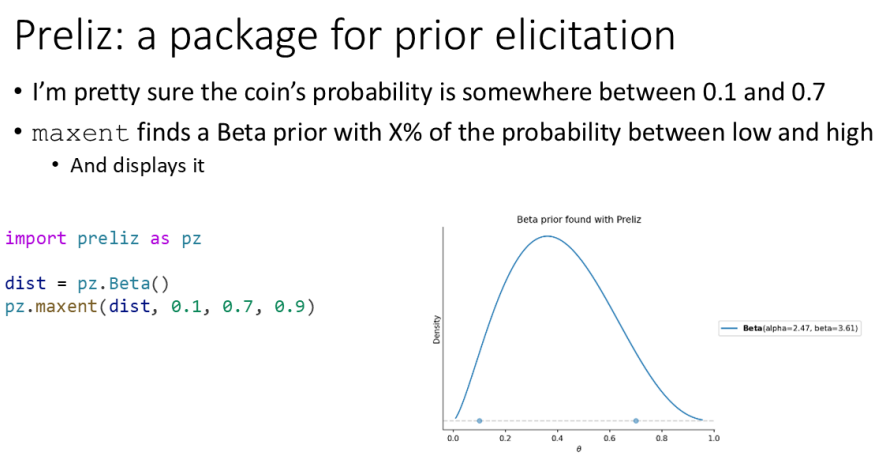


Figure 12: Slide 67 — Preliz: a package for prior elicitation

2.12 — Next week

The lecture closes by explaining why **grid calculations** are useful for learning but do not scale well to models with many parameters. This motivates the move to **sampling** methods and to PyMC, which will allow the course to fit richer models without evaluating the posterior on an infeasibly large grid.

Sampling as an alternative to grid computation

- Grid computation breaks down in high dimensions
- The better option:
  - More information where there is more probability
  - Sampling!

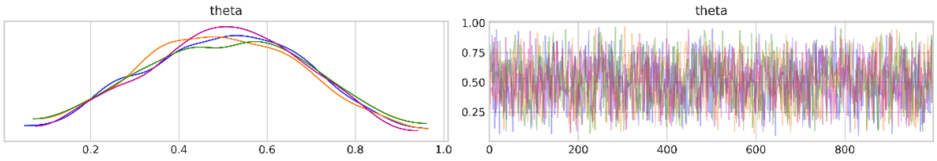


Figure 13: Slide 71 — Sampling as an alternative to grid computation

Lecture 3 Review Handout

Statistics 367-1-4361

Lecture Overview

Lecture 3 moves from analytic posterior calculation to **computational Bayesian inference**. It introduces graphical models and probabilistic programming languages, then explains why we sample from posteriors and how PyMC stores those draws. The lecture next uses posterior draws for visualization, point estimation, and HDIs. It then shifts to a frequentist lens, covering repeated-sampling ideas, estimators, maximum likelihood, and confidence intervals as an alternative way to quantify uncertainty.

Lecture Sections

3.1 — Graphical models

This section introduces graphical models as compact representations of how a probabilistic model is structured. The main goal is not aesthetics but clarity: a graph should show what counts as data, what counts as a distribution, and how parameters connect to observations. The slides also emphasize that there is no single correct notation, only clearer or less clear ones.

New Definitions and Terms

- **Graphical model** — a diagram showing the structure of a probabilistic model
- **Node** — a graphical element representing data, a distribution, or an equation
- **Arrow** — a graphical indication that one quantity helps determine another
- **DAG** — a directed acyclic graph, one common style for representing probabilistic models
- **Plate notation** — a graphical convention for repeated structure

Figure

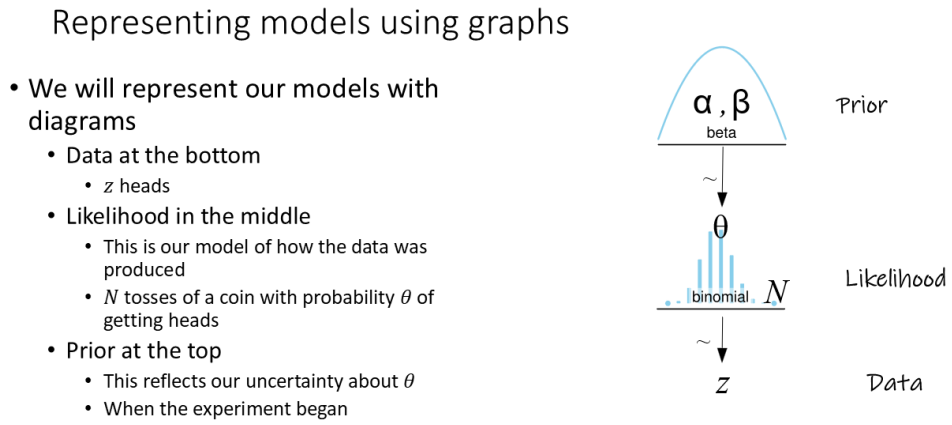


Figure 1: Slide 4 — Representing models using graphs

3.2 — Probabilistic programming languages

This section explains why Bayesian inference became computational rather than purely analytic. As models became more realistic, exact posterior calculation often became impossible, so probabilistic programming languages were developed to let us specify models and then compute with them. PyMC is introduced as the course PPL, and the lecture shows that the same model can be written as equations, a graph, or code.

New Definitions and Terms

- **Probabilistic programming language (PPL)** — a programming system for specifying probabilistic models and carrying out inference
- **PyMC** — the Python PPL used in the course
- **Graphical visualizer** — a tool for displaying the dependency structure of a PyMC model

Figure

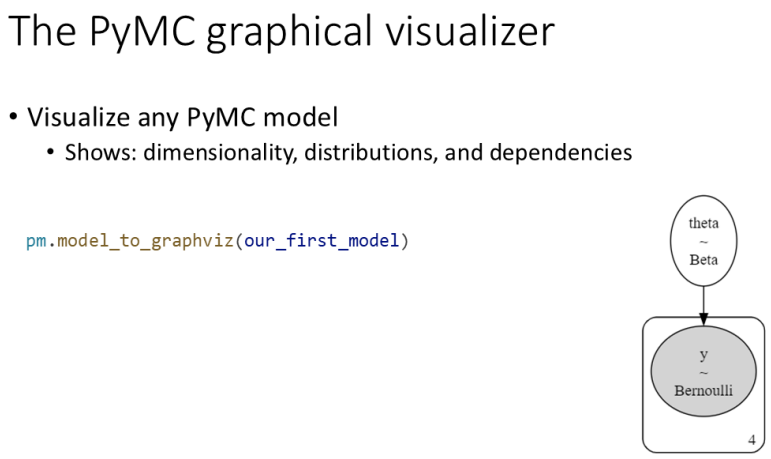


Figure 2: Slide 11 — The PyMC graphical visualizer

3.3 — Sampling

This section gives the main computational motivation for the lecture: many posteriors are too hard to calculate analytically and too large to approximate well by a grid. Sampling solves this by generating draws from the posterior distribution, with more draws giving a more stable approximation. The section also distinguishes **draws** from **chains**, and warns that the word *sample* is overloaded in statistics.

New Definitions and Terms

- **Draw** — one set of parameter values generated from the posterior distribution
- **Chain** — a sequence of posterior draws generated by one run of the sampler
- **Posterior sampling** — generating draws that approximate the posterior distribution
- **InferenceData object** — the object PyMC returns to store posterior draws and related outputs

Figure

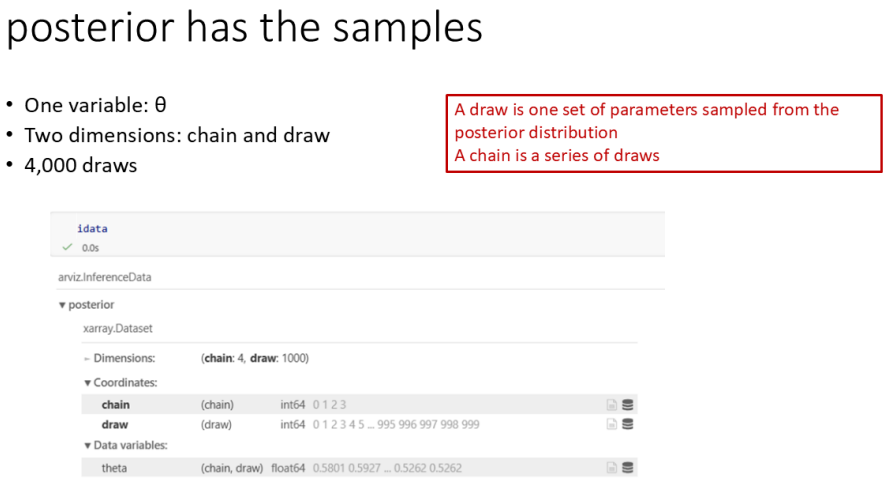


Figure 3: Slide 19 — posterior has the samples

3.4 — Plotting the posterior distribution

This section introduces ArviZ as the main plotting and summary tool for posterior draws. The emphasis is practical: once we have posterior draws, we need ways to inspect them visually and summarize them numerically. The trace plot and posterior plot are presented as the first standard views of sampled Bayesian results.

New Definitions and Terms

- **ArviZ** — the package used for analyzing and plotting Bayesian posterior results
- **Trace plot** — a plot of sampled draws across chains and iterations
- **Posterior plot** — a plot summarizing the posterior distribution of a parameter

Figure

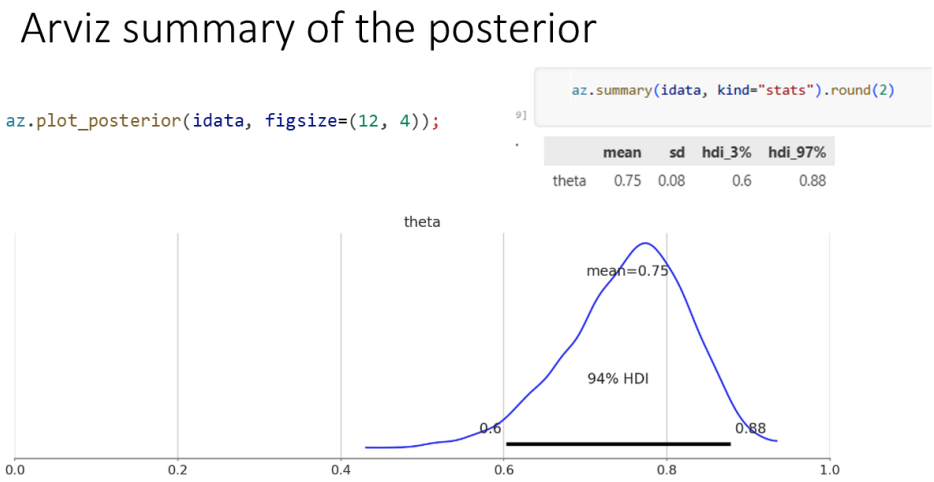


Figure 4: Slide 24 — Arviz summary of the posterior

3.5 — Estimation

This section asks how to turn a posterior distribution into a single best guess. Using simulated data, the lecture compares different estimators and shows that they can differ in both **bias** and **variance**. The main message is that point estimation is a summary of the posterior, not a replacement for the full posterior.

New Definitions and Terms

- **Estimator** — a rule for producing a numerical guess from data or from a posterior
- **Bias** — systematic deviation of an estimator from the true value
- **Variance** — the amount an estimator changes across repeated datasets
- **Bias-variance tradeoff** — the fact that lower bias can come with higher variance, or vice versa
- **Posterior mean** — the average of the posterior distribution
- **Posterior mode** — the value with the highest posterior density

Figure

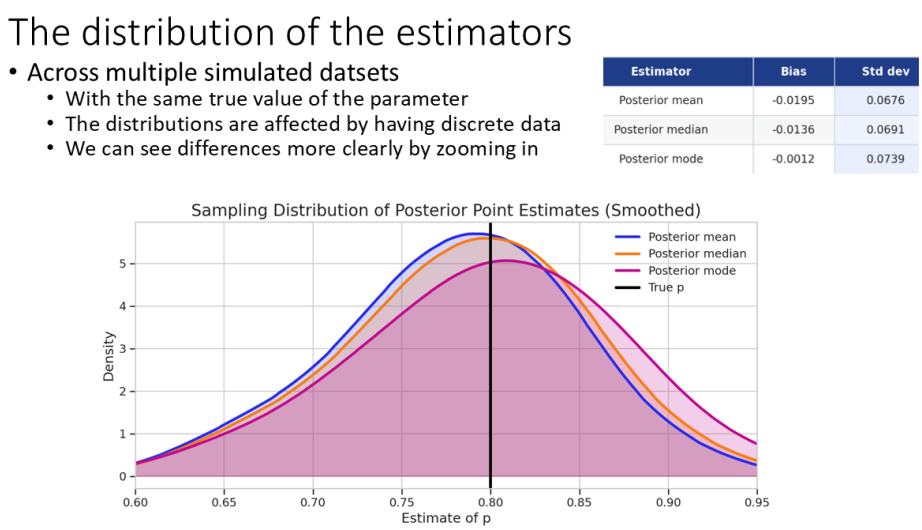


Figure 5: Slide 29 — The distribution of the estimators

3.6 — HDI

This section introduces the **highest density interval** as a compact summary of posterior uncertainty. An HDI contains a fixed amount of posterior probability while prioritizing the most credible values, so in the unimodal univariate case it is the shortest interval with that probability mass. The lecture also stresses that HDI probability and repeated-sampling coverage are not the same thing.

New Definitions and Terms

- **HDI** — the highest density interval; an interval containing the most credible posterior values
- **Coverage** — the fraction of repeated datasets for which an interval would contain the true parameter
- **Disconnected HDI** — an HDI that splits into separate regions when the posterior is multi-peaked

Figure

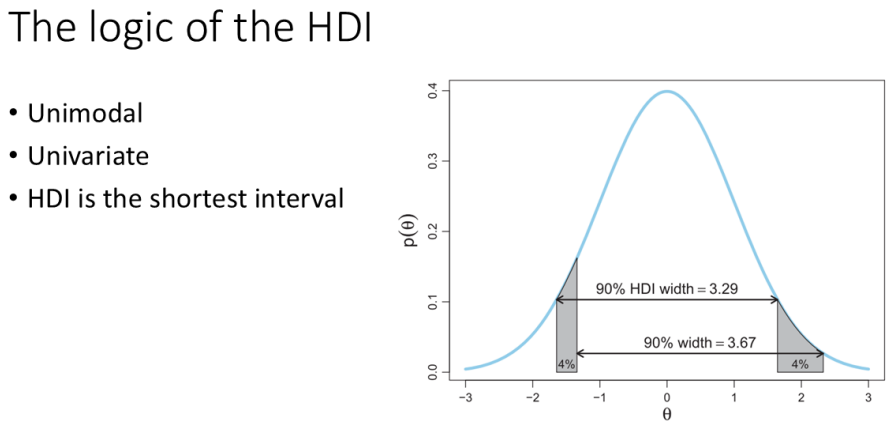


Figure 6: Slide 36 — The logic of the HDI

3.7 — Frequentist vs Bayesian philosophies

This section presents the conceptual split between Bayesian and frequentist reasoning. The Bayesian view treats data as updating our uncertainty about parameters, while the frequentist view treats uncertainty as arising from repeated random datasets generated by a procedure. The key point is that these approaches are answering different questions, not merely using different notation.

New Definitions and Terms

- **Bayesian uncertainty** — uncertainty about parameters or models given the data and assumptions
- **Frequentist uncertainty** — uncertainty defined through repeated sampling behavior of a procedure
- **Procedure-based inference** — inference judged by how a method behaves across repeated uses

Figure

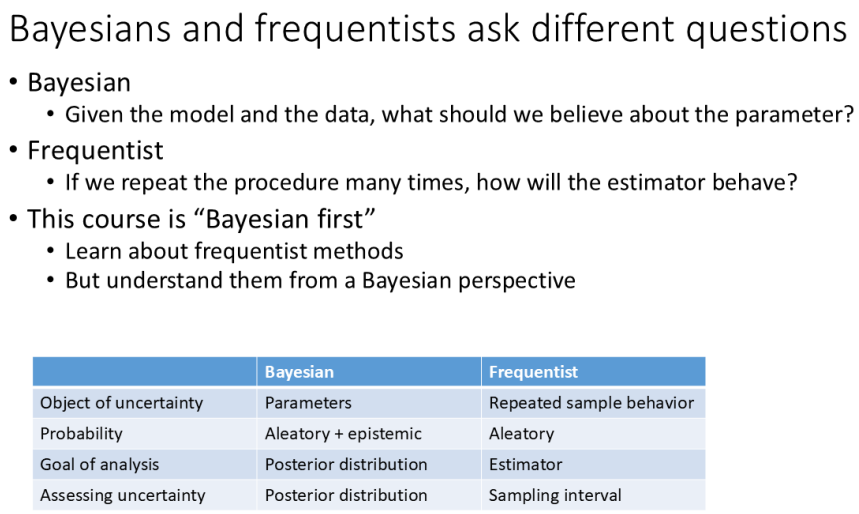


Figure 7: Slide 42 — Bayesians and frequentists ask different questions

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3.8 — Frequentist estimation

This section develops the frequentist idea of an estimator as a statistic whose quality is judged across repeated experiments. The lecture compares several estimators of a mean and shows that their behavior depends strongly on data quality, sample size, and outliers. The sampling distribution is the central object here because it tells us how an estimator behaves under replication.

New Definitions and Terms

- **Statistic** — any quantity calculated from data
- **Estimator** — a statistic used to guess a real parameter value
- **Sampling distribution** — the distribution of an estimator across repeated datasets
- **Reduced mean** — a mean computed after trimming extreme observations

Figure

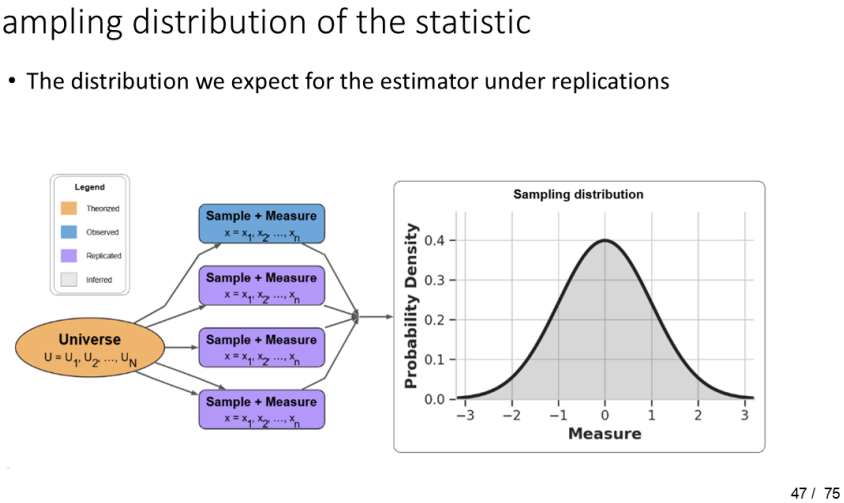


Figure 8: Slide 47 — Sampling distribution of the statistic

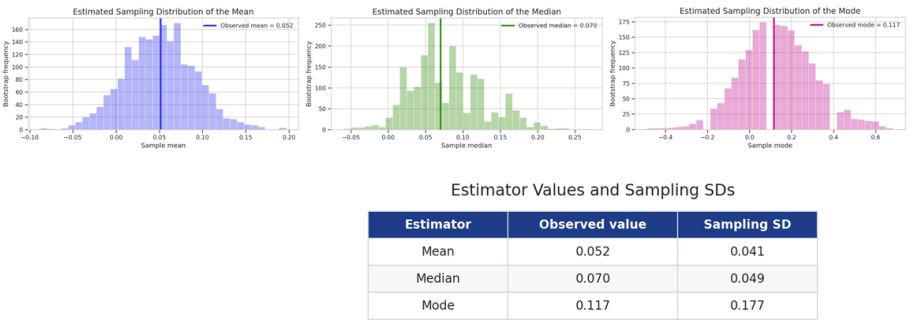


Figure 9: Slide 50 — Sampling distribution for estimator uncertainty

3.9 — Maximum likelihood

This section introduces the **maximum likelihood estimate** as a model-based frequentist estimator. The MLE chooses the parameter value that makes the observed data most plausible under the likelihood. The lecture then interprets MLE from a Bayesian perspective: it resembles taking a flat prior, using the likelihood as the posterior shape, and then collapsing that distribution to a single point.

New Definitions and Terms

- **Maximum likelihood estimate (MLE)** — the parameter value that maximizes the likelihood
- **Consistency** — the property that an estimator approaches the true value as sample size grows
- **Efficiency** — low variance relative to other estimators, especially in large samples
- **Model misspecification** — using a model that does not match the real data-generating process

Figure

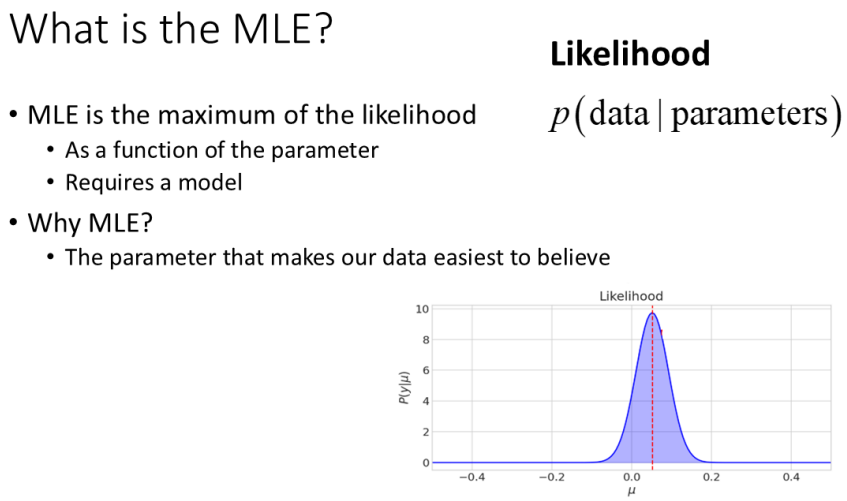


Figure 10: Slide 52 — What is the MLE?

3.10 — Confidence intervals

This section defines the confidence interval through the sampling distribution, not through posterior probability. The lecture builds lower and upper tail statistics, then defines a confidence interval as the set of parameter values for which the observed estimate is not too unusual. The slides end by contrasting CI logic with HDI logic and by warning against the very common mistake of treating a confidence interval as a probability statement about the parameter.

New Definitions and Terms

- **Confidence interval (CI)** — an interval-valued procedure with a stated repeated-sampling coverage rate
- **Tail statistic** — a boundary defined by a tail probability in the sampling distribution
- **Coverage probability** — the long-run proportion of intervals that contain the true parameter

Formulas

$[\ \mathrm{p}(\mathrm{L}(\hat{y}) \quad H(\hat{y})) = 0.96]$

A 96% confidence interval is defined by lower and upper bounds whose repeated-sampling procedure contains the true parameter 96% of the time.

Figure

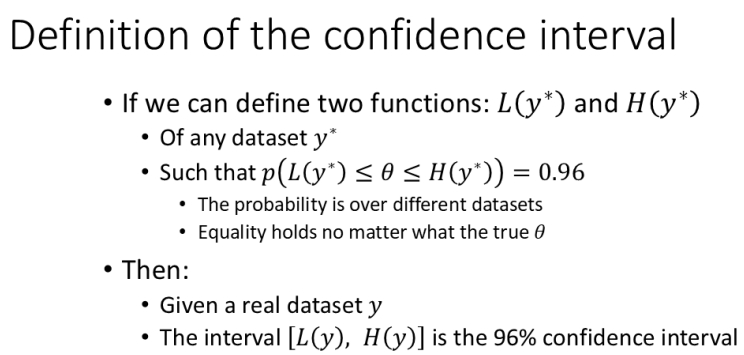


Figure 11: Slide 60 — Definition of the confidence interval

3.11 — Next time

The lecture closes by pointing out that drawing posterior samples is itself a hard problem. If there is no direct random-number generator for the posterior, we need an algorithm that moves through parameter space in a way that matches posterior probability. That motivates the next lecture on MCMC.

New Definitions and Terms

- **MCMC** — a family of algorithms for generating draws from difficult posterior distributions

---

Lecture 4 Review Handout

Statistics 367-1-4361

Lecture Overview

Lecture 4 expands the simple Bayesian pipeline into a full **Bayesian workflow**. The lecture expands our idea of Bayesian data anlysis to include a series of tests for the validity of the workout. These include prior predictive testing, sampling diagnostics, and postrior predictive testing. Each serves a different purpose. The lecture spends the bulk of the time on how sampling works and what sampling diagnostics to look at. It introduces specific sampling algorithsm: rejection sampling, MCMC, Metropolis, HMC, and Nuts. It describes specific diagonstics including the trace plot, Rhat, the autocorrelation plot, ESS, MCSE, and divergences. Different diagnostics reflect different failure modes of the sampling algoirthm. The lecture ends by returning to workflow validation: prior predictive checks, diagnostics, and posterior predictive checks each test a different failure mode.

Lecture Sections

4.1 — Bayesian workflow

This section introduces the central idea of the lecture: Bayesian analysis is not just **data → model → sample → posterior → inference**. Each stage can fail, so the workflow requires explicit checks before posterior inference should be trusted. The three major checks introduced here are model coherence, sampling validity, and model fit.

New Definitions and Terms

- **Bayesian workflow** — an iterative process for building, checking, sampling, and revising statistical models
- **Sampling validity** — whether the algorithm actually sampled from the posterior distribution
- **Model fit** — whether the fitted model can generate data resembling the observed data

- **Prior predictive sampling** — simulating parameters from the prior and data from the likelihood before fitting the model
- **Posterior predictive sampling** — simulating new data from parameter values drawn from the posterior
- **Iterative workflow** — the process of revising data procedures, priors, models, parameterizations, or sampler settings when checks fail

Figure

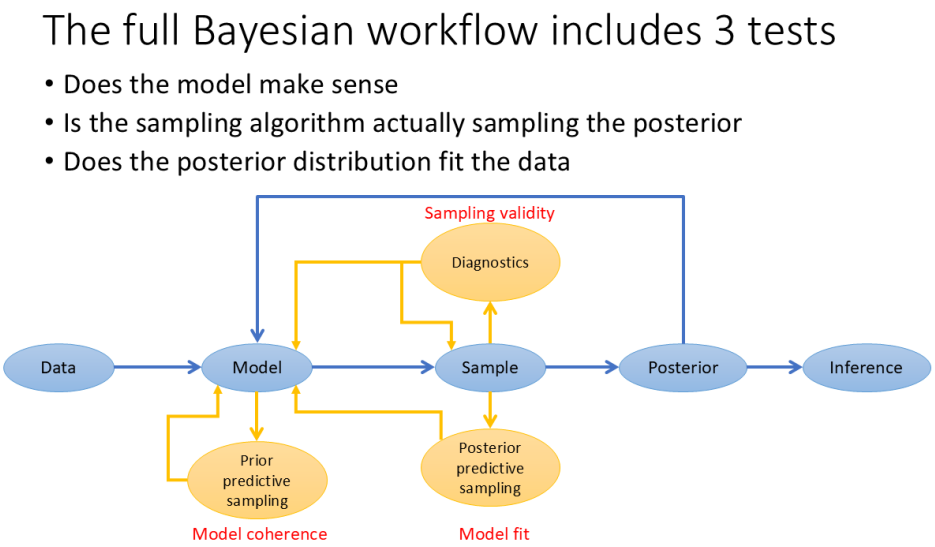


Figure 1: Slide 6 — The full Bayesian workflow includes 3 tests

---

4.2 — Sampling challenges

This section explains why `pm.sample()` cannot always be treated as a simple random-number generator. In conjugate models, such as the beta-binomial model, the posterior belongs to a known distribution family. But with a non-Beta prior such as a truncated normal prior on a coin probability, the posterior is only proportional to likelihood times prior, and ordinary distribution samplers are no longer enough. Rejection sampling is introduced as a simple general method for sampling a hard target distribution by sampling from an easier proposal distribution.

New Definitions and Terms

- **Truncated normal prior** — a normal prior restricted to a bounded interval, here used for $ heta [0,1]$
- **Rejection sampling** — sampling from an easier proposal distribution and accepting only some proposed draws
- **Proposal function** — the easy-to-sample distribution used to generate candidate draws
- **Target distribution** — the distribution the chain is designed to sample from, usually the posterior
- **Acceptance rate** — the fraction of proposed draws kept by the sampler

Formulas

$$p(\theta \mid y) \propto p(y \mid \theta)p(\theta)$$

The posterior is proportional to the likelihood times the prior. The proportional form is enough for many samplers, even when the normalizing constant is unknown.

$$Mq(x) \geq p(x)$$

In rejection sampling, the scaled proposal must cover the target distribution.

$$\text{accept if } u < \frac{p(x)}{Mq(x)}$$

A candidate draw is kept with probability equal to the target height divided by the scaled proposal height.

Figure

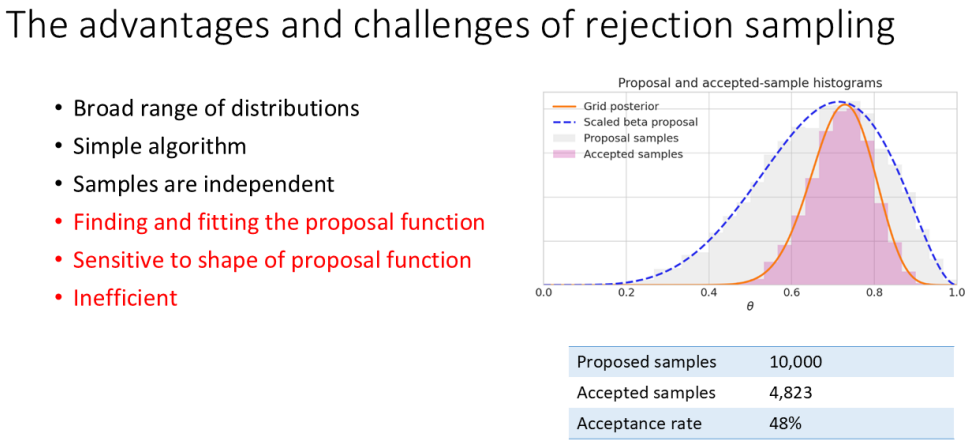


Figure 2: Slide 14 — The advantages and challenges of rejection sampling

---

4.3 — Monte-Carlo Markov Chain

This section introduces MCMC as a way to turn the global proposal problem of rejection sampling into a local walk through the posterior. Each new proposal is centered on the previous accepted draw, making proposals easier to construct and often easier to accept. The cost is that the draws are no longer independent, so the chain must explore the full posterior well enough before we can trust it.

New Definitions and Terms

- **MCMC** — Monte Carlo Markov Chain; a family of algorithms that generate dependent draws from a target distribution
- **Local proposal** — a proposed next draw centered near the current draw
- **Connected support** — the requirement that the sampler can move through the full region where the posterior has probability
- **Warm-up / burn-in** — early draws discarded before the chain has reached useful sampling behavior
- **Convergence** — practical evidence that the chains are exploring the same posterior distribution

Figure

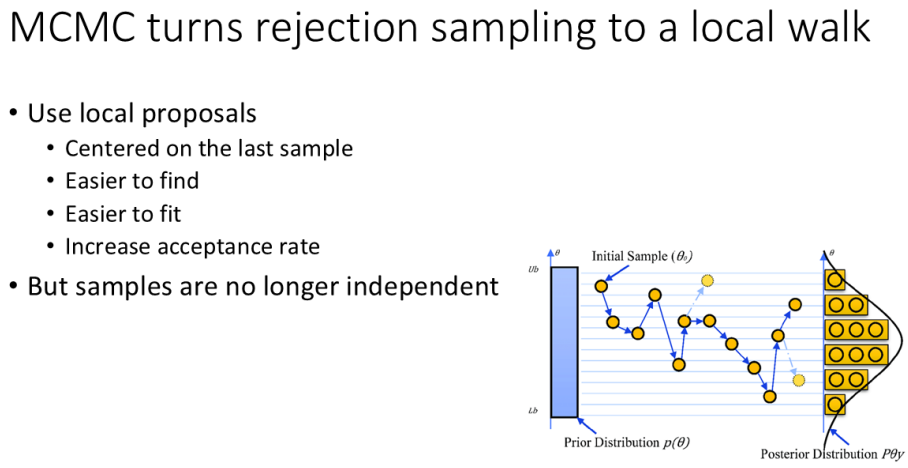


Figure 3: Slide 17 — MCMC turns rejection sampling to a local walk

---

4.4 — Metropolis

This section develops the Metropolis algorithm as the simplest MCMC sampler. The algorithm proposes one move at a time, always accepts moves to higher probability, and sometimes accepts moves to lower probability. The key intuition is that Metropolis is a sampler, not an optimizer: it must spend more time in high-probability regions while still exploring lower-probability regions often enough to reproduce the full target distribution.

New Definitions and Terms

- **Metropolis algorithm** — an MCMC algorithm that proposes local moves and accepts them according to a probability ratio
- **Metropolis-Hastings** — a generalization of Metropolis that corrects for asymmetric proposal distributions
- **Acceptance probability** — the probability of accepting a proposed move
- **Detailed balance** — equality of probability flow between two states in both directions under the target distribution

Formulas

$$a(x, x') = \min\left(1, \frac{p(x')}{p(x)}\right)$$

For symmetric proposals, the Metropolis acceptance probability depends only on the ratio of target probabilities.

$$\pi(x)T(x \to x') = \pi(x')T(x' \to x)$$

Detailed balance means the probability flow from x to x' equals the flow from x' to x when the chain has reached the target distribution.

$$a(x, x') = \min\left(1, \frac{p(x')q(x \mid x')}{p(x)q(x' \mid x)}\right)$$

Metropolis-Hastings adjusts the acceptance rule when the proposal is asymmetric.

Figure

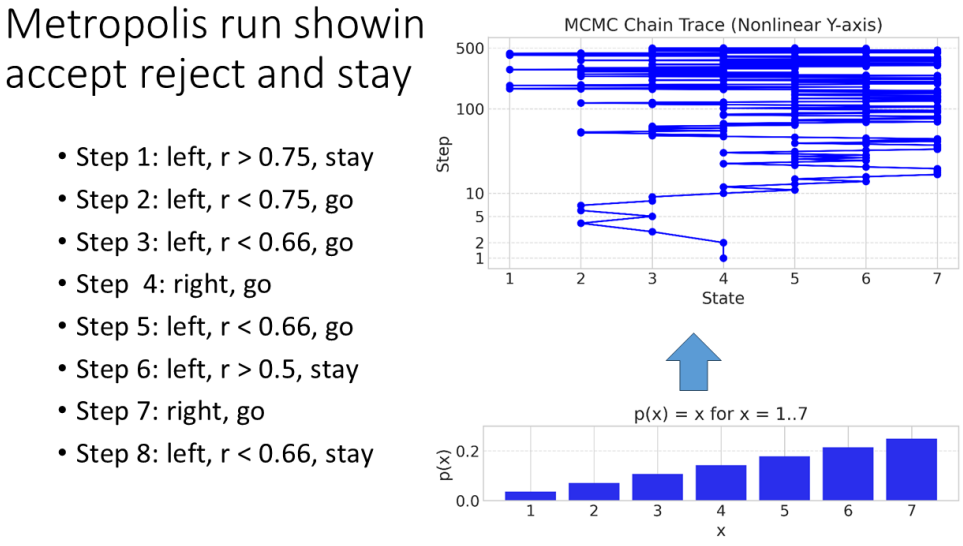


Figure 4: Slide 23 — Metropolis run showing accept reject and stay

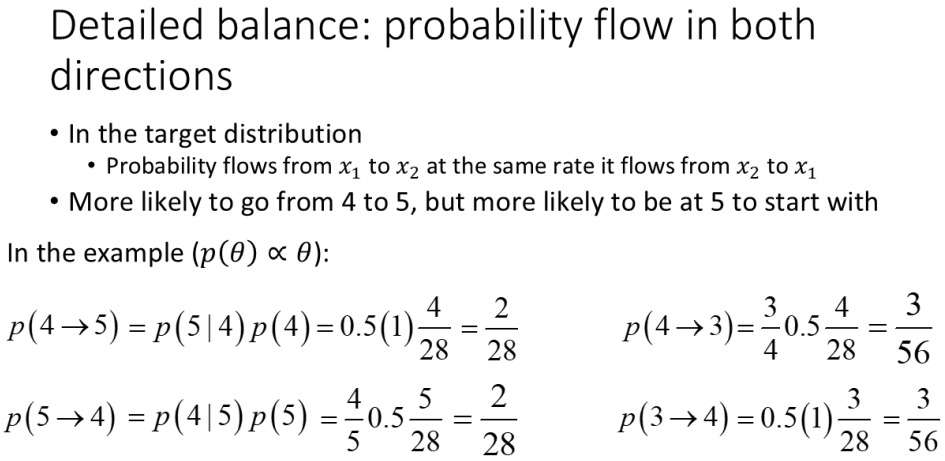


Figure 5: Slide 28 — Detailed balance: probability flow in both directions

---

4.5 — Sampling diagnostics

This section separates **sampling diagnostics** from model checking. Diagnostics ask whether the sampler successfully approximated the posterior of the specified model; they do not tell us whether the model is scientifically sensible or whether it fits the observed data. The main diagnostic themes are whether chains mix with one another and whether within-chain samples carry enough independent information.

New Definitions and Terms

- **Sampling diagnostics** — checks of whether the sampling algorithm approximately sampled the posterior
- **Mixing** — the degree to which chains overlap and explore the same posterior region
- **Trace plot** — a plot of draws across iterations, used to inspect chain behavior
- **Stuck chain** — a chain that remains trapped in one region instead of exploring the posterior
- $\widehat{R}$ — a convergence diagnostic comparing between-chain variation to within-chain variation
- **Autocorrelation** — dependence between nearby draws in a chain
- **ESS** — effective sample size; the number of independent draws with roughly equivalent information content

- **MCSE** — Monte Carlo standard error; uncertainty in posterior summaries caused by finite sampling

Formulas

$$\text{MCSE} \approx \frac{\text{sd}(\theta)}{\sqrt{\text{ESS}}}$$

Monte Carlo uncertainty gets smaller when the posterior is sampled more effectively.

Figure

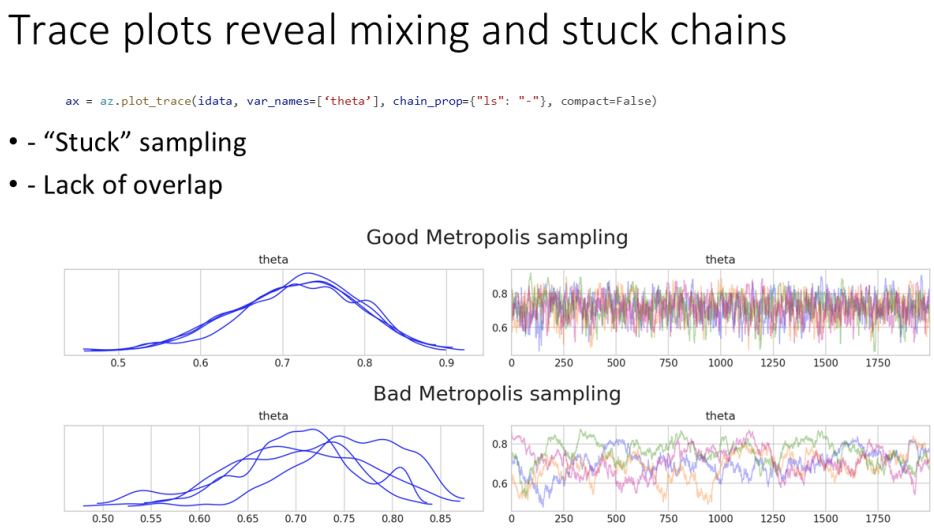


Figure 6: Slide 39 — Trace plots reveal mixing and stuck chains

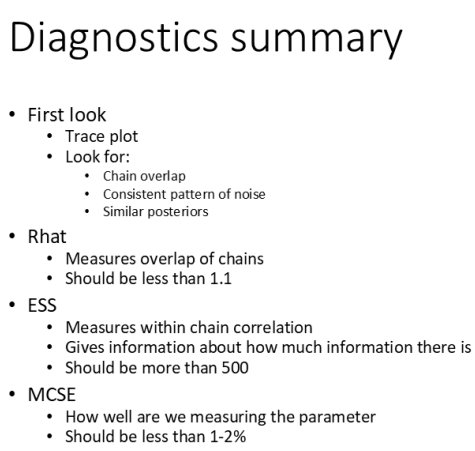


Figure 7: Slide 44 — Diagnostics summary

4.6 — HMC and NUTS

This section explains why simple random-walk proposals become inefficient in higher-dimensional and correlated posteriors. HMC uses gradients to build proposals that move through high-probability regions more efficiently. The key conceptual shift is that HMC is not doing gradient ascent toward the mode; in high dimensions, most probability mass is in the **typical set**, not at the point of highest density.

HMC simulates a physical system with momentum and leapfrog integration, then applies a Metropolis acceptance step. NUTS automates the difficult choice of HMC path length by stopping before the trajectory turns back on itself.

New Definitions and Terms

- **HMC** — Hamiltonian Monte Carlo; an MCMC method using gradients and simulated physical dynamics to propose efficient moves
- **Typical set** — the region where most posterior probability mass lies
- **Probability mass** — the combination of probability density and volume
- **Momentum** — the auxiliary quantity that controls HMC movement through parameter space
- **Leapfrog integration** — the numerical method used to approximate HMC trajectories

- **Target acceptance probability** — tuning parameter controlling the tradeoff between slower, more careful proposals and faster sampling
- **NUTS** — No-U-Turn Sampler; an HMC variant that automatically chooses path length
- **Maximum tree depth** — NUTS tuning parameter controlling the maximum number of trajectory steps

Figure

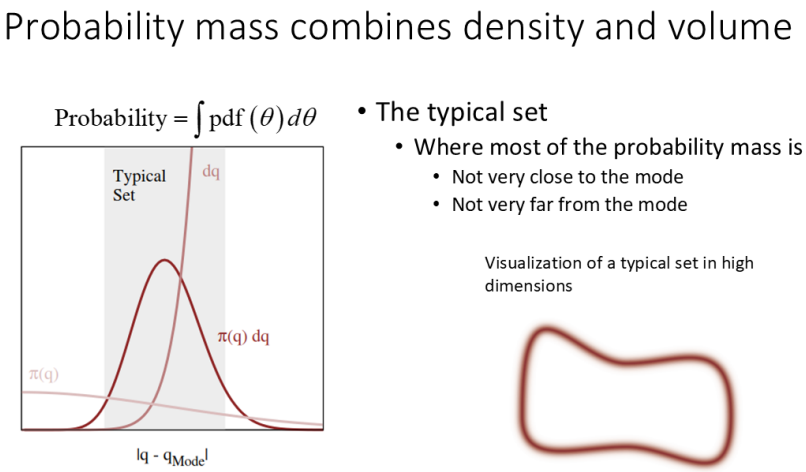


Figure 8: Slide 52 — Probability mass combines density and volume

HMC proposals stay in the typical set

- HMC moves perpendicular to the gradient
  - Like a satellite
  - Not towards the mode
  - Not away from it



Figure 9: Slide 53 — HMC proposals stay in the typical set

NUTS automates HMC path length selection

- NUTS: No U-Turn Sampling
  - A variant of HMC
  - Explores multiple possible jumps in two direcitons
  - Default state-of-the-art sampler in PPCs
- Parameters
  - Target acceptance probability
    - Controls step size
    - Higher gives slower but better sampling
  - Maximum tree depth
    - Controls the number of steps
    - Higher gives slower but better sampling

HMC with u turns

Figure 10: Slide 59 — NUTS automates HMC path length selection

4.7 — Divergences: HMC diagnostics

This section introduces divergences as a specific diagnostic for HMC and NUTS. A divergence occurs when the approximate physical simulation fails badly enough that the sampler detects an unreliable trajectory. In lecture terms, the satellite has either flown away from the stable orbit or crashed into the gravity well; in practical terms, posterior samples from a run with divergences cannot simply be trusted.

New Definitions and Terms

- **Divergence** — a detected failure of the HMC/NUTS trajectory simulation
- **Energy error** — change in the simulated Hamiltonian energy caused by numerical approximation failure
- **Problematic region** — an area of parameter space where the sampler repeatedly diverges
- **Reparameterization** — rewriting the model to make the posterior easier to sample

Figure

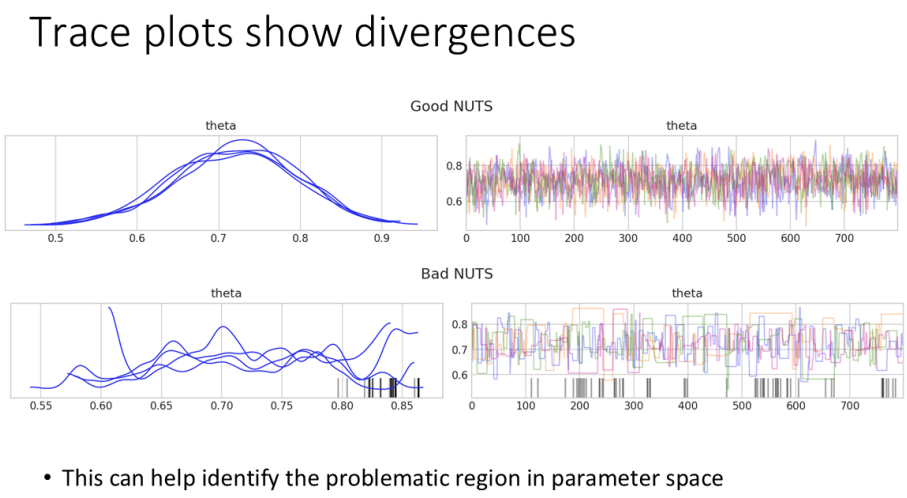


Figure 11: Slide 64 — Trace plots show divergences

4.8 — PyTensor

This section explains where PyMC gets the mathematical objects needed by modern samplers. The user writes a probabilistic model in PyMC, but HMC and NUTS require numerical functions for the log probability density and its gradients. PyTensor provides this computational backend by turning symbolic model variables into probability calculations and computational graphs.

New Definitions and Terms

- **PyTensor** — PyMC's computational backend for symbolic probability calculations and gradients
- **Symbolic variable** — a variable that represents a mathematical quantity before numerical values are supplied
- **Computational graph** — a graph representation of the operations needed to compute model probabilities
- **Log joint probability density** — the log probability of parameters and observed data together
- **Gradient of the log probability** — the derivative information used by HMC/NUTS to build efficient proposals
- **Transformed parameter** — an internal representation of a constrained parameter used by PyMC during sampling

Formulas

$$\log p(\theta, y) = \log p(\theta) + \log p(y \mid \theta)$$

The log joint probability combines the prior and likelihood terms.

$$\nabla_\theta \log p(\theta, y)$$

The gradient tells the sampler how log probability changes as parameter values change.

Figure

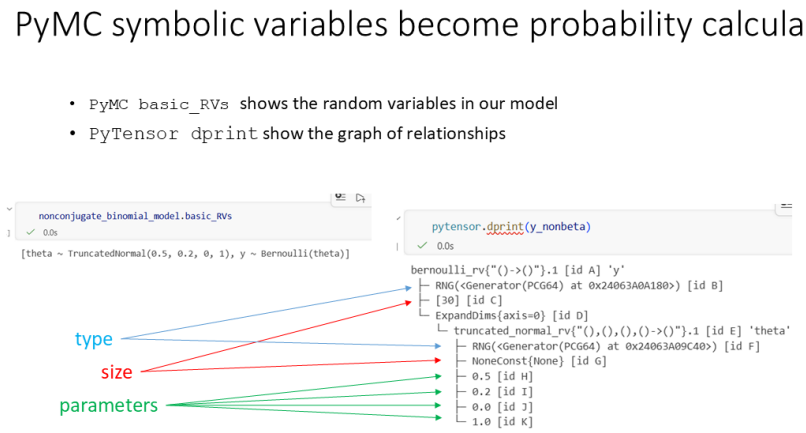


Figure 12: Slide 69 — PyMC symbolic variables become probability calculations

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4.9 — Workflow validation

This final section returns to the full Bayesian workflow and clarifies what each check is testing. Prior predictive checks test whether the priors and likelihood together generate plausible data before seeing the observed dataset. Diagnostics test whether sampling succeeded. Posterior predictive checks test whether the fitted model can generate data that resemble the observed data.

The important workflow message is that failures have different meanings. A failed prior predictive check points back to the model or priors. Failed diagnostics point to the sampler, parameterization, or model complexity. A failed posterior predictive check points to model fit. Only after these checks does the posterior become a serious basis for inference.

New Definitions and Terms

- **Predictive sampling** — generating data by first drawing parameters and then drawing data conditional on those parameters

Figure

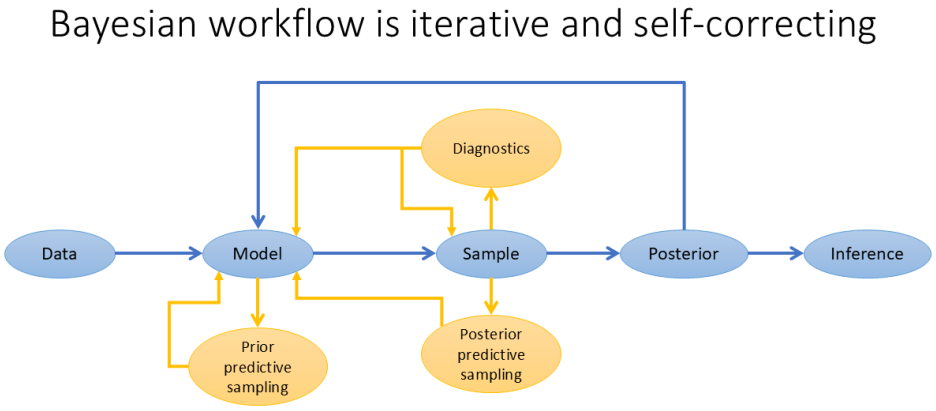


Figure 13: Slide 73 — Bayesian workflow is iterative and self-correcting

Each failure mode has different solutions

Step	Test	Solutions
Data collection	Data validation procedures	Improve methodology Develop protocols Document processes
Propose model	Generate graph Take samples	Test dimensions Debug
Sample prior predictive	Prior predictive plots	Simplify model Change priors
Sample posterior	Trace plots Rhat ESS MCSE Divergences	Improve sampling Change initialization Reparameterize model Simplify model Change priors
Sample posterior predictive	Posterior predictive plots Bayesian p values	Change model

87

Figure 14: Slide 87 — Each failure mode has different solutions

Lecture 5 Review Handout

Statistics 367-1-4361

Lecture Overview

Lecture 5 moves from simple coin-flip models to data with **group structure**, using hospital death rates as the running example. The lecture compares pooled, unpooled, and hierarchical models, showing how hierarchical models share information across groups while still allowing group-level differences. It then uses the Bayesian workflow to fit and check a hierarchical beta-binomial model, interpret shrinkage, diagnose funnel geometry, and extend the same logic to multi-level models. The lecture ends by summarizing both the benefits and the pitfalls of hierarchical modeling.

Lecture Sections

5.1 — Grouped data

This section introduces the problem that motivates the whole lecture: most biomedical data are not just a single flat sample. Measurements are often grouped by hospital, subject, session, animal, cell culture, treatment, prosthesis, day, or trial. The hospital cardiac-arrest example shows the central question: apparent differences between hospitals may reflect real structure, sampling noise, or both.

New Definitions and Terms

- **Grouped data** — data in which observations are organized into meaningful groups, such as hospitals, subjects, sessions, or treatment groups
- **Group structure** — the organization of data into levels that may affect how observations should be modeled

Figure

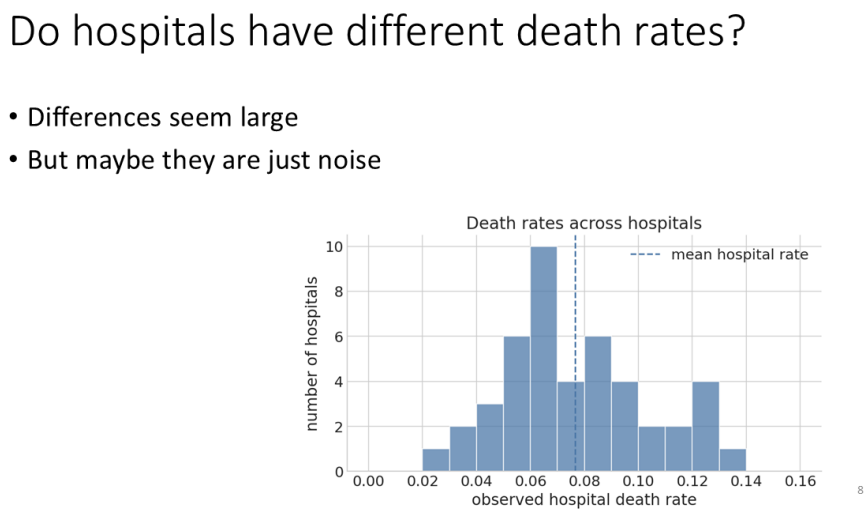


Figure 1: Slide 8 — Do hospitals have different death rates?

5.2 — Pooled model

A pooled model collapses the hospital structure and estimates one death-rate parameter from all cases together. This gives a simple baseline model: one probability of death is used for all hospitals. The workflow then asks whether the prior predictive data are plausible and whether the posterior predictive data reproduce the variability seen across hospitals.

New Definitions and Terms

- **Pooled model** — a model that ignores group structure and estimates a single parameter from all observations together
- **Pooling** — combining data across groups so that the groups do not have separate parameters

Formulas

$$\theta \sim \text{Beta}(\alpha_\omega, \beta_\omega)$$

The pooled model has one death-rate parameter, shared across all hospitals.

$$z_i \sim \text{Binomial}(n_i, \theta)$$

Each hospital's observed deaths are modeled using the same probability parameter.

Figure

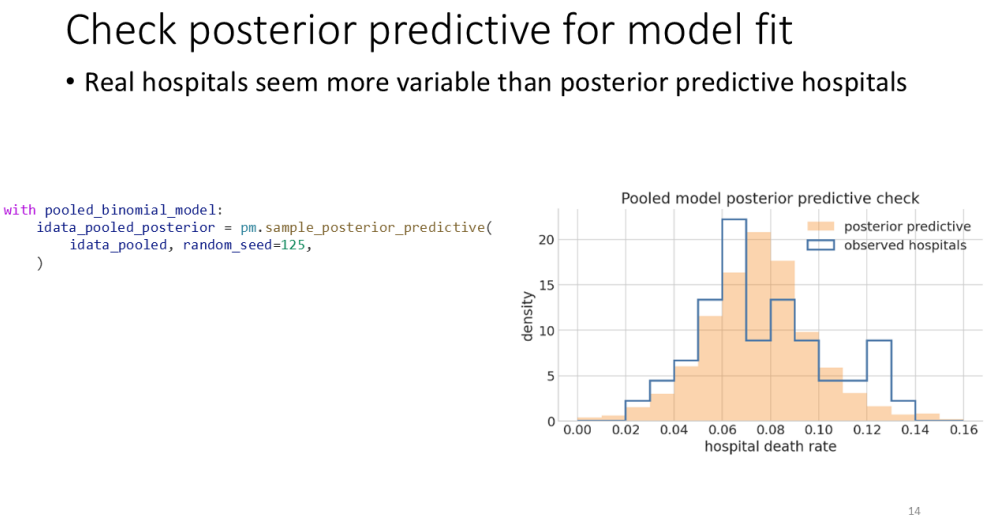


Figure 2: Slide 14 — Check posterior predictive for model fit

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5.3 — Unpooled model

An unpooled model gives each hospital its own probability parameter. This captures hospital-to-hospital differences, but it does not allow hospitals to inform each other. The posterior plots show the resulting increase in variability and uncertainty for individual hospital parameters.

New Definitions and Terms

- **Unpooled model** — a model that estimates separate parameters for each group without sharing information across groups
- **No information sharing** — a modeling structure in which data from one group do not affect the parameter estimates for other groups

Formulas

$$\theta_i \sim \text{Beta}(\alpha_\omega, \beta_\omega)$$

The unpooled model assigns a separate death-rate parameter to each hospital.

$$z_i \sim \text{Binomial}(n_i, \theta_i)$$

Each hospital's observed deaths depend only on its own parameter.

Figure

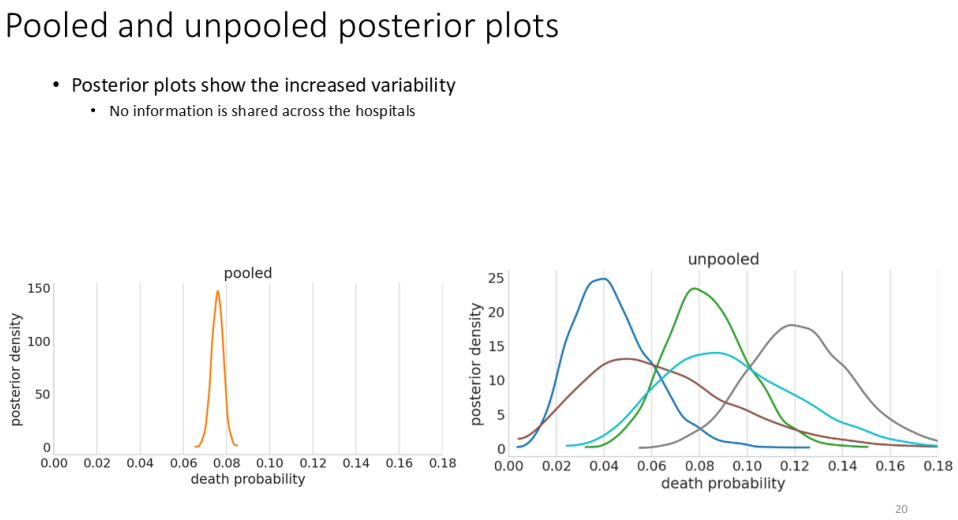


Figure 3: Slide 20 — Pooled and unpooled posterior plots

5.4 — Hierarchical model

The hierarchical model makes the parameters of the hospital-level prior into unknown quantities. These new parameters are estimated from the data, so each hospital parameter is influenced by its own data and by the other hospitals. This produces information sharing while still preserving hospital-level differences.

New Definitions and Terms

- **Hierarchical model** — a model in which parameters for lower-level groups are themselves modeled as coming from a higher-level distribution
- **Hyperparameter** — a parameter of a prior distribution, such as the population-level mean or concentration governing hospital-level parameters
- **Hyperprior** — a prior distribution placed on a hyperparameter
- **Information sharing** — the way group-level parameter estimates are influenced by data from other groups through shared hyperparameters

Formulas

$$\mu_\theta \sim \text{Beta}(\mu_{\mu_\theta}, \nu_{\mu_\theta})$$

The overall mean death rate is treated as an unknown hyperparameter.

$$\nu_\theta \sim \text{HalfNormal}(\sigma_{\nu_\theta})$$

The similarity of hospital death rates is also treated as an unknown hyperparameter.

$$\theta_i \sim \text{Beta}(\mu_\theta, \nu_\theta)$$

Hospital-specific death rates are drawn from a shared population distribution.

Figure

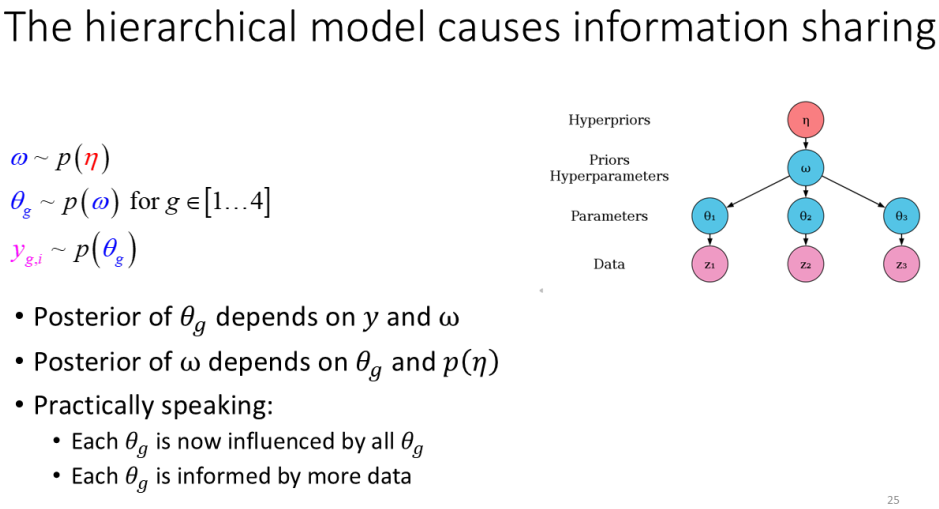


Figure 4: Slide 25 — The hierarchical model causes information sharing

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5.4 — Parameterizing Beta

This section shows that the same Beta distribution can be written using different parameterizations. The lecture first connects the Beta distribution to successes and failures, then rewrites it in terms of a mean and a concentration, and finally in terms of a mean and width. The hierarchical model uses the mean/concentration parameterization because it separates the center of the hospital distribution from the similarity of hospitals.

New Definitions and Terms

- **Concentration** — a Beta-distribution parameter controlling how tightly values cluster around the mean
- **Mean/concentration parameterization** — a way of writing the Beta distribution using a mean parameter and a concentration parameter
- **Mean/width parameterization** — a way of writing the Beta distribution using a mean parameter and a spread parameter

Formulas

$$\text{Beta}(\alpha, \beta)$$

The familiar parameterization treats α as success-like and β as failure-like.

$$\text{Beta}(\mu, \nu)$$

The lecture uses μ as the mean and ν as the concentration.

Figure

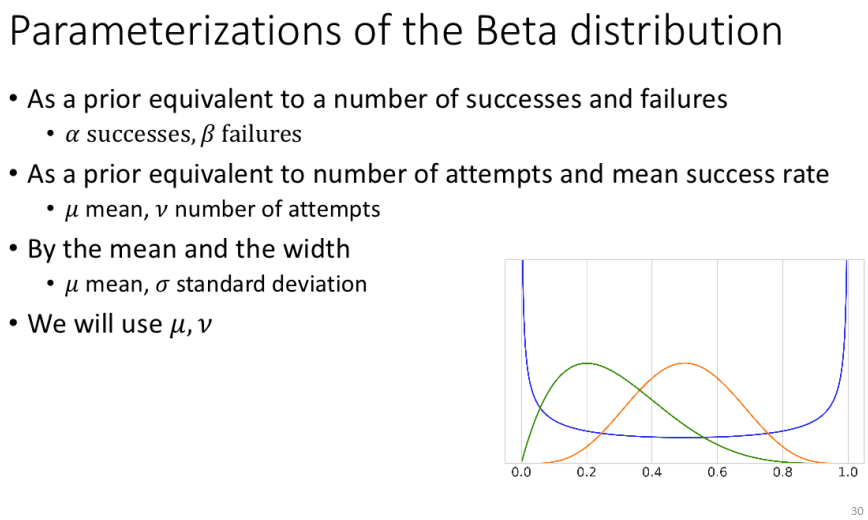


Figure 5: Slide 30 — Parameterizations of the Beta distribution

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5.5 — Coding hierarchical models

This section translates the graphical hierarchy into PyMC code. The data likelihood sits at the bottom, hospital-level parameters sit above the data, and hyperparameters with hyperpriors sit above the hospital-level parameters. The code also shows how the same model can be converted into a graphical model, making the hierarchy visible.

New Definitions and Terms

- **Model hierarchy** — the layered structure from data, to group-level parameters, to hyperparameters, to hyperpriors
- **Graphical hierarchy** — a model diagram that makes the levels of a hierarchical model explicit

Figure

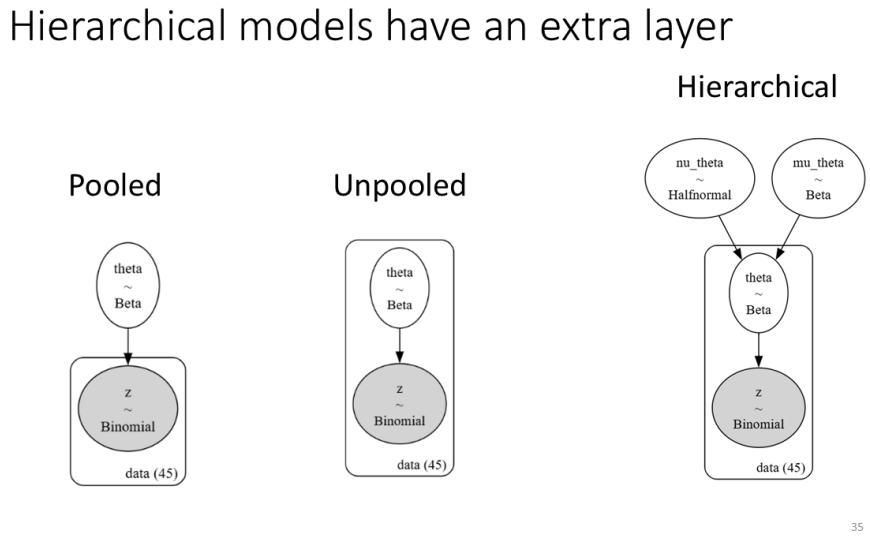


Figure 6: Slide 35 — Hierarchical models have an extra layer

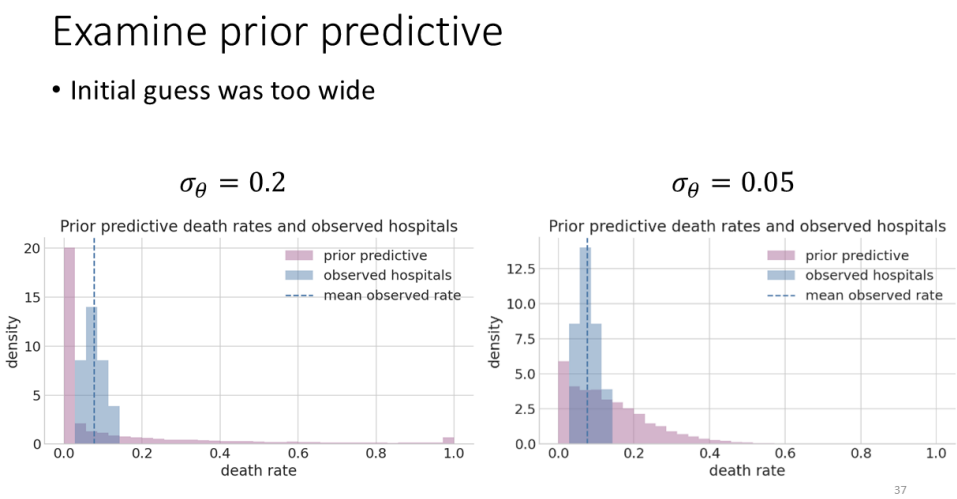


Figure 7: Slide 37 — Examine prior predictive

5.6 — Fitting hierarchical models

This section applies the workflow after fitting the hierarchical model. The lecture checks summary diagnostics, trace plots for hyperparameters, trace plots for selected hospital parameters, pair plots, divergences, and posterior predictive simulations. Only after these checks look reasonable is the posterior treated as interpretable.

New Definitions and Terms

- **Pair plot** — a plot of joint posterior draws for two parameters, used both for interpretation and for diagnosing difficult posterior geometry
- **Joint draw** — one posterior draw containing a self-consistent set of values for all parameters in the model

Figure

Pair plots show that parameters are drawn jointly

- Every dot is one draw
  - Each draw contains all parameters
  - Parameters are self-consistent
    - Within a draw
- Pair plots are also used for diagnosis
  - Divergences can reveal problematic regions of the posterior
  - Strong correlations may mean the model is not well specified

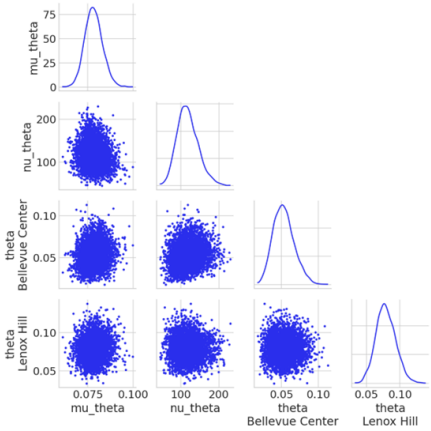
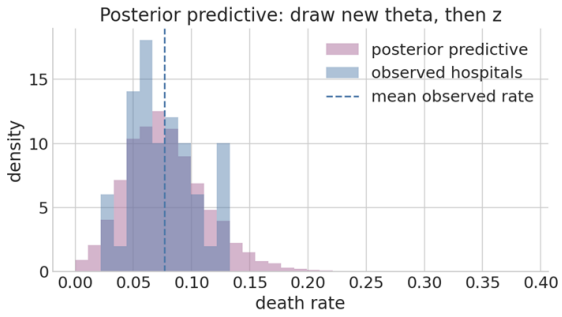


Figure 8: Slide 43 — Pair plots show that parameters are drawn jointly

Posterior predictive data is similar to real data

- For every sample:
  - Use sampled μ_θ, ν_θ
  - Draw new θ_i
  - For each new θ_i
    - Draw a new z_i



44

Figure 9: Slide 44 — Posterior predictive data is similar to real data

5.7 — The posterior distribution

This section interprets the fitted hierarchical posterior. Compared with the pooled model, the hierarchical model has less certainty about the population mean because the mean reflects variation across hospitals, not just thousands of individual cases. Compared with the unpooled model, hospital-specific estimates are more certain and more similar because the hospitals inform one another through the hyperparameters.

New Definitions and Terms

- **Shrinkage** — movement of group-level estimates toward the population mean caused by information sharing in a hierarchical model
- **Regression toward the mean** — the tendency for extreme group estimates to move closer to the overall mean when group-level uncertainty is modeled
- **HDI width shrinkage** — reduction in posterior interval width, especially for groups with less data, due to hierarchical information sharing

Figure



Figure 10: Slide 50 — Shrinkage is regression towards the mean

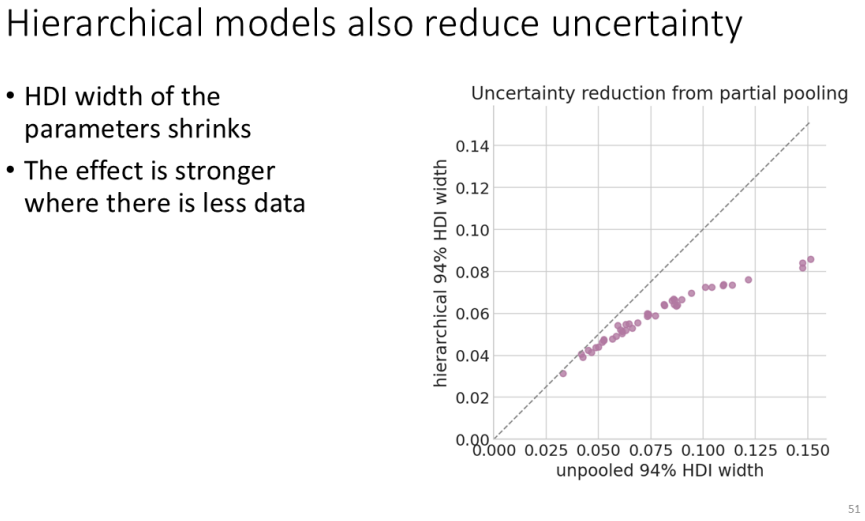


Figure 11: Slide 51 — Hierarchical models also reduce uncertainty

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5.8 — Funnels

Hierarchical models can produce difficult posterior geometry. A funnel appears when an unknown scale parameter controls how widely unknown location parameters can vary. When the scale becomes very small or very large, the shape of the posterior can become hard for NUTS to explore with one step size.

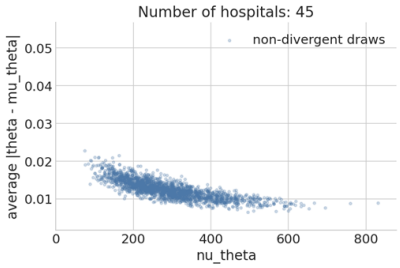
New Definitions and Terms

- **Funnel** — posterior geometry in which changes in a scale parameter strongly change the available range of other parameters
- **Location parameter** — a parameter describing the center of a distribution, such as a mean, median, or mode
- **Scale parameter** — a parameter describing the spread of a distribution, such as a standard deviation, variance, or interquartile range
- **Centered parameterization** — a hierarchical parameterization in which group parameters are drawn directly from a distribution controlled by unknown hyperparameters
- **Non-centered parameterization** — a reparameterization that separates standardized group variation from the unknown scale parameter

Figure

Our hierarchical model has a funnel

- As ν_θ gets large
  - θ get closer and closer together



55

Figure 12: Slide 55 — Our hierarchical model has a funnel

When to worry about funnels

- An unknown **scale** is used as the scale of an unknown **location**
- The **scale** parameter can go to 0
  - The data is weak
  - The prior allows 0

59

Figure 13: Slide 59 — When to worry about funnels

5.9 — Multi-level models

This section extends the same hierarchical logic to models with more than one level of grouping. The hospital data can be grouped by borough or by hospital type, and the slides build a borough-level model step by step: likelihood, hospital parameters, borough hyperparameters, and hyperpriors. The key idea is that Bayesian models can represent nested structure rather than forcing all variation into a single level.

New Definitions and Terms

- **Multi-level model** — a hierarchical model with more than one level of group structure
- **Nested hierarchy** — a structure in which lower-level units belong to intermediate groups, which themselves belong to a higher-level population
- **Group index** — an index variable used in code to connect each observation or group-level parameter to the correct higher-level group
- **Grand mean** — the population-level mean governing the distribution of group-level means

Formulas

$$\mu_{\theta,g} \sim \text{Beta}(\mu_{\mu_\theta}, \nu_{\mu_\theta})$$

Each borough has its own mean death rate, drawn from a higher-level distribution.

$$\nu_{\theta,g} \sim \text{HalfNormal}(\sigma_{\nu_\theta})$$

Each borough can also have its own within-borough similarity parameter.

$$\theta_i \sim \text{Beta}(\mu_{\theta,g[i]}, \nu_{\theta,g[i]})$$

Each hospital's death rate is tied to the borough to which the hospital belongs.

Figure

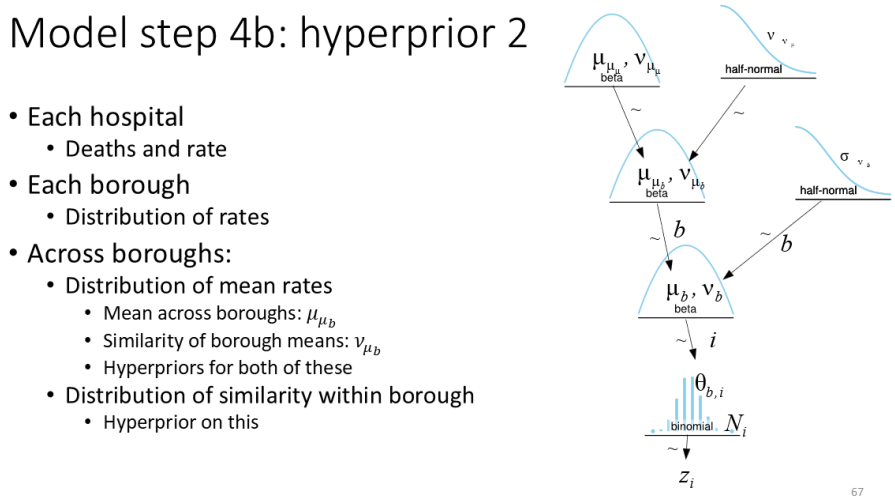


Figure 14: Slide 67 — Model step 4b: hyperprior 2

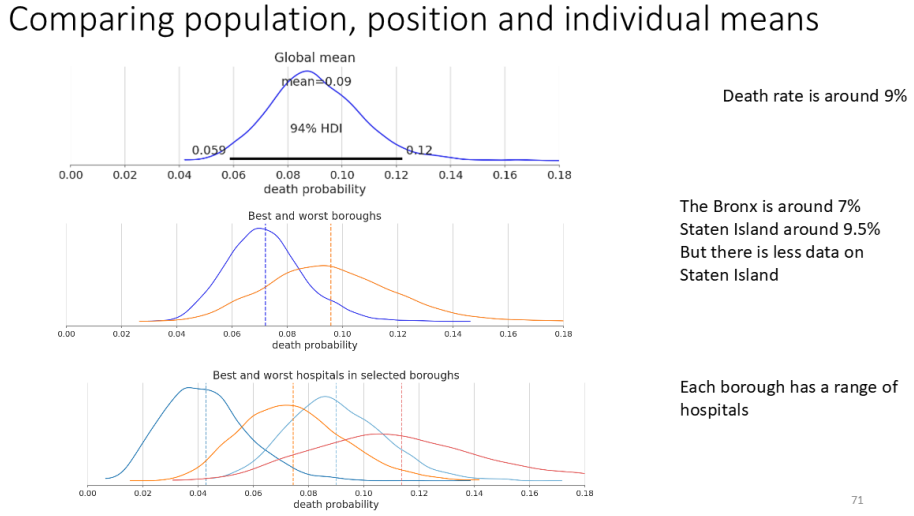


Figure 15: Slide 71 — Comparing population, position and individual means

5.10 — Summing up / looking forward

The lecture closes by emphasizing the tradeoff in hierarchical modeling. Hierarchical models can represent structured data, balance population and group-level information, and provide more robust estimates when some groups have little data. The cost is greater computational and interpretational complexity, including a stronger need for prior predictive checks, sampling diagnostics, posterior predictive checks, and careful multi-level interpretation.

Figure

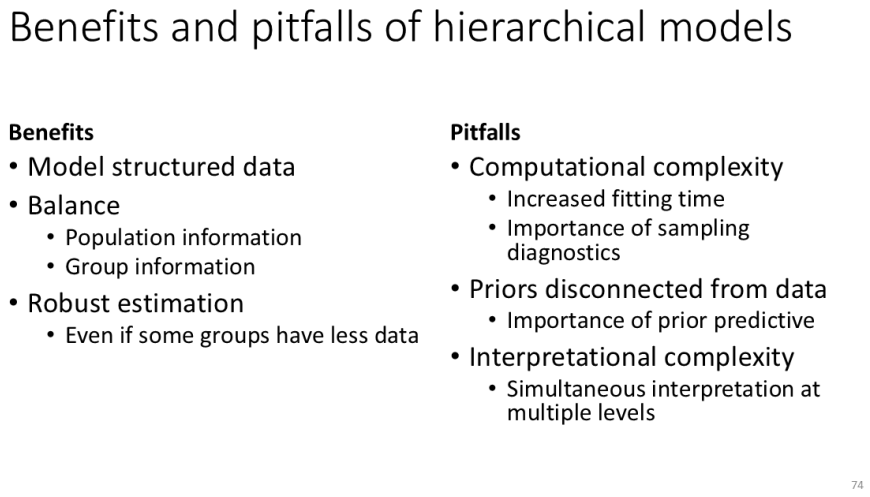


Figure 16: Slide 74 — Benefits and pitfalls of hierarchical models

Lecture 6 Review Handout

Statistics 367-1-4361

Lecture Overview

Lecture 6 introduces hypothesis testing as a way to decide whether data contradict, support, or fail to resolve a null model. The lecture starts by distinguishing raw and normalized effect sizes, then compares Savage-Dickey, Bayes factors, and HDI/ROPE tests as Bayesian approaches to hypothesis testing. It then introduces significance, type I and type II errors, power, and pre-hoc and post-hoc power analysis. The lecture ends by connecting these ideas to frequentist tests and to a two-group hospital example using PCI and non-PCI groups.

Lecture Sections

6.1 — Effect sizes

This section introduces the null model and defines effect size as the difference between reality and the null hypothesis. Raw effect sizes are expressed in the original units of the data and depend on domain knowledge. Normalized effect sizes are designed to be comparable across experiments or domains, but they can hide practical meaning.

New Definitions and Terms

- **Null model** — a default or naïve model, often assuming no effect, no difference, no relationship, or no influence
- **Null hypothesis** — the hypothesis that the null model is true
- **Effect size** — the difference between reality and the null hypothesis
- **Raw effect size** — an effect size expressed in the actual units of the data
- **Normalized effect size** — an effect-size statistic designed to be comparable across experiments and domains
- **True effect size** — the effect size in reality
- **Measured effect size** — the effect size estimated from data
- **Cohen's h** — a normalized effect size for comparing proportions or probabilities

Formulas

$$h = 2\arcsin\sqrt{\theta} - 2\arcsin\sqrt{\theta_0}$$

Cohen's h transforms a difference between probabilities into a normalized effect-size scale.

Figure

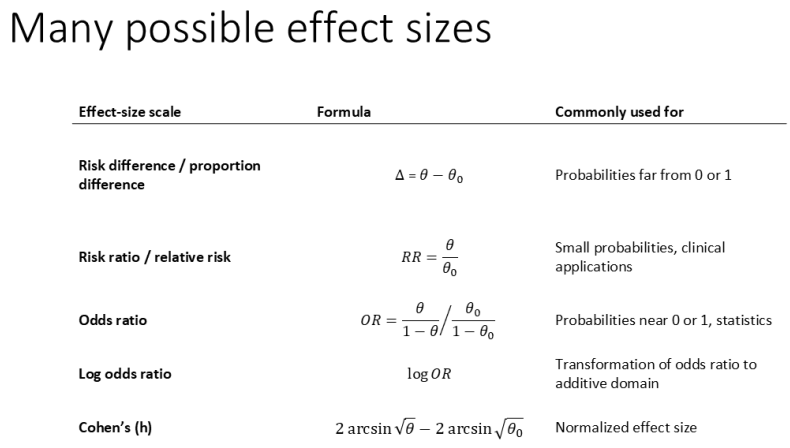


Figure 1: Slide 8 — Many possible effect sizes

6.2 — Hypothesis tests

This section defines hypothesis testing as a way to evaluate whether the null hypothesis remains credible after seeing the data. The key contrast is between effect size and evidence against the null: an estimated effect size can be large or small regardless of sample size, but a hypothesis test depends strongly on the amount of data. Savage-Dickey is introduced as a Bayesian density-ratio method for measuring evidence for or against a reference value.

New Definitions and Terms

- **Hypothesis test** — a procedure for deciding whether data support rejecting, accepting, or withholding judgment about a hypothesis
- **Savage-Dickey density ratio** — the ratio of posterior and prior density at a reference value
- **Reference value** — the parameter value being tested, usually the null value
- **Evidence against the null** — data making the null value less credible than it was under the prior
- **Evidence for the null** — data making the null value more credible than it was under the prior

Formulas

$$BF_{01} = \frac{p(\theta_0 \mid y)}{p(\theta_0)}$$

The Savage-Dickey density ratio compares posterior and prior density at the null value. A larger posterior density at the null value supports the null; a smaller posterior density supports rejecting it.

Figure

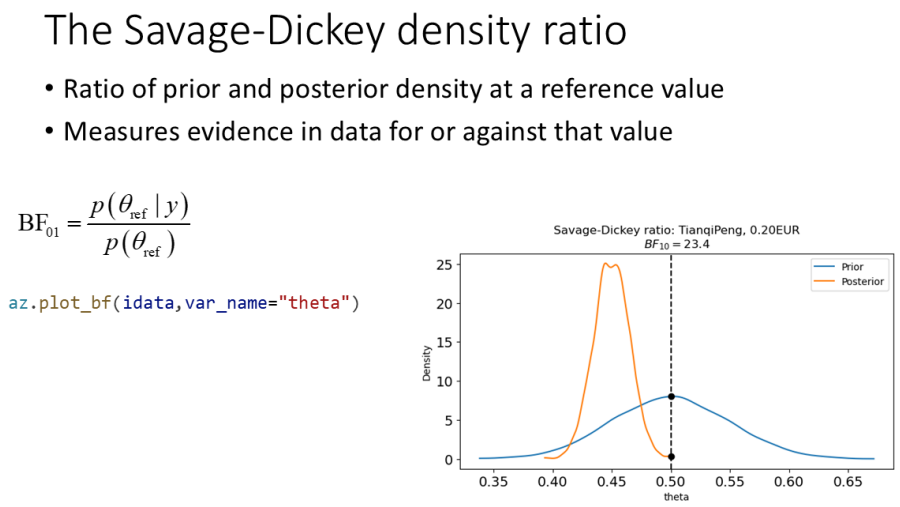


Figure 2: Slide 14 — The Savage-Dickey density ratio

Savage-Dickey depends on effect size and sample size

person	coin	N	$\hat{\theta}$	BF_{10}	BF_{01}
EJ	50CZK	200	0.44	1.46	0.68
TianqiPeng	0.20EUR	1000	0.45	23.38	0.04
davidV	0.10EUR	3660	0.48	2.47	0.40

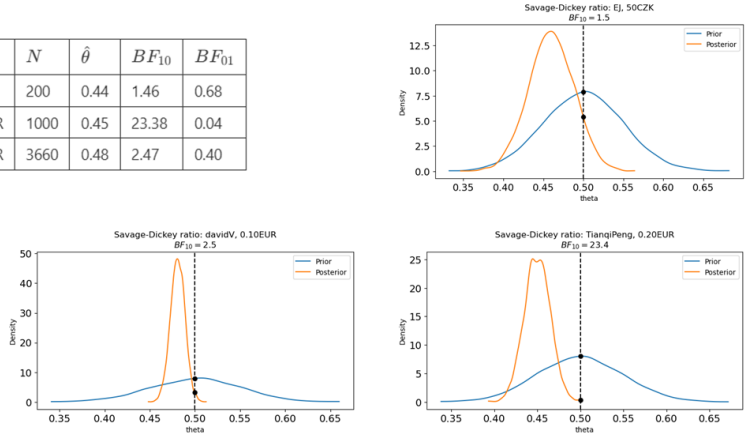


Figure 3: Slide 16 — Savage-Dickey depends on effect size and sample size

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6.3 — Baye's factor

This section places Bayes factors inside posterior odds. A Bayes factor is the part of the posterior odds that comes from the data, after separating out the prior odds between models. The lecture emphasizes that Savage-Dickey is a Bayes factor only under specific conditions, and that Bayes factors can be difficult to calculate and interpret.

New Definitions and Terms

- **Posterior odds** — the relative posterior probability of two models
- **Prior odds** — the relative prior probability of two models before seeing data
- **Bayes factor** — the part of the posterior odds contributed by the data
- **Marginal likelihood** — the likelihood of the data under a model after averaging over that model's parameters
- **Point null** — a null hypothesis that assigns one exact parameter value to the null model

Formulas

$$\frac{p(m_1 \mid y)}{p(m_2 \mid y)} = \frac{p(y \mid m_1)}{p(y \mid m_2)} \frac{p(m_1)}{p(m_2)}$$

Posterior odds equal the Bayes factor times the prior odds.

$$p(y \mid m) = \int d\theta \; p(y \mid \theta, m) p(\theta \mid m)$$

The likelihood under a model is averaged over the parameters of that model.

Figure

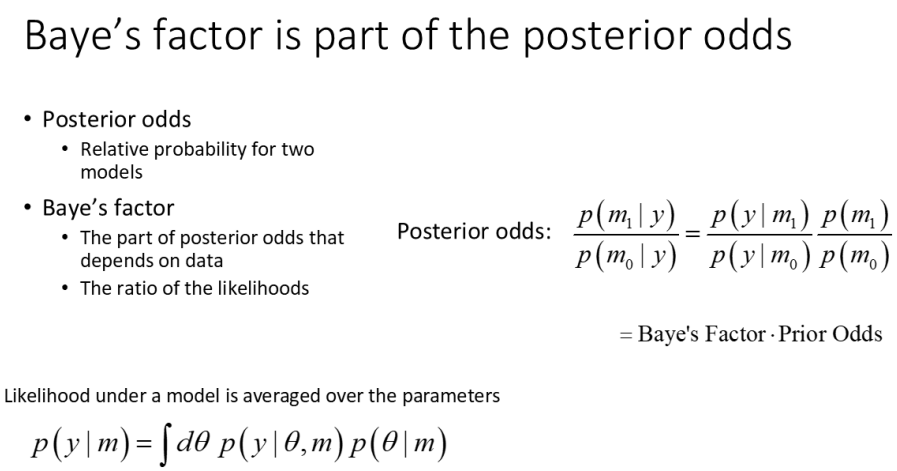


Figure 4: Slide 20 — Baye's factor is part of the posterior odds

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6.4 — ROPE and HDI test

This section motivates interval null hypotheses by arguing that exact point-null models can be unrealistic. The ROPE defines a range of parameter values that count as practically equivalent to the null. The HDI/ROPE test then compares the posterior interval to this practical-null region.

New Definitions and Terms

- **ROPE** — region of practical equivalence; parameter values where the null model is practically true
- **Practical equivalence** — a situation where an effect may not be exactly zero but is too small to matter
- **HDI/ROPE test** — a hypothesis test based on the overlap between the posterior HDI and the ROPE
- **No decision** — the result when the posterior interval and ROPE overlap without one clearly containing the other

Formulas

$$p(\theta \in ROPE) < \alpha$$

If the posterior mass in the ROPE is small enough, the null model is rejected.

Figure

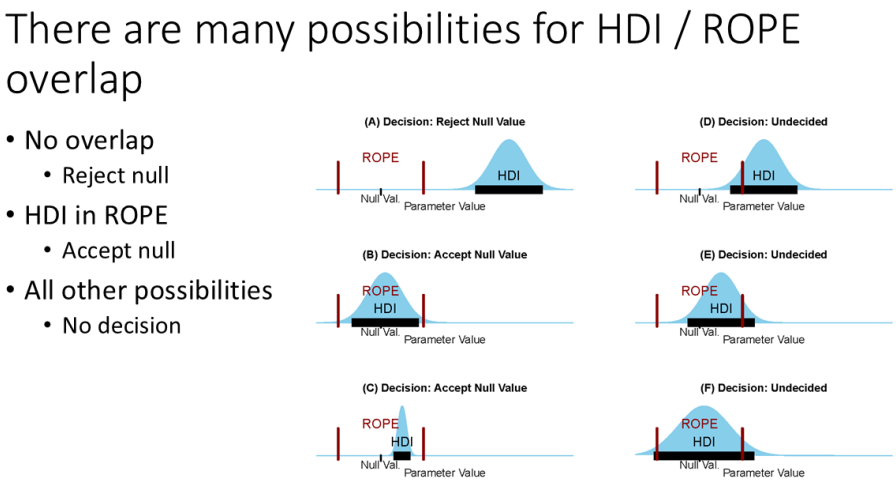


Figure 5: Slide 28 — There are many possibilities for HDI / ROPE overlap

6.5 — Significance

This section introduces significance from the perspective of type I error. A significant result is not just a posterior statement about one dataset; it is tied to the long-run probability of falsely rejecting a true null model. The lecture uses repeated simulated datasets under the null to ask whether the HDI/ROPE test is calibrated, conservative, or liberal.

New Definitions and Terms

- **Type I error** — falsely rejecting a true null model
- **error** — another name for type I error
- **Significance** — the probability of type I error for a hypothesis test
- **Significance level** — the chosen threshold α used to control type I error
- **Calibrated test** — a test whose type I error rate matches the target significance level
- **Conservative test** — a test with type I error probability smaller than α
- **Liberal test** — a test with type I error probability larger than α
- **Equivalence** — evidence that the parameter lies inside the ROPE

Formulas

$$p(\text{type I error}) < \alpha$$

A significance threshold controls the probability of falsely rejecting a true null model.

$$p(\theta \in ROPE) > \alpha$$

Equivalence means that enough posterior mass is inside the ROPE.

Figure

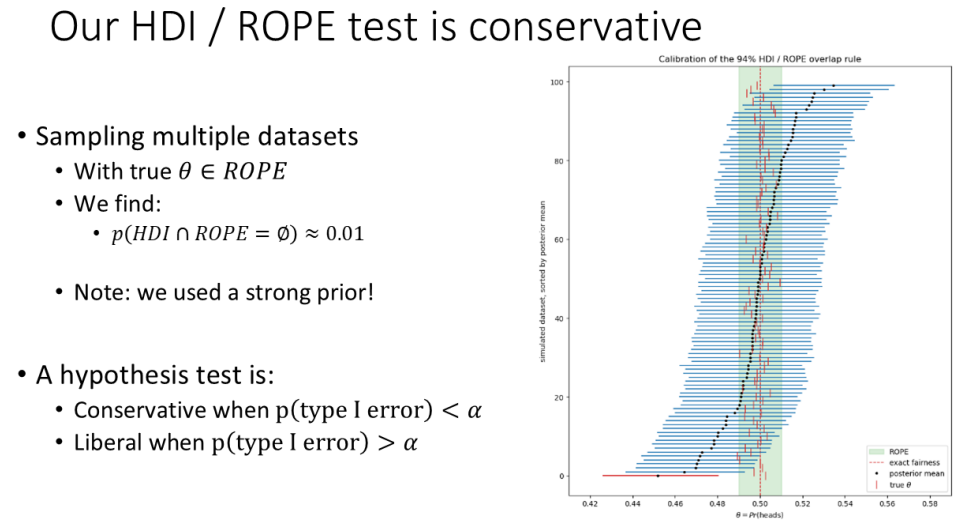


Figure 6: Slide 35 — Our HDI / ROPE test is conservative

6.6 — Basics of statistical reporting

This section shifts from analysis to reporting. Statistical reporting depends on the expectations of the context: papers, grants, pre-registrations, ethical approval, medical reporting, engineering applications, and patents may all have different norms or requirements. At the most basic level, the report should say what was measured, what was estimated, how uncertain the estimate is, what the effect size is, and whether the effect is significant.

New Definitions and Terms

- **Statistical reporting** — communicating the measurements, estimates, uncertainty, effect sizes, and hypothesis-test results expected in a scientific or applied context
- **Reporting standard** — a formal rule or informal norm for what statistical information must be reported
- **Parameter estimate** — the reported estimate of the model parameter or quantity of interest

Figure

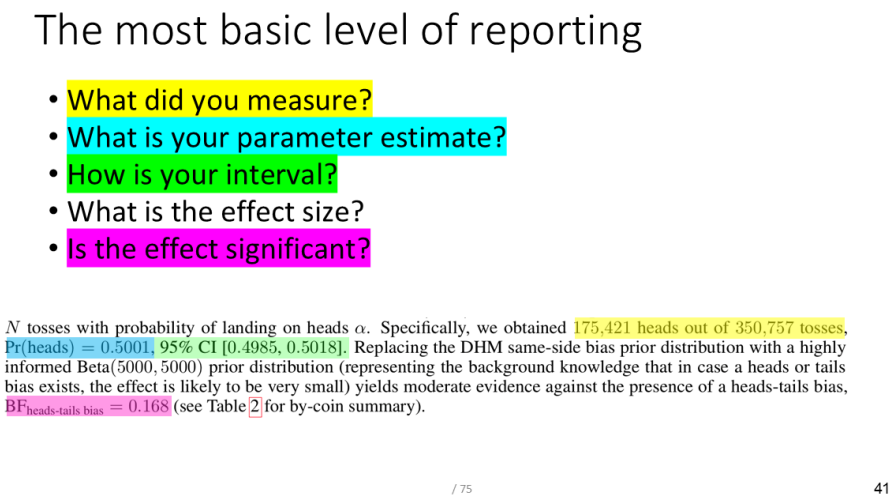


Figure 7: Slide 41 — The most basic level of reporting

6.7 — Power

This section introduces type II error and statistical power. Power is the probability that a test will identify a real effect when the null model is false. The lecture emphasizes three determinants of power: the true effect size, the amount of data, and the threshold used for significance.

New Definitions and Terms

- **Type II error** — failing to reject a false null model
- **error** — another name for type II error
- **Power** — the probability of successfully identifying a true effect
- **More powerful test** — a test with a smaller probability of type II error

Formulas

$$\text{power} = 1 - \beta$$

Power increases when the probability of type II error decreases.

Figure

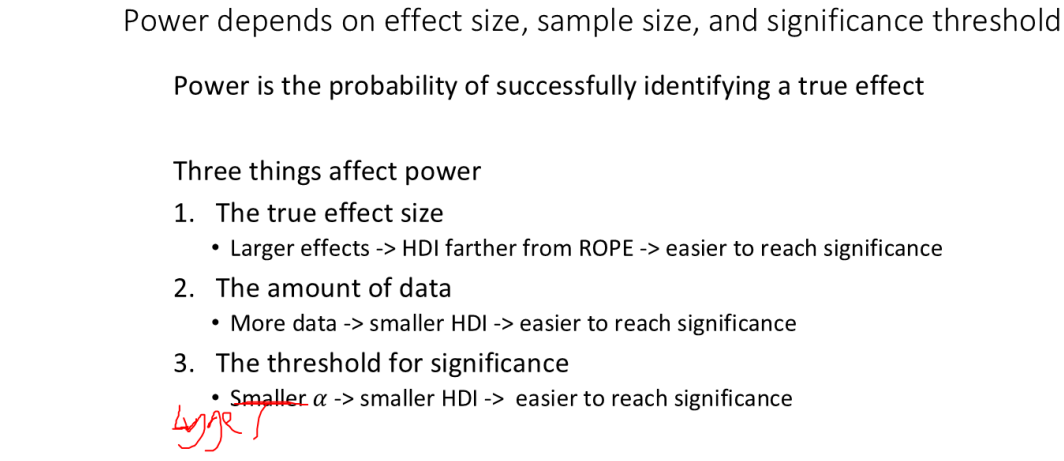


Figure 8: Slide 44 — Power depends on effect size, sample size, and significance threshold

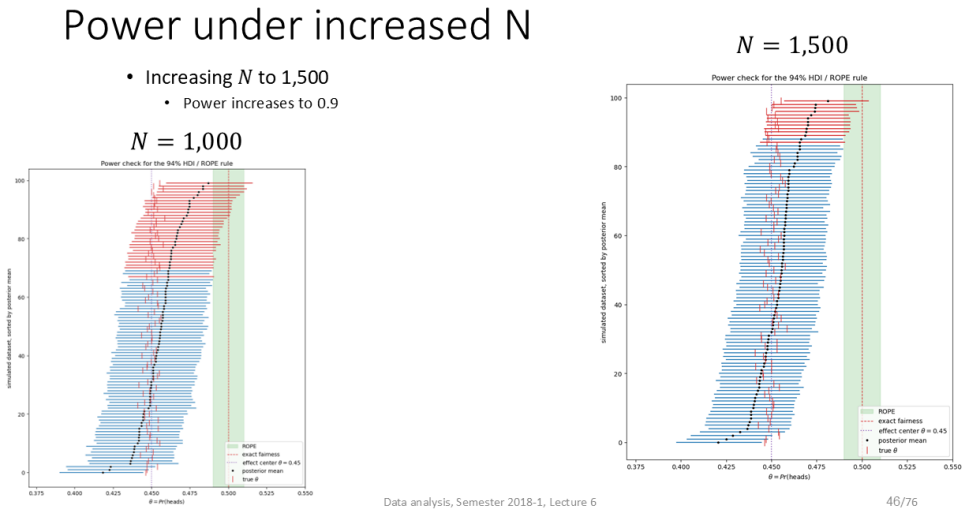


Figure 9: Slide 46 — Power under increased N

6.8 — Pre-hoc and post-hoc power

This section distinguishes two uses of power analysis. Pre-hoc power analysis asks how much data are needed to reach a target power for an expected or minimally meaningful effect. Post-hoc power analysis asks, after the experiment has already been run, how sensitive the experiment was and how large an effect would have needed to be to reach the target power.

New Definitions and Terms

- **Pre-hoc power analysis** — power analysis used before collecting data to choose an appropriate sample size
- **Post-hoc power analysis** — power analysis used after collecting data to evaluate the sensitivity of the completed experiment
- **Minimal effect size** — the smallest effect that would be meaningful enough to detect
- **Sensitivity** — the size of effect an experiment could reliably identify

Formulas

$$\text{minimal effect size} = \frac{1}{2} \times \text{small effect size}$$

The lecture gives this as a rule of thumb for choosing a minimal effect size.

$$ROPE = \frac{1}{2} \times \text{minimal effect size}$$

The lecture gives this as a rule of thumb for choosing the ROPE.

Figure

Pre-hoc power: choosing N to get enough power

- How big a sample do we need?
- We want a power of 0.9
  - For very small effects
    - Cohen's h = 0.1
- How big a dataset do we need?
- Simulate!
  - Measure power
    - As a function of N
  - Find N
    - Where power > 0.9
    - $N \approx 1{,}600$

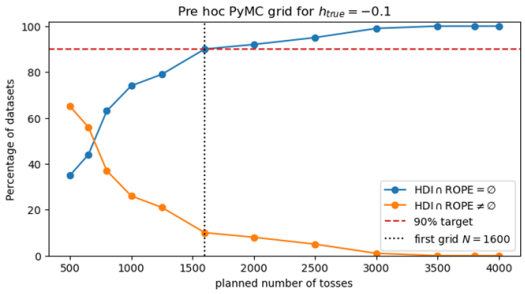


Figure 10: Slide 49 — Pre-hoc power: choosing N to get enough power

Post-hoc power: how big an effect do we need?

- We've done the experiment
  - N is fixed
  - $\hat{\theta}$ is known
- We want to know how sensitive our experiment is
  - If the effect had been small
    - Would it be significant?
- Simulate!
  - Measure power
    - As a function of effect size
  - Find smallest effect size
    - Where power > 0.9

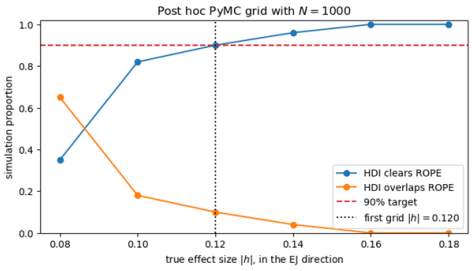


Figure 11: Slide 52 — Post-hoc power: how big an effect do we need?

6.9 — Frequentist methods

This section introduces the frequentist hypothesis-test structure using a statistic whose sampling distribution is known under the null hypothesis. The test asks whether the statistic calculated from the observed data falls in the tail of that null sampling distribution. The lecture then compares frequentist tests with Bayesian HDI/ROPE tests and emphasizes that different tests can disagree.

New Definitions and Terms

- **Statistic** — a quantity calculated from data and used for inference
- **Sampling distribution** — the distribution of a statistic across repeated datasets generated under a model
- **z statistic** — a standardized statistic that is approximately normally distributed under the null when the sample size is large
- **p value** — the probability, under the null model, of getting data as extreme as the observed data or more extreme
- **Null sampling distribution** — the distribution of the statistic when the null model is true

Formulas

$$\hat{\theta} = \frac{y}{N}$$

The observed proportion is the number of successes divided by the total number of trials.

$$z = \frac{\hat{\theta} - \theta_0}{\sqrt{\frac{1}{N}\theta_0(1-\theta_0)}}$$

For large N under the null hypothesis, the z statistic is approximately normally distributed.

$$z \sim N(0, 1)$$

The normal approximation allows the observed statistic to be compared with the null sampling distribution.

Figure

Frequentist hypothesis tests use a statistic

- Find a statistic
  - Whose sampling distribution is known
    - Under the null hypothesis
  - Test whether calculating the statistic on our data is consistent
    - With the null sampling distribution
    - That means:
      - Is measured statistic
      - Is in tail
      - Of null sampling distribution

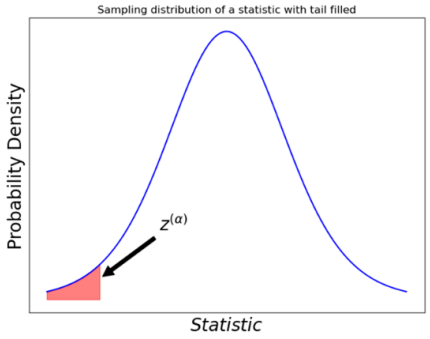


Figure 12: Slide 55 — Frequentist hypothesis tests use a statistic

Different hypothesis tests don't always agree

- TianqiPeng was significant for z test and HDI / ROPE
- EJ was not significant for either z test or HDI / ROPE
- DavidV was significant for z test but not for HDI / ROPE

	person	coin	N	heads proportion	difference from 0.5	z statistic	normal approx p-value
0	EJ	50CZK	200	0.440	-0.060	-1.70	0.08969
1	TianqiPeng	0.20EUR	1000	0.445	-0.055	-3.48	0.0005042
2	davidV	0.10EUR	3660	0.480	-0.020	-2.38	0.0173

Figure 13: Slide 58 — Different hypothesis tests don't always agree

6.10 — Comparing two groups

This section applies the lecture's ideas to comparing outcomes in hospitals with and without PCI capabilities. The hierarchical model from the previous lecture is extended by adding deterministic quantities for the PCI and non-PCI means and their difference. The effect sizes are very small, the HDI overlaps the ROPE, and the lecture uses both Bayesian and frequentist approaches to think about effect size, significance, and sensitivity.

New Definitions and Terms

- **PCI** — percutaneous coronary intervention, used here to define one hospital group
- **Deterministic parameter** — a model quantity calculated from other parameters and stored in the `idata` object
- **Group comparison** — estimating and testing a difference between two groups
- **Difference parameter** — the parameter representing the contrast between PCI and non-PCI groups

Formulas

```
mu_nonpci = pm.Deterministic("mu_nonpci", mu_type[0])
mu_pci = pm.Deterministic("mu_pci", mu_type[1])
mu_difference = pm.Deterministic("mu_difference", mu_pci - mu_nonpci)
```

Deterministic parameters store calculated quantities of interest in the inference data object.

$$\hat{\theta}_i = \frac{y_i}{N_i}, \quad i = 1, 2$$

Each group has its own observed proportion.

$$\hat{\theta}_{12} = \frac{y_1 + y_2}{N_1 + N_2}$$

The pooled estimate is used in the frequentist normal approximation for comparing two groups.

$$\frac{1}{N_{12}} = \frac{1}{N_1} + \frac{1}{N_2}$$

N_{12} combines the uncertainty from the two group estimates.

$$z = \frac{\hat{\theta}_1 - \hat{\theta}_2}{\sqrt{\frac{\hat{\theta}_{12}(1-\hat{\theta}_{12})}{N_{12}}}}$$

The two-group frequentist test is based on the same normal-approximation logic introduced for one proportion.

Figure

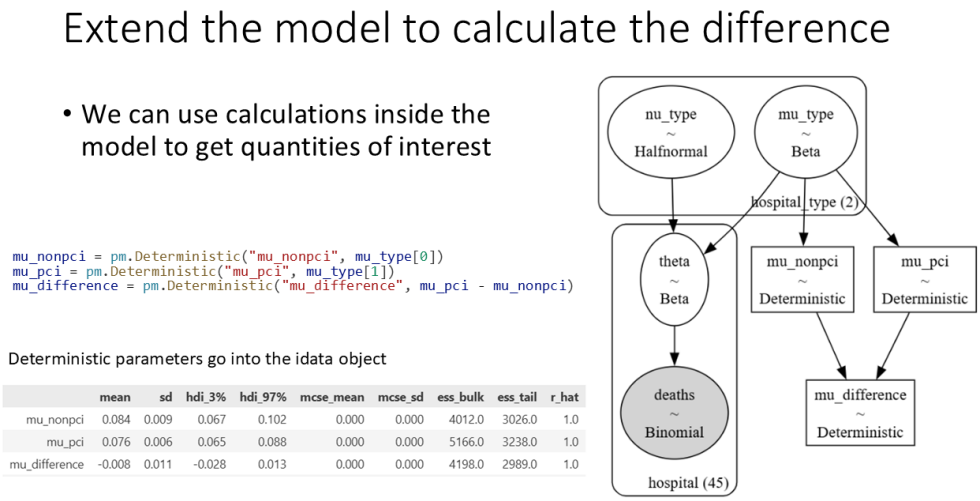


Figure 14: Slide 65 — Extend the model to calculate the difference

Comparing HDI to ROPE does not allow conclusions

- HDI intersects with ROPE
- ROPE does not contain HDI

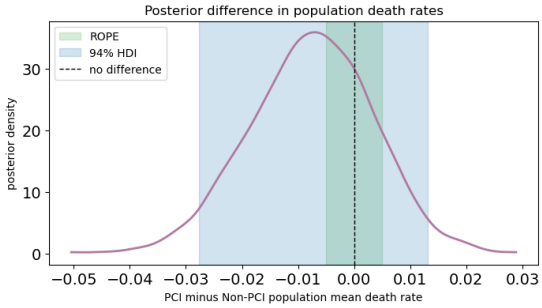


Figure 15: Slide 67 — Comparing HDI to ROPE does not allow conclusions

Frequentist hypothesis test for comparison of two groups

- Based on same normal approximation

$$\hat{\theta}_i = \frac{y_i}{N_i} \quad i = 1,2$$

$$\hat{\theta}_{12} = \frac{y_1 + y_2}{N_1 + N_2}$$

$$\frac{1}{N_{12}} = \frac{1}{N_1} + \frac{1}{N_2}$$

$$z = \frac{\hat{\theta}_1 - \hat{\theta}_2}{\sqrt{\hat{\theta}_{12}\left(1 - \hat{\theta}_{12}\right)\frac{1}{N_{12}}}} = -1.32$$

Figure 16: Slide 70 — Frequentist hypothesis test for comparison of two groups

Lecture 7 Review Handout

Statistics 367-1-4361

Lecture Overview

Lecture 7 introduces normal models for continuous biomedical data. The lecture moves from the one-parameter binary models used earlier in the course to likelihoods with both a center and a spread. It then applies the Bayesian workflow to a normal model for normalized whole brain volume, extends the model to compare age groups, and introduces raw and normalized effect sizes for normal models. The lecture ends by distinguishing unpaired and paired comparisons, showing how outliers motivate robust Student's t likelihoods, and connecting Bayesian paired analyses to classical paired tests.

Lecture Sections

7.1 — Continuous data

This section explains why continuous measurements require different likelihoods from the binary and count data used earlier in the course. Continuous biomedical variables come with units, have both a middle and a spread, and may be bounded or multimodal. The normal likelihood is introduced as the basic model for continuous data, with one parameter for the population center and one for the population variability.

New Definitions and Terms

- **Continuous data** — measurements represented on a numerical scale with units, such as reaction time, blood pressure, or normalized brain volume
- **Normal model** — a model for continuous data using a normal likelihood, usually parameterized by a mean and a standard deviation
- **Mean parameter** — the parameter representing the center of the modeled population
- **Scale parameter** — the parameter representing the spread of the modeled population

Formulas

$$y_i \sim \text{Normal}(\mu, \sigma)$$

Each observed continuous value is modeled as a draw from a normal distribution with population mean μ and population standard deviation σ.

$$\mu \sim \text{Normal}(\mu_\mu, \sigma_\mu), \qquad \sigma \sim \text{HalfNormal}(\sigma_\sigma)$$

The mean can have a normal prior, while the standard deviation needs a positive prior such as a half-normal.

Figure

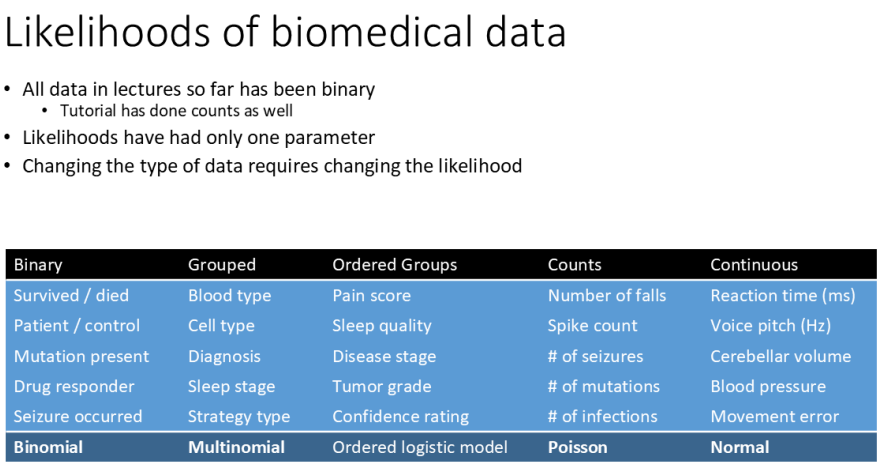


Figure 1: Slide 4 — Likelihoods of biomedical data

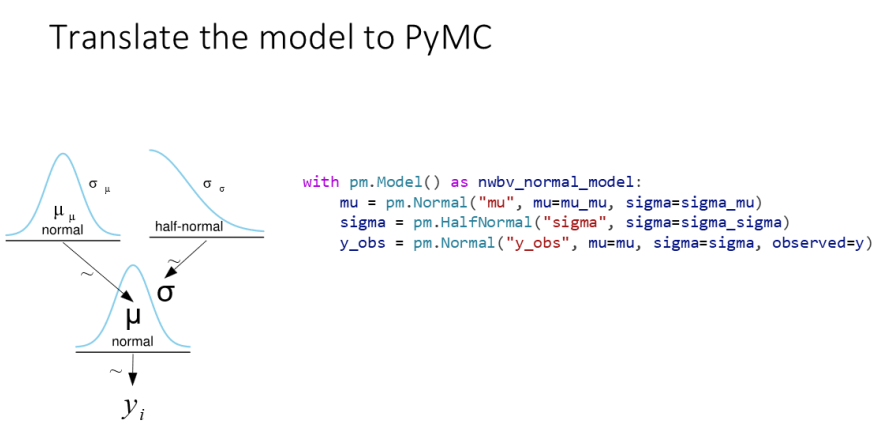


Figure 2: Slide 8 — Translate the model to PyMC

7.2 — Normal model workflow

This section applies the Bayesian workflow to the normal model including prior predictive check before seeing the data. After sampling, trace plots and diagnostics are used to check sampling quality, and posterior predictive checks are used to evaluate model fit. Bayesian p values are introduced as a way to quantify how well posterior predictive data resemble the observed data.

New Definitions and Terms

- **Cumulative distribution** — a representation showing the probability of values less than or equal to each value on the x-axis
- **Bayesian p value** — the posterior predictive probability that a simulated statistic is as extreme as, or more extreme than, the observed statistic
- **Posterior predictive sampling distribution** — the distribution of simulated summaries generated from posterior predictive datasets

Formulas

$$p_B = P\left(T(y^{rep}) \geq T(y) \mid y\right)$$

A Bayesian p value compares a statistic calculated on posterior predictive datasets, $T(y^{rep})$, with the same statistic calculated on the observed data, $T(y)$.

Figure

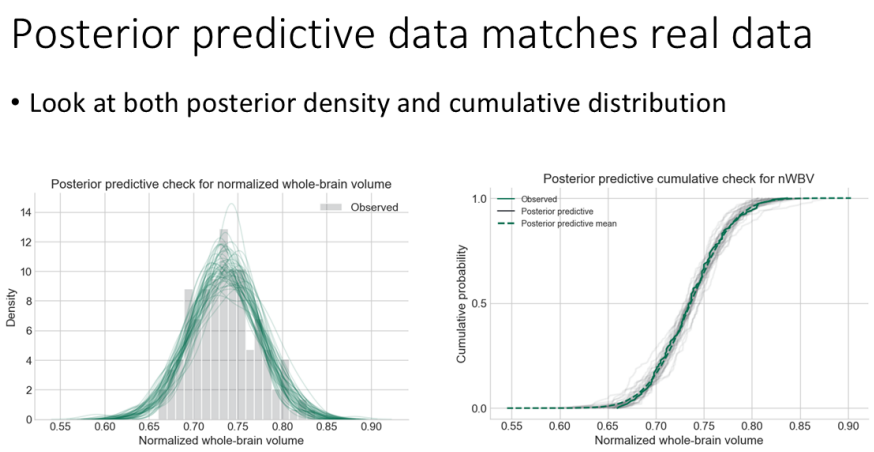


Figure 3: Slide 16 — Posterior predictive data matches real data

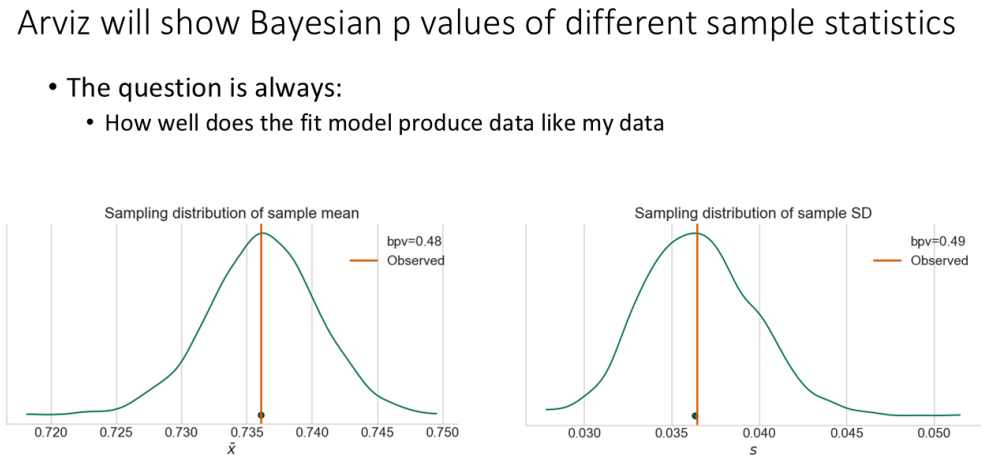


Figure 4: Slide 20 — Arviz will show Bayesian p values of different sample statistics

---

7.3 — Comparing across ages

This section extends the normal model of brian volume to compare brain volumes of two age groups. The model is a pooled / unpooled: each group has its own mean, while they use a shared standard deviation. The main quantity of interest is the difference between the means of the two groups.

New Definitions and Terms

- **Group index** — an integer or categorical variable used to select the correct group-specific parameter for each observation
- **Shared standard deviation** — an assumption that both groups have the same population spread
- **`pm.Data`** — a PyMC data object that is stored in the model and can be updated later

Formulas

$$y_i \sim \text{Normal}(\mu_{g_i}, \sigma)$$

Each observation uses the mean corresponding to its group g_i, while sharing the same standard deviation σ.

$$\Delta_\mu = \mu_1 - \mu_0$$

The difference between group means is stored as a deterministic parameter so it can be examined directly in the posterior.

Figure

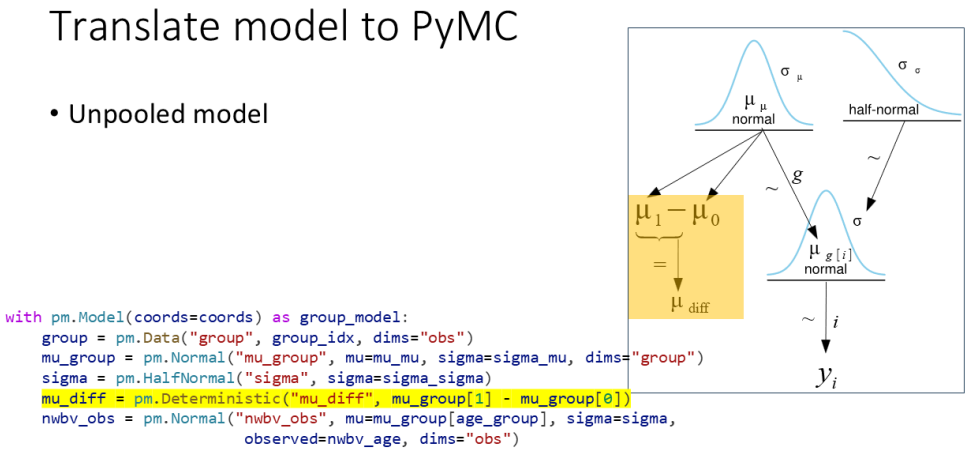


Figure 5: Slide 26 — Translate model to PyMC

7.4 — Effect sizes for normal models

This section distinguishes raw and normalized effect sizes for a normal two-group comparison. The raw effect size is the difference in measured units, here percent total intracranial volume. The normalized effect size is Cohen's d, which divides the mean difference by a standard deviation and therefore depends on the variance structure assumed by the model.

New Definitions and Terms

- **Cohen's d** — a normalized mean difference, here calculated as a posterior deterministic quantity
- **Student's t distribution** — a distribution with heavier tails than the normal distribution, controlled by a degrees-of-freedom parameter
- **t test** — a frequentist test comparing a mean difference or standardized mean difference to a t sampling distribution

Formulas

$$\Delta_\mu = \mu_1 - \mu_0$$

The raw effect size is the difference between the two group means, measured in the original units.

$$d = \frac{\Delta_\mu}{\sigma}$$

Cohen's d is the mean difference divided by the standard deviation. In this lecture's model, the denominator is the single shared σ.

Figure

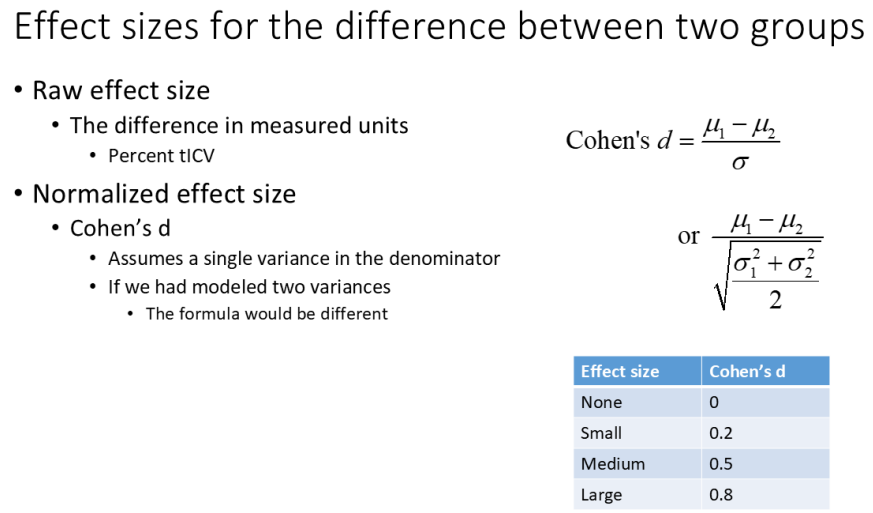
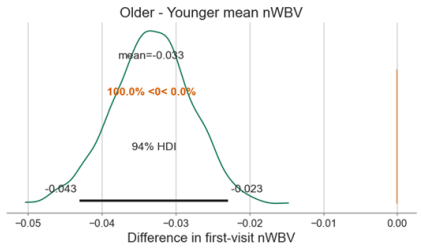
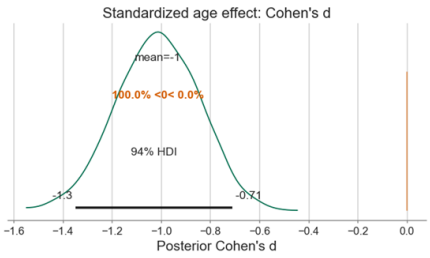


Figure 6: Slide 33 — Effect sizes for the difference between two groups

Both raw and normalized effect sizes have distributions





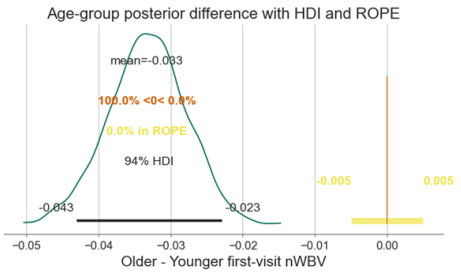

```python
with pm.Model(coords=coords) as group_model:
    group = pm.Data("group", group_idx, dims="obs")
    mu_group = pm.Normal("mu_group", mu=mu_mu, sigma=sigma_mu, dims="group")
    sigma = pm.HalfNormal("sigma", sigma=sigma_sigma)
    mu_diff = pm.Deterministic("mu_diff", mu_group[1] - mu_group[0])
    cohen_d = pm.Deterministic("cohen_d", mu_diff / sigma)
    nwbv_obs = pm.Normal("nwbv_obs", mu=mu_group[age_group], sigma=sigma, observed=nwbv_age, dims="obs")
```

Figure 7: Slide 34 — Both raw and normalized effect sizes have distributions

A frequentist test for comparing two groups

- The t test
  - Compares frequentist Cohen's d
    - To the t distribution
    - With the correct degrees of freedom

Bayesian hypothesis test for comparison



Statistic	Value
Younger mean	0.753686
Older mean	0.720662
Younger SD	0.035784
Older SD	0.029551
Pooled SD	0.032605
Cohen's d (Older - Younger)	-1.012819
Degrees of freedom	148.000000
94% two-sided |t| threshold	1.895319
Two-sided p-value	0.000000

Figure 8: Slide 39 — A frequentist test for comparing two groups

7.5 — Paired comparisons

This section contrasts unpaired comparisons between groups with paired comparisons within the same individuals. In the MRI example, each participant has several scans, so the relevant question can be asked as the change in normalized brain volume from the first to the last visit. Pairing removes much of the between-subject variability and can reveal a consistent change that is harder to see in an unpaired comparison.

New Definitions and Terms

- **Paired comparison** — a comparison made within the same subjects or units across two conditions or times
- **Unpaired comparison** — a comparison made between separate groups of observations
- **Within-subject change** — the difference between two measurements from the same subject
- **Between-subject variability** — variability among people that can hide consistent within-person effects
- **Outlier** — an observation that is poorly explained by the fitted model and can strongly influence a normal model

Formulas

$$\text{change}_i = y_{i,\text{last}} - y_{i,\text{first}}$$

The paired analysis converts repeated measurements into one change value per subject.

$$\text{change}_i \sim \text{Normal}(\mu_{change}, \sigma_{change})$$

The paired normal model is just the normal model applied to the vector of within-subject changes.

Figure

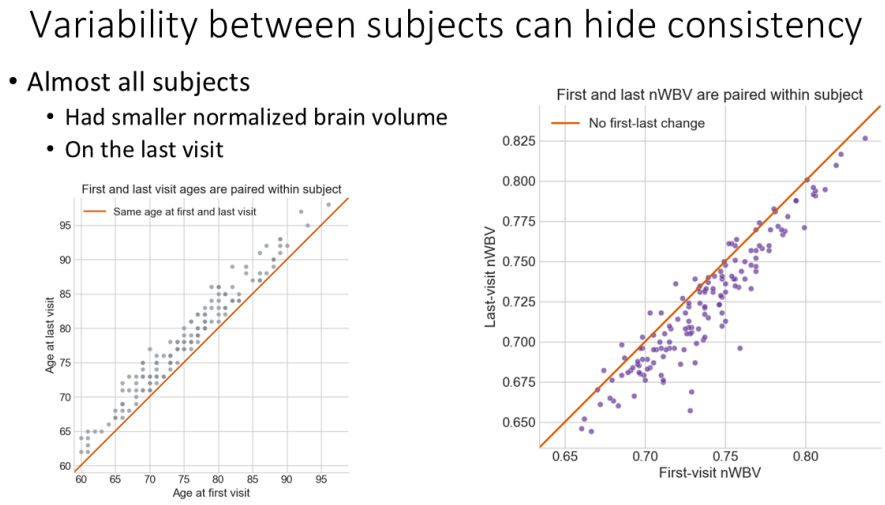


Figure 9: Slide 43 — Variability between subjects can hide consistency

---

7.6 — Robust models

This section uses the failure of the paired normal model to motivate a robust likelihood. Normal likelihoods are sensitive to outliers because they assign very low probability to observations far from the mean. Replacing the normal likelihood with a Student's t likelihood adds a degrees-of-freedom parameter, allowing the model to have thicker tails when the data require them.

New Definitions and Terms

- **Robust model** — a model whose conclusions are less sensitive to outliers or deviations from ideal assumptions
- **Student's t likelihood** — a continuous likelihood with heavier tails than a normal likelihood
- **Degrees of freedom parameter** — the parameter ν controlling the tails of the Student's t distribution
- **Normality parameter** — the lecture's interpretation of ν: small values imply heavier tails, while large values converge toward a normal model
- **Thick tails** — a distributional shape that assigns more probability to extreme values

Formulas

$$\text{change}_i \sim t_\nu(\mu_{change}, \sigma_{change})$$

The robust paired model replaces the normal likelihood with a Student's t likelihood.

$$\nu = \nu_{min} + \nu_{excess}, \qquad \nu_{excess} \sim \text{Exponential}(\lambda)$$

Adding a lower bound keeps the tails from becoming unrealistically wide and helps avoid sampling problems.

Figure

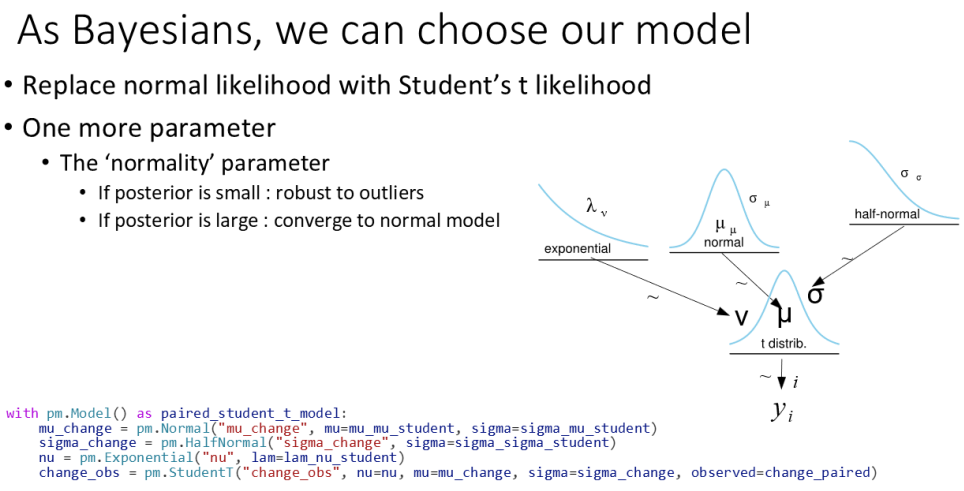


Figure 10: Slide 50 — As Bayesians, we can choose our model

7.7 — Effect sizes for paired comparisons

This section returns to effect sizes after fitting the robust paired model. The raw paired effect is the mean change, and the normalized effect is the paired Cohen's d. The lecture also shows that deterministic quantities can be added to the posterior after sampling by operating directly on XArray objects in the `idata` object.

New Definitions and Terms

- **Paired raw effect** — the posterior distribution of the mean within-subject change
- **Paired Cohen's d** — the mean within-subject change divided by the standard deviation of the within-subject changes
- **XArray DataArray** — the object type used inside ArviZ inference data for posterior variables
- **Broadcasting** — automatic alignment of array operations across compatible dimensions
- **Wilcoxon signed-rank test** — a frequentist paired test that can be used when a normal likelihood is not appropriate

Formulas

$$d_{paired} = \frac{\mu_{change}}{\sigma_{change}}$$

The normalized paired effect size is calculated from posterior variables.

```
idata.posterior["cohen_d"] = (
    idata.posterior["mu_change"] / idata.posterior["sigma_change"]
)
```

A deterministic posterior quantity can be added after sampling if it is just a calculation on sampled posterior variables.

Figure

Compare paired and unpaired analyses

- We can add a calculation to the posterior if we didn't sample it

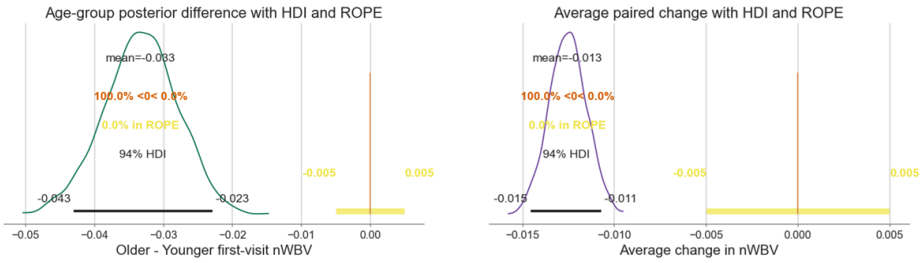


Figure 11: Slide 59 — Compare paired and unpaired analyses

Tests on paired data

- The t-test assumes that the likelihood is normal
  - Not appropriate for non-normal data
- The Wilcoxon signed-rank test can be used

Quantity	Value
Sample mean of paired difference	-0.013400
Sample SD of paired difference	0.013600
Cohen's dz	-0.987500
Degrees of freedom	149.000000
94% two-sided |t| threshold	1.895200
Two-sided p-value	0.000000

Quantity	Value
W statistic	603.500000
94% two-sided W lower threshold	4402.770200
Two-sided p-value	0.000000

Figure 12: Slide 61 — Tests on paired data

7.8 — Next week: linear models

This final section connects normal models to the next topic: linear models. Up to this point, the course has mostly modeled one outcome variable at a time, sometimes with grouping variables. Linear models extend this framework to relationships among variables, allowing questions about how one measurement changes with another.

Figure

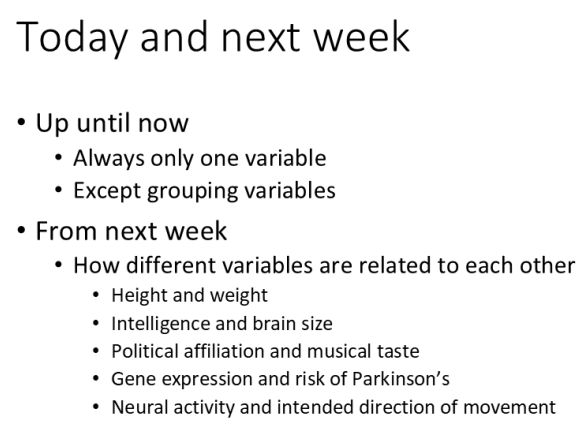


Figure 13: Slide 65 — Today and next week

Lecture 8 Review Handout

Statistics 367-1-4361

Lecture Overview

Lecture 8 extends normal models from group comparisons to regression models. The key change is that each observation now has its own mean, defined as a linear function of one or more independent variables. The lecture uses age, normalized whole-brain volume, and MMSE to show how Bayesian linear models are specified, sampled, diagnosed, interpreted, and translated into effect sizes. It ends by connecting the Bayesian regression model to frequentist regression estimators, intervals, R^2, and the F statistic.

Lecture Sections

8.1 — Bivariate data

This opening section places regression in the larger course sequence. Earlier lectures developed probability, Bayesian updating, sampling, workflow, binomial models, hierarchical models, and normal models. Linear regression now extends the same Bayesian logic to **relationships among variables**, using brain-volume data with age, MMSE, and estimated intracranial volume as motivating examples.

New Definitions and Terms

- **Bivariate data** — paired measurements on two variables for each observational unit
- **Regression model** — a model for how a dependent variable changes with one or more independent variables

Figure

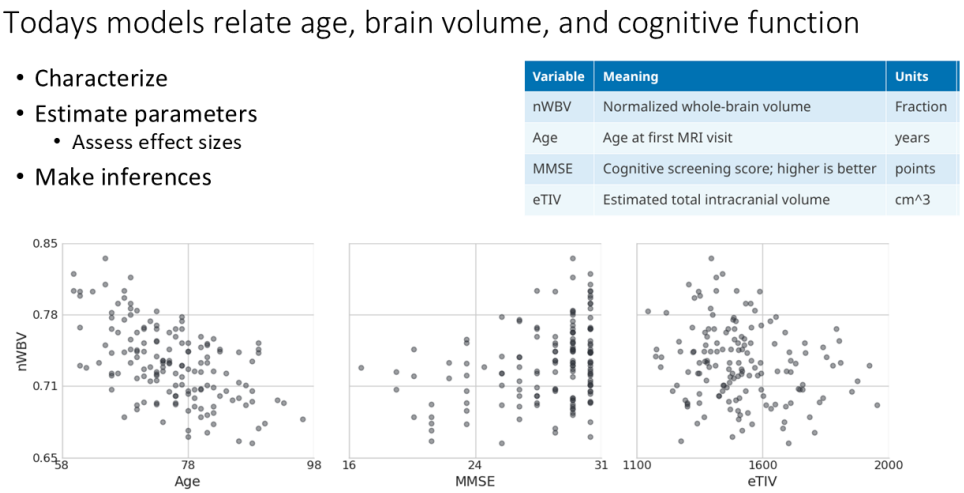


Figure 1: Slide 5 — Todays models relate age, brain volume, and cognitive function

---

8.2 — The bivariate model

This section builds the simple linear regression model as a direct extension of the two-group normal model. In the two-group model, each observation belonged to one of two means; in the bivariate model, each observation has its own mean. The mean is a linear function of the independent variable, while all observations share a residual standard deviation.

The model is asymmetric: x is treated as known precisely, while y is treated as measured with noise. This asymmetry may represent a plausible causal direction, as in age affecting brain volume, but it can also be a modeling convention when the causal interpretation is less clear.

New Definitions and Terms

- **Independent variable** — the predictor variable, treated as known precisely in the regression model
- **Dependent variable** — the outcome variable, modeled as noisy conditional on the predictor
- **Linear model** — a model in which the expected value of the outcome is a linear function of predictors

- **DAG** — a directed acyclic graph used to represent dependency or causal assumptions among variables

Formulas

$$\mu_i = \beta_0 + \beta_1 x_i$$

Each observation has its own expected value, determined by the regression line.

$$y_i \sim \text{Normal}(\mu_i, \sigma)$$

The observed outcome is modeled as noisy around that expected value.

$$\beta_0 \sim \text{Normal}(\mu_0, \sigma_0), \qquad \beta_1 \sim \text{Normal}(\mu_1, \sigma_1), \qquad \sigma \sim \text{HalfNormal}(\sigma_\sigma)$$

The priors are placed on the intercept, slope, and residual standard deviation.

Figure

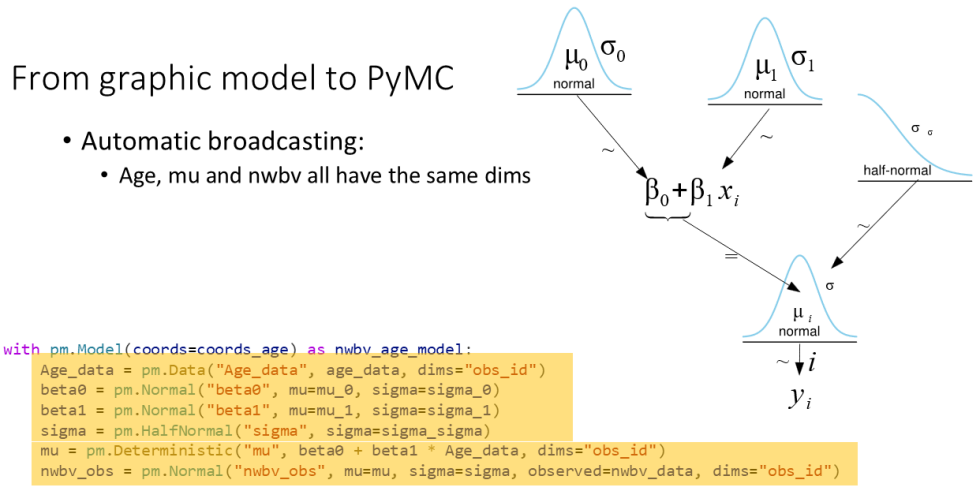


Figure 2: Slide 12 — From graphic model to PyMC

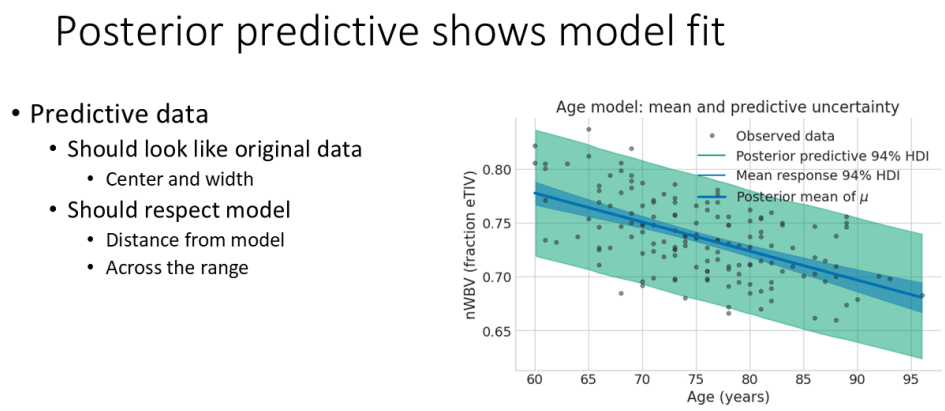


Figure 3: Slide 18 — Posterior predictive shows model fit

8.3 — Posterior regression parameters

This section interprets the posterior distribution of the simple regression model. Each posterior draw contains a possible intercept, slope, and residual standard deviation, and therefore defines a possible regression line. Uncertainty is smallest near the middle of the observed data and larger near the edges, where fewer observations constrain the line.

The main scientific quantity is the slope: in the age-nWBV model, the posterior suggests a loss of roughly 1% nWBV every 3 years. The intercept is mathematically part of the line, but because it corresponds to predicted nWBV at age 0, it is an extrapolation and is not automatically scientifically meaningful.

New Definitions and Terms

- **Posterior regression line** — a line defined by one posterior draw of the regression parameters
- **Raw effect size** — the slope expressed in the original units of the outcome and predictor
- **Standardized slope** — the slope after converting variables to standard-deviation units
- **Correlation coefficient** — in simple linear regression, the standardized slope is equal to r

Formulas

Raw slope:

$$\beta_1 = \frac{\Delta y}{\Delta x}$$

The raw effect size is the change in the dependent variable per unit change in the independent variable.

Standardized slope:

$$\beta_1^z = \frac{s_x}{s_y}\beta_1 = r$$

After standardization, the slope is dimensionless and equals the correlation coefficient in the bivariate case.

Figure

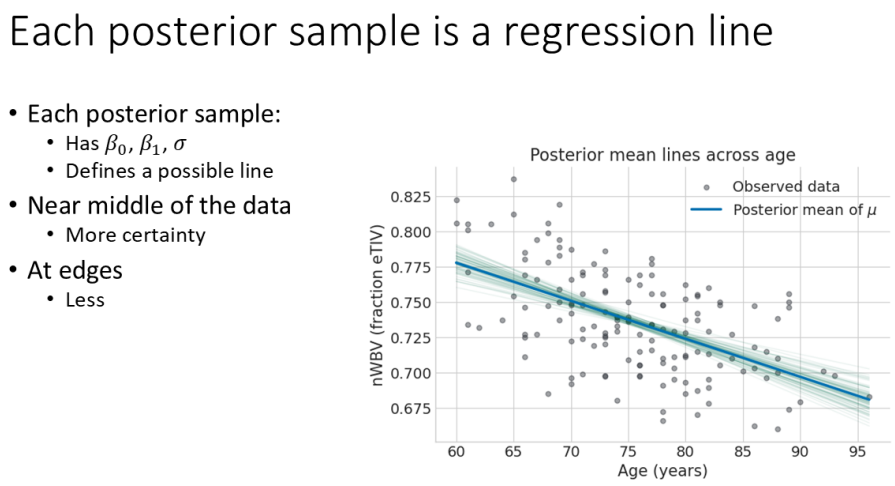


Figure 4: Slide 21 — Each posterior sample is a regression line

---

8.4 — Centered and uncentered data

This section shows that centering the independent variable changes the meaning of the intercept without changing the fitted line. In the uncentered model, the intercept is the expected outcome when age is 0. In the centered model, the intercept is the expected outcome when age is equal to the sample mean, which is usually more interpretable and often improves sampling.

Centering is therefore a reparameterization, not a different scientific model. The posterior for the slope and the final fit are unchanged, but the posterior for the intercept becomes narrower and more meaningful.

New Definitions and Terms

- **Centered variable** — a variable with its sample mean subtracted
- **Standardized variable** — a variable centered and divided by its sample standard deviation
- **Reparameterization** — rewriting the same model using different parameters
- **Centered intercept** — the expected outcome when the original predictor is at its mean

Formulas

Centered and standardized variables:

$$x_i^c = x_i - \bar{x}, \qquad x_i^z = \frac{x_i - \bar{x}}{s_x}$$

Regression with a centered predictor:

$$\mu_i = \beta_0^c + \beta_1^c x_i^c$$

Relationship between uncentered and centered parameters:

$$\beta_1^c = \beta_1, \qquad \beta_0 = \beta_0^c - \beta_1 \bar{x}$$

The slope is unchanged, while the intercept is relabeled to refer to the mean predictor value rather than zero.

Figure

Centering and standardization remove means and units

		Random variable		Data
The variable: x		$E[x] = \mu$ $\text{var}[x] = \sigma^2$		$\frac{1}{N}\sum_{i=1}^{N} x_i = \bar{x}$ $\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2 = s^2$
Centered: x^c	$x^c = x - \mu$	$E[x^c] = 0$ $\text{var}[x^c] = \sigma^2$	$x_i^c = x_i - \bar{x}$	$\frac{1}{N}\sum_{i=1}^{N} x_i^c = 0$ $\frac{1}{N-1}\sum_{i=1}^{N}(x_i^c - \bar{x})^2 = s^2$
Standardized: x^z	$x^z = \frac{x - \mu}{\sigma}$	$E[x^z] = 0$ $\text{var}[x^z] = 1$	$x_i^z = \frac{x_i - \bar{x}}{s}$	$\frac{1}{N}\sum_{i=1}^{N} x_i^z = 0$ $\frac{1}{N-1}\sum_{i=1}^{N}(x_i^z - \bar{x})^2 = 1$

Figure 5: Slide 30 — Centering and standardization remove means and units

Intercept has different meaning in the two models

- Scientific question
  - Which do you care about?
- Statistical question
  - Which produces stable inference?
  - Because we can convert between them

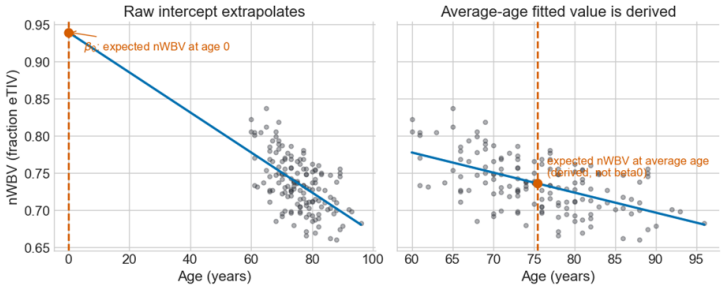


Figure 6: Slide 34 — Intercept has different meaning in the two models

8.5 — Multivariate regression

This section generalizes the bivariate model to include more than one independent variable. Age and MMSE are modeled together as predictors of nWBV, so each slope describes an effect while holding the other predictor fixed. The regression model still uses a normal likelihood, but the expected value now depends on a vector of predictors.

The lecture also emphasizes that the regression model does not assume that the independent variables themselves are normally distributed. MMSE is ordinal, bounded, and non-Gaussian, but this does not violate the normal regression likelihood because the likelihood is a model of the dependent variable conditional on the predictors.

New Definitions and Terms

- **Multivariate regression** — regression with more than one independent variable
- **Conditional effect** — the effect of one predictor when the other predictors are held fixed
- **Data matrix** — the matrix containing all predictor columns used to compute the model mean

Formulas

Multivariate regression mean:

$$\mu_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i}$$

For the lecture example, the predictors are centered age and centered MMSE.

Matrix form:

$$\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$$

The data matrix $\mathbf{X}$ holds the independent variables, and $\boldsymbol{\beta}$ holds the corresponding regression coefficients.

Figure

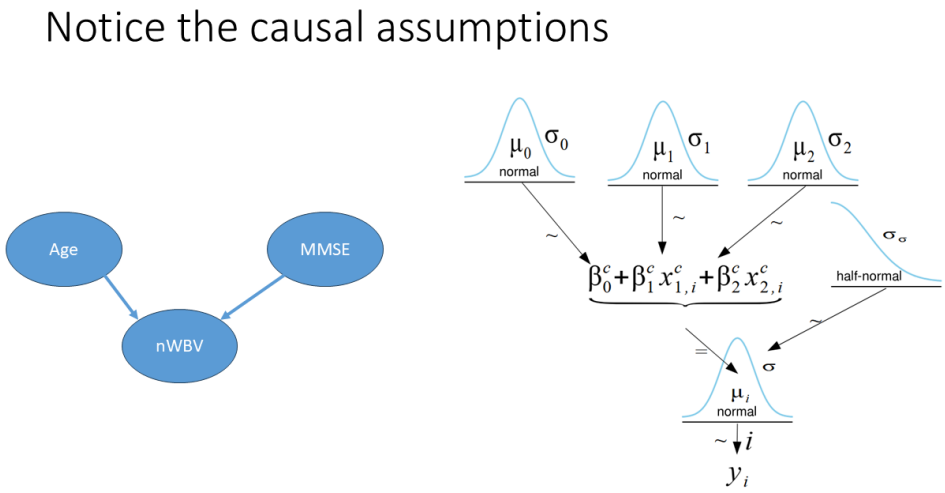


Figure 7: Slide 45 — Notice the causal assumptions

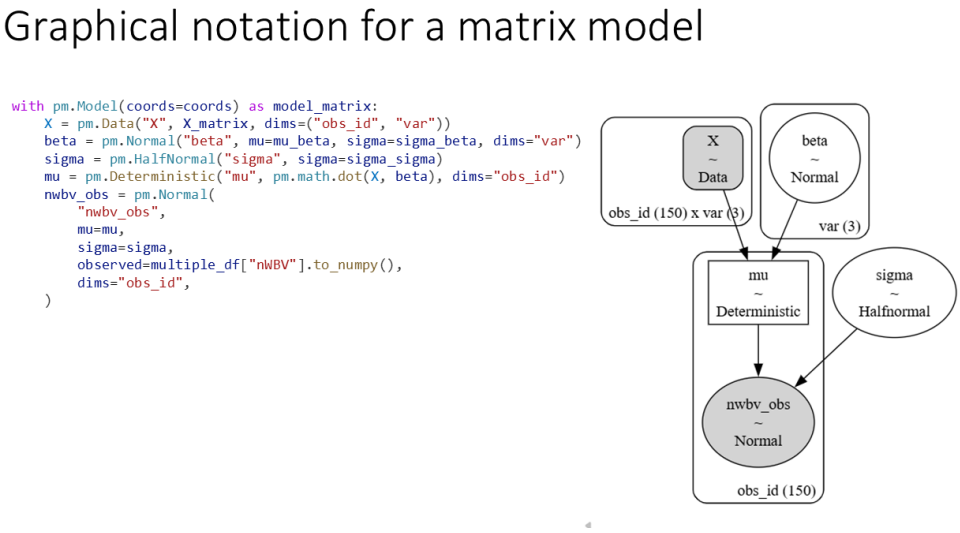


Figure 8: Slide 48 — Graphical notation for a matrix model

8.6 — Multivariate regression posterior

This section interprets the posterior of the multivariate regression. The age coefficient gives the effect of age when MMSE is held fixed, while the MMSE coefficient gives the effect of MMSE when age is held fixed. In the lecture example, nWBV decreases with age and increases with MMSE, conditional on the other variable.

Because age and MMSE have different units, the raw slopes are not directly comparable. Standardized slopes put them on a common scale, and Bayesian R^2 summarizes how much of the outcome variance is explained by the model.

New Definitions and Terms

- **Standardized conditional slope** — a multivariate regression coefficient expressed in standard-deviation units
- **Bayesian R^2** — a posterior distribution for the proportion of variance explained by the model
- **Variance explained** — the variance in the model mean relative to the total modeled variance

Formulas

Standardized slope for predictor j:

$$\beta_j^z = \frac{s_{x_j}}{s_y}\beta_j$$

This allows slopes for predictors with different units to be compared on a common scale.

Bayesian R^2:

$$R^2 = \frac{\text{var}()}{\text{var}() + {}^2}$$

The numerator is the variance explained by the model mean, and the denominator adds the residual variance.

Figure

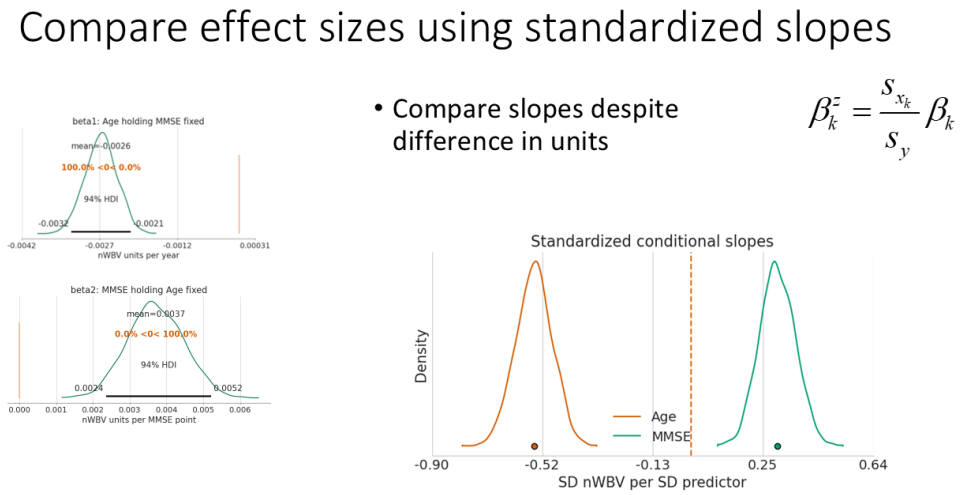


Figure 9: Slide 57 — Compare effect sizes using standardized slopes

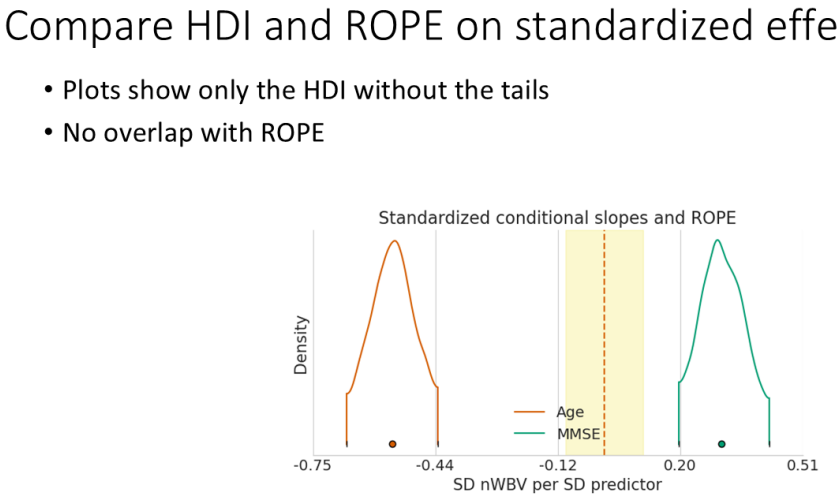


Figure 10: Slide 58 — Compare HDI and ROPE on standardized effects

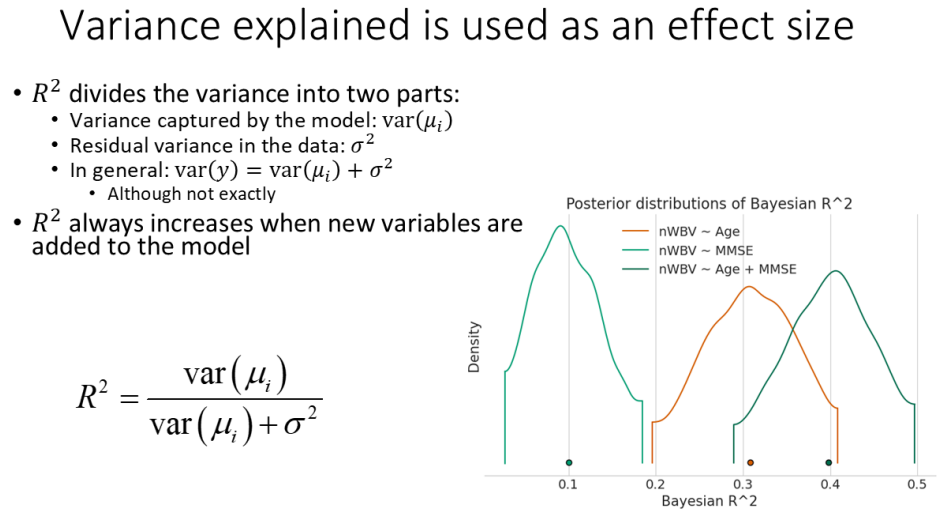


Figure 11: Slide 59 — Variance explained is used as an effect size

---

8.7 — Frequentist methods

This section translates the Bayesian multivariate regression model into the corresponding frequentist regression machinery. The frequentist model uses estimators, usually written with Roman letters, instead of posterior random variables written with Greek letters. The regression coefficients are estimated by minimizing the sum of squared residuals, and the sampling distribution of the estimators is used to construct intervals and hypothesis tests.

The key Bayesian contrast is that frequentist regression intervals and tests are based on sampling distributions of estimators, while the Bayesian analysis works directly with posterior distributions of parameters and derived quantities. The same model structure appears in both approaches, but the interpretation of uncertainty differs.

New Definitions and Terms

- **Estimator** — a statistic calculated from data and used as a point estimate of a parameter
- **Ordinary least squares** — the regression estimator obtained by minimizing summed squared residuals
- **Standard error** — the standard deviation of the estimator's sampling distribution
- **Regression sum of squares** — variability explained by the fitted model relative to the mean

- **Residual sum of squares** — variability not explained by the fitted model
- **Degrees of freedom** — counts used to scale sums of squares into variance estimates
- F **statistic** — a ratio of model variance to residual variance used for frequentist hypothesis tests

Formulas

Frequentist regression notation:

$$\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}, \qquad \hat{\mathbf{y}} = \mathbf{X}\mathbf{b}$$

The parameter vector $\boldsymbol{\beta}$ is replaced by an estimator vector $\mathbf{b}$.

Least-squares estimator:

$$\mathbf{b} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

Residual variance estimate:

$$s^2 = \frac{(\mathbf{y} - \mathbf{X}\mathbf{b})^T(\mathbf{y} - \mathbf{X}\mathbf{b})}{N - (k+1)}$$

Sampling distribution of the coefficient estimators:

$$\mathbf{b} \sim \text{Normal}(\boldsymbol{\beta}, \Sigma_b), \qquad \Sigma_b = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$$

Frequentist R^2:

$$R^2 = \frac{\text{var}(\hat{\mathbf{y}})}{\text{var}(\mathbf{y})}$$

Sums of squares:

$$SS_{reg} = (\hat{\mathbf{y}} - \bar{y})^T(\hat{\mathbf{y}} - \bar{y}), \qquad SS_{res} = (\mathbf{y} - \hat{\mathbf{y}})^T(\mathbf{y} - \hat{\mathbf{y}})$$

Degrees of freedom:

$$df_{reg} = k, \qquad df_{res} = N - (k+1)$$

F statistic:

$$F = \frac{MS_{reg}}{MS_{res}} = \frac{SS_{reg}/df_{reg}}{SS_{res}/df_{res}} \sim F(df_{reg}, df_{res})$$

Under the null model, the model variance and residual variance should be approximately equal, so the ratio should be close to 1.

Figure

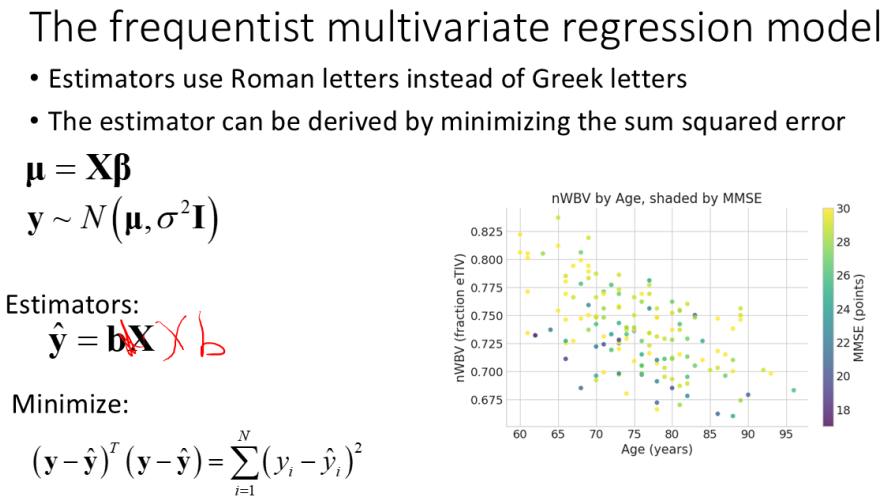


Figure 12: Slide 61 — The frequentist multivariate regression model

Compare frequentist to Bayesian intervals

$df = N - k + 1$

term	estimate	SE	standardized slope	t	p	95% CI lower	95% CI upper
b0	0.7361	0.0023	—	318.985	<0.0001	0.7315	0.7406
b1: Age_c	-0.0027	0.0003	-0.551	-8.675	<0.0001	-0.0033	-0.0021
b2: MMSE_c	0.0038	0.0008	0.310	4.873	<0.0001	0.0023	0.0053

Predictor	Raw effect mean	Standardized effect mean	Std. 94% HDI lower	Std. 94% HDI upper	Posterior in ROPE	HDI overlaps ROPE
Age	-0.0026	-0.547	-0.664	-0.429	0.000	no
MMSE	0.0037	0.302	0.192	0.424	0.001	no

Figure 13: Slide 65 — Compare frequentist to Bayesian intervals

Define some useful statistics

- The total variability in the model
  - The sum squared distance between the model and the mean
- The total variability not explained by the model
  - The squared distance between the data and the model

$$SS_{\text{reg}} = \left(\hat{\mathbf{y}} - \bar{y}\right)^T \left(\hat{\mathbf{y}} - \bar{y}\right) = \sum_{i=1}^{N} \left(\hat{y}_i - \bar{y}\right)^2$$

$$SS_{\text{res}} = \left(\mathbf{y} - \hat{\mathbf{y}}\right)^T \left(\mathbf{y} - \hat{\mathbf{y}}\right) = \sum_{i=1}^{N} \left(y_i - \hat{y}_i\right)^2$$

Figure 14: Slide 67 — Define some useful statistics

8.8 — Next time

The next lecture extends regression by combining continuous predictors with categorical variables. This connects the regression framework developed here to ANOVA-style models and group comparisons.

Lecture 9 Review Handout

Statistics 367-1-4361

Lecture Overview

Lecture 9 extends the regression framework to categorical predictors, multiple factors, and interactions. The main idea is that group comparisons and ANOVA-style models can be written as regression models using indicator variables and Wilkinson notation. The lecture uses the Palmer penguins data to introduce categorical predictors and interactions, then uses birth-weight data to show how modeling choices differ for continuous and categorical predictors. It ends by connecting Bayesian model effect sizes to frequentist sums-of-squares measures: R^2, partial R^2, η^2, and Cohen's f.

Lecture Sections

9.1 — Regression notation

This section introduces Wilkinson notation as a compact grammar for specifying regression models. The dependent variable appears on the left side of the formula, while independent variables and their combinations appear on the right. The penguin dataset provides the main running example, with body mass modeled using continuous predictors such as flipper length and categorical predictors such as species and sex.

New Definitions and Terms

- **Wilkinson notation** — a formula grammar for writing regression models, such as `body_mass_g ~ flipper_length_mm`

Formulas

$$y \sim x_1 + x_2$$

The left side gives the dependent variable, and the right side gives the predictors used to explain it.

Figure

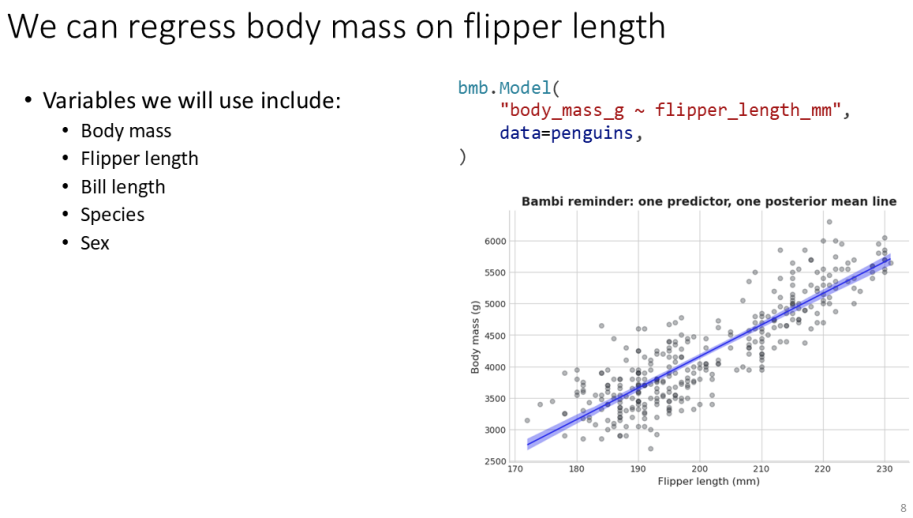


Figure 1: Slide 8 — We can regress body mass on flipper length

---

9.2 — Categorical predictors

This section shows that a categorical predictor can be written as a regression model by using indicator variables. In the penguin example, Adelie is the reference species, and the other species are represented by dummy variables. The intercept estimates the mean for the reference group, while the species coefficients estimate differences from that reference group.

New Definitions and Terms

- **Categorical predictor** — a predictor whose values are group labels rather than numerical measurements
- **Indicator variable** — a variable that equals 1 when an observation belongs to a category and 0 otherwise
- **Dummy coding** — representing categories with indicator variables
- **Reference coding** — dummy coding in which one group is represented by the intercept and all other groups are represented as differences from it
- **Effects coding** — coding in which dummy variables sum to 0 across groups
- **Cell means coding** — coding in which each group has its own parameter and there is no special reference intercept

Formulas

For species coded relative to Adelie penguins:

$$I_{1,i} = \begin{cases} 0 & \text{if species}_i = \text{Adelie} \\ 1 & \text{if species}_i = \text{Chinstrap} \\ 0 & \text{if species}_i = \text{Gentoo} \end{cases} \qquad I_{2,i} = \begin{cases} 0 & \text{if species}_i = \text{Adelie} \\ 0 & \text{if species}_i = \text{Chinstrap} \\ 1 & \text{if species}_i = \text{Gentoo} \end{cases}$$

The model mean becomes:

$$\mu_i = \beta_0 + \beta_1 I_{1,i} + \beta_2 I_{2,i}$$

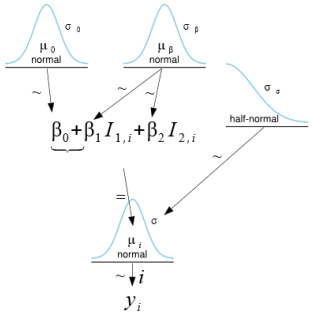
Equivalently:

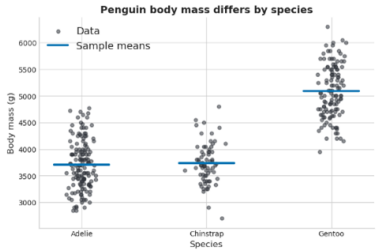
$$\mu_i = \begin{cases} \beta_0 & \text{if species}_i = \text{Adelie} \\ \beta_0 + \beta_1 & \text{if species}_i = \text{Chinstrap} \\ \beta_0 + \beta_2 & \text{if species}_i = \text{Gentoo} \end{cases}$$

Figure

We can write categorical variables as a regression

- Use indicator variables
- Get a different combination of parameters for each group
- In Wilkinson notation
  - body_mass ~ species





$$I_{1,i} = \begin{cases} 0 \text{ if species}_i = \text{Adelie} \\ 1 \text{ if species}_i = \text{Chinstrap} \\ 0 \text{ if species}_i = \text{Gentoo} \end{cases}$$

$$I_{2,i} = \begin{cases} 0 \text{ if species}_i = \text{Adelie} \\ 0 \text{ if species}_i = \text{Chinstrap} \\ 1 \text{ if species}_i = \text{Gentoo} \end{cases}$$

$$\mu_i = \begin{cases} \beta_0 & \text{if species}_i = \text{Adelie} \\ \beta_0 + \beta_1 & \text{if species}_i = \text{Chinstrap} \\ \beta_0 + \beta_2 & \text{if species}_i = \text{Gentoo} \end{cases}$$

12

Figure 2: Slide 12 — We can write categorical variables as a regression

Coefficients are differences from Adelie penguins

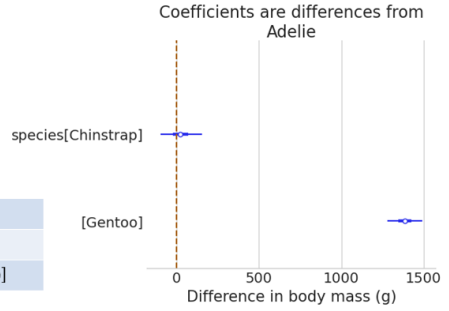
- Gentoo penguins seem bigger

```
coef_summary = az.summary(idata_mass_species, var_names=["species"])
```

	mean	sd	hdi_3%	hdi_97%
species[Chinstrap]	25.296	67.623	-98.350	153.913
species[Gentoo]	1385.146	57.599	1279.879	1491.911

Group differences as coefficient differences

Chinstrap vs Adelie	species[Chinstrap]
Gentoo vs Adelie	species[Gentoo]
Chinstrap vs Gentoo	species[Gentoo] – species[Chinstrap]



17

Figure 3: Slide 17 — Coefficients are differences from Adelie penguins

9.3 — Two categorical predictors

This section extends categorical regression to two predictors: species and sex. Bambi automatically creates dummy variables for both factors, so the model can be written with the same Wilkinson grammar as a continuous regression. The additive model assumes that the sex difference is the same for all species and that species differences are the same for both sexes.

New Definitions and Terms

- **Factor** — a categorical variable in a regression formula
- **Level** — one possible value of a factor
- **Additive model** — a model in which predictors contribute separate terms but do not change each other's effects
- **Main effect** — the contribution of one predictor in an additive model

Formulas

An additive categorical model with species and sex:

$$\mu_i = \beta_0 + \sum_k \beta_{\text{species},k} I_{\text{species},k,i} + \beta_{\text{sex}} I_{\text{sex},i}$$

For sex coded as male relative to female:

$$I_{\text{sex},i} = \begin{cases} 1 & \text{if male} \\ 0 & \text{if female} \end{cases}$$

For a male and female penguin from the same species:

$$\mu_i - \mu_j = \beta_{\text{sex}}$$

This is the assumption that the sex difference is constant across species.

Figure

The additive model makes strong assumptions

- Sex differences are the same for all species

$$\mu_i = \beta_0 + \sum_k \beta_{\text{species},k} I_{\text{species},k,i} + \beta_{\text{sex}} I_{\text{sex},i} \qquad I_{\text{sex},i} = \begin{cases} 1 \text{ if male} \\ 0 \text{ if female} \end{cases}$$

Two dummy variables for species
One dummy variable for sex

For the difference between any male penguin i and female penguin j from the same species:

$$\mu_i - \mu_j = \left(\beta_0 + \sum_k \beta_{\text{species},k} I_{\text{species},k,i} + \beta_{\text{sex}}\right) - \left(\beta_0 + \sum_k \beta_{\text{species},k} I_{\text{species},k,j}\right)$$

$$= \beta_{\text{sex}} \qquad \text{because } I_{\text{species},k,i} = I_{\text{species},k,j} \text{ for all } k$$

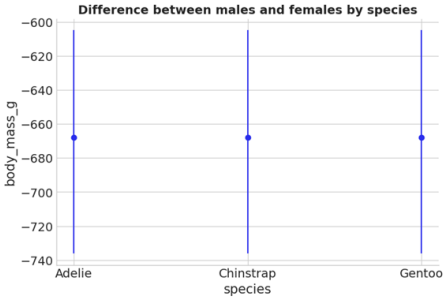
21

Figure 4: Slide 21 — The additive model makes strong assumptions

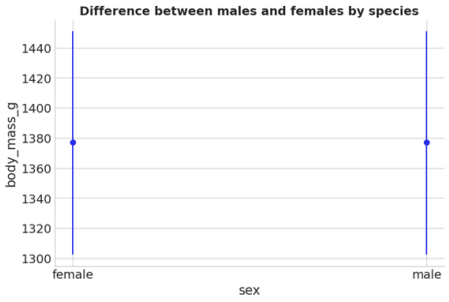
No interaction between sex and species in the additive model

•

```
bmb.interpret.plot_comparisons(model=model, idata=idata,
                               contrast={"sex": ["male", "fema
                               conditional="species")
```




```
bmb.interpret.plot_comparisons(model=model, idata=idata,
                               contrast={"species": ["Adelie", "Gentoo"]},
                               conditional="sex")
```



23

Figure 5: Slide 23 — No interaction between sex and species in the additive model

9.4 — Interactions

This section introduces interactions as changing slopes. In an additive model with two continuous predictors, each predictor has a fixed slope. In an interaction model, the slope of one predictor depends on the value of the other predictor, which is represented by a product term in the regression mean.

New Definitions and Terms

- **Interaction** — a model term allowing the effect of one predictor to depend on another predictor
- **Interaction term** — a product term, such as $x_1 x_2$, added to the model
- **Correlated predictors** — predictors that carry overlapping information, making individual coefficients harder to interpret

Formulas

Additive model:

$$\mu_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

Interaction model:

$$\mu_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$$

Wilkinson notation:

$$y \sim x_1 + x_2 + x_1 : x_2$$

or equivalently:

$$y \sim x_1 * x_2$$

The `:` operator adds only the interaction term, while `*` adds the main effects and the interaction.

Figure

Interactions mean changing slopes

- For two continuous predictors
  - The additive model has a fixed slope for each one
- In an interaction model:
  - the slope of one variable depends on the value of the other
  - Written as an additive model with an interaction term

Additive model $\mu_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2$

Interaction model

$$\mu_i = \beta_0 + \left(\beta'_{1,0} + \beta'_{1,2} x_2\right) x_1 + \left(\beta'_{2,0} + \beta'_{2,1} x_1\right) x_2$$

$$= \beta_0 + \beta'_{1,0} x_1 + \beta'_{2,0} x_2 + \left(\beta'_{1,2} + \beta'_{2,1}\right) x_1 x_2$$

$$= \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$$

25

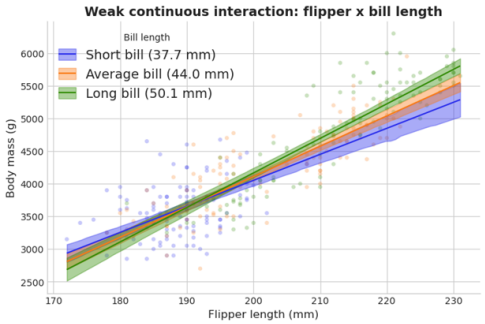
Figure 6: Slide 25 — Interactions mean changing slopes

Fit a Bambi model to see the changing slope

- Specify the interaction model
  - And automatically get changing slope

```
bmb.Model(
    "body_mass_g ~ flipper_length_mm * bill_length_mm",
    data=penguins_cont,
)

bmb.interpret.plot_predictions(
    model, idata,
    conditional={
        "flipper_length_mm": np.linspace(
            penguins_cont["flipper_length_mm"].min(),
            penguins_cont["flipper_length_mm"].max(),
            80,
        ),
        "bill_length_mm": bill_level_values,
    }
)
```



28

Figure 7: Slide 28 — Fit a Bambi model to see the changing slope

9.5 — Interacting categorical variables

This section applies interaction logic to categorical variables. Using `0 + species:sex` switches to a cell-means representation in which each species-sex combination receives its own mean. In the penguin data, the interaction is visible because the male-female body-mass difference is smaller for Chinstrap penguins and larger for Gentoo penguins.

The section also introduces the ToothGrowth example, where dose and supplement type interact. The posterior plot suggests that treating dose categorically may sometimes be more appropriate than forcing it to behave as a continuous linear predictor.

New Definitions and Terms

- **Categorical interaction** — an interaction between two categorical predictors
- **Cell mean** — the mean for one combination of factor levels

Formulas

Cell-means interaction model:

$$y \sim 0 + \text{species} : \text{sex}$$

This gives a separate parameter for each combination of species and sex.

Mixed categorical-continuous interaction:

$$\text{len} \sim \text{dose} * \text{supp}$$

This allows the effect of dose to differ between supplement types.

Figure



Figure 8: Slide 32 — The interaction is visible by eye

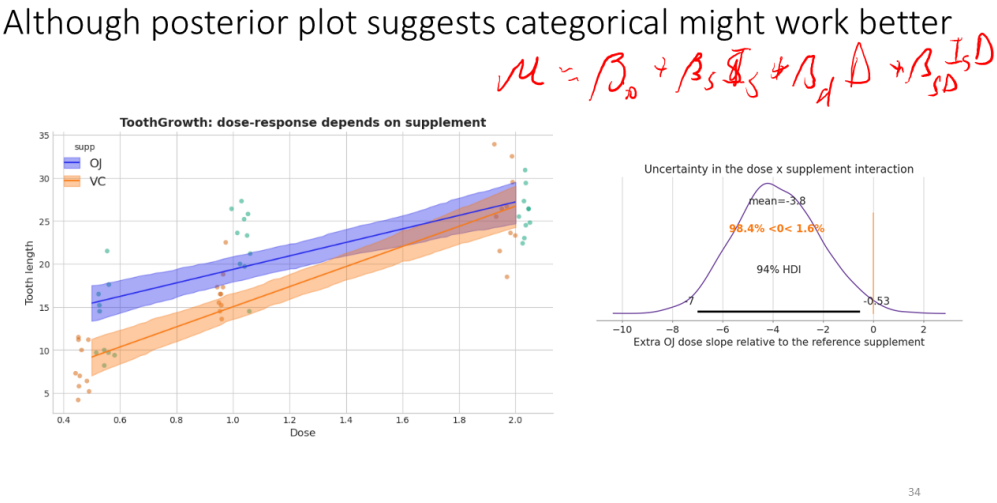


Figure 9: Slide 34 — Although posterior plot suggests categorical might work better

9.6 — Categorical vs continuous predictors

This section uses birth weight, gestation, and smoking to show that modeling choices are substantive, not just technical. A continuous gestation model assumes that a one-week change has the same effect at all gestational ages. A categorical gestation model relaxes that assumption, but it also creates unbalanced categories and wider posterior intervals.

The birth-weight example shows that an additive model suggests a possible smoking-related decrease, while the interaction model is less decisive. The categorical version clarifies that effects may vary with gestational term, but the HDIs also show that the available data are not enough to resolve every category-specific effect.

New Definitions and Terms

- **Continuous predictor** — a predictor treated as numerical, with changes interpreted per unit
- **Unbalanced categories** — categories with unequal sample sizes

Formulas

Additive birth-weight model:

$$\text{birth_weight} \sim \text{gestation} + \text{smoking}$$

Interaction birth-weight model:

$$\text{birth_weight} \sim \text{gestation} * \text{smoking}$$

Categorical gestation model:

$$\text{birth_weight} \sim 0 + \text{gestation_category} : \text{smoking}$$

Figure

Fit an additive model and see a possible difference

- Smoking introduces a decrease of 175 g (HDI: -260 to 50)

	mean	sd	hdi_3%	hdi_97%
Intercept	-1452.914	237.598	-1891.673	-1013.963
gestation_weeks	122.558	6.107	111.409	134.027
smoking[smoker]	-175.865	47.776	-261.325	-83.772
sigma	490.352	10.710	470.265	510.394

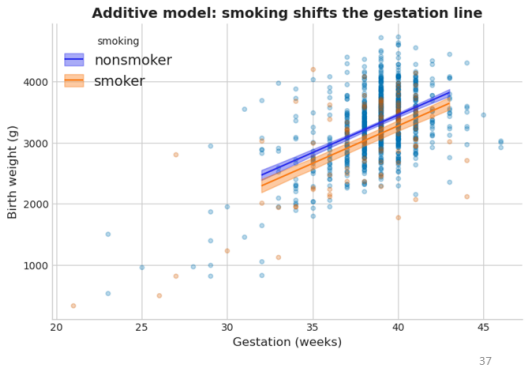


Figure 10: Slide 37 — Fit an additive model and see a possible difference

Effects may vary with gestational term

- HDIs suggest we don't have enough data
  - Should we repeat the study in a country with more smoking?
- Categorical coding clarifies the picture
  - In this case

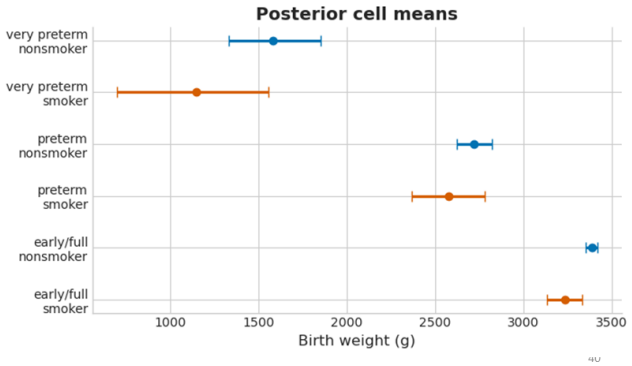


Figure 11: Slide 40 — Effects may vary with gestational term

9.7 — Model effect sizes

This section asks how much explanatory power a model or model term has. Bayesian R^2 is framed as variance in the model mean relative to the total modeled variance: explained variation plus residual noise. Partial R^2, η^2, and Cohen's f then measure the contribution of individual predictors or factors by comparing the full model to versions in which terms are left out.

The important distinction is the denominator. Partial R^2 asks how much of the remaining unexplained variance is explained by adding a term. η^2 compares the added variance to a total-variance scale, while Cohen's f expresses the added effect relative to the residual standard deviation.

New Definitions and Terms

- **Partial** R^2 — variance explained by one term relative to what remains unexplained after considering other terms
- η^2 — variance explained by a factor relative to a total-variance scale
- **Cohen's** f — added explained variation expressed in units of residual variation

Formulas

Bayesian R^2:

$$\text{var}(y) = \text{var}(\mu_i) + \sigma^2$$

$$R^2 = \frac{\text{var}(\mu_i)}{\text{var}(\mu_i) + \sigma^2}$$

Reduced means for a two-predictor interaction model:

$$\mu_i^{x_1} = \beta_0 + \beta_2 x_{2,i} + \beta_{12} x_{1,i} x_{2,i}$$

$$\mu_i^{x_2} = \beta_0 + \beta_1 x_{1,i} + \beta_{12} x_{1,i} x_{2,i}$$

$$\mu_i^{x_1 x_2} = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i}$$

Partial R^2 for a term:

$$R^2_{x_1} = \frac{\text{var}(\mu_i^{x_1} - \mu_i)}{\text{var}(\mu_i^{x_1} - \mu_i) + \sigma^2}$$

Eta-squared for a term:

$$\eta^2_{x_1} = \frac{\text{var}(\mu_i^{x_1} - \mu_i)}{\text{var}(\mu_i) + \sigma^2}$$

Cohen's f for a term:

$$f = \frac{\text{std}(\mu_i^{x_1} - \mu_i)}{\sigma}$$

Figure

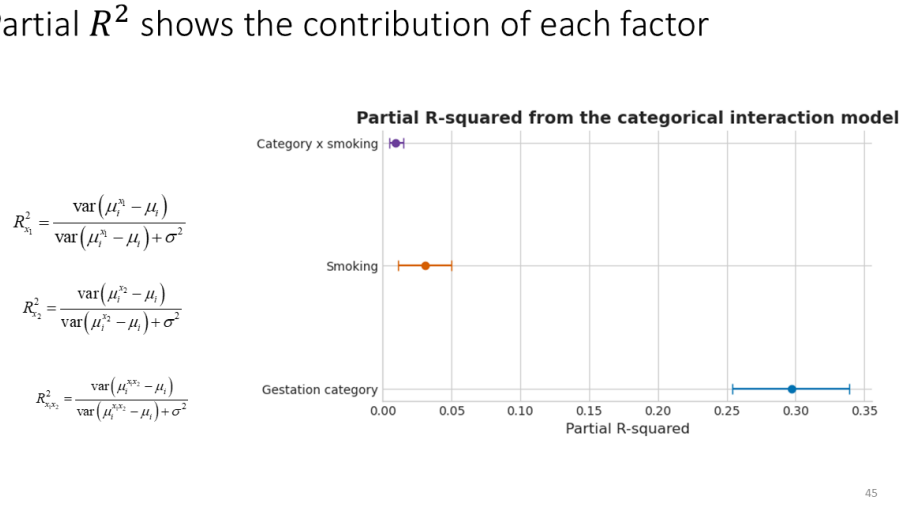


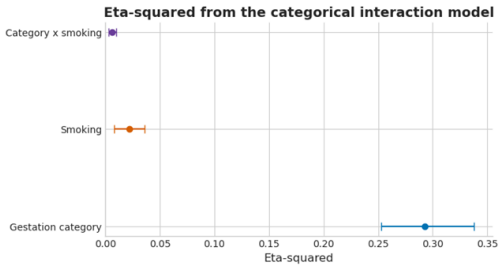
Figure 12: Slide 45 - Partial R-squared from the categorical interaction model

η^2 measures variance explained by a factor relative to its full model

- R^2: Variance explained out of what was left
  - After considering all other facgors
- η^2: Variance explained relative to total variance

$$R^2_{x_1} = \frac{\text{var}\left(\mu_i^{x_1} - \mu_i\right)}{\text{var}\left(\mu_i^{x_1} - \mu_i\right) + \sigma^2}$$

$$\eta^2_{x_1} = \frac{\text{var}\left(\mu_i^{x_1} - \mu_i\right)}{\text{var}\left(\mu_i\right) + \sigma^2}$$



46

Figure 13: Slide 46 - Eta-squared from the categorical interaction model

---

9.8 — Frequentist translation

This section translates the Bayesian effect-size discussion into frequentist sums of squares. Frequentist R^2, partial R^2, η^2, and Cohen's f are calculated using fitted values, residual sums of squares, and total sums of squares. The key idea is still model comparison: the contribution of a term is measured by how much the predictions change when that term is removed.

Partial R^2 normalizes the added explained sum of squares by added explained variation plus residual variation. Partial η^2 normalizes added explained variation by the total variance for that factor plus noise. Cohen's f uses the same added explained variation but expresses it on a residual-standard-deviation scale.

New Definitions and Terms

- **Sum of squares** — a squared-deviation measure of variation
- **Added explained sum of squares** — the increase in explained variation when a term is included
- **Frequentist** R^2 — explained sum of squares divided by total sum of squares

Formulas

Residual and regression sums of squares:

$$SS_{\text{res}} = \sum_i (y_i - \hat{y}_i)^2$$

$$SS_{\text{reg}} = \sum_i (\hat{y}_i - \bar{y})^2$$

$$SS_{\text{tot}} = \sum_{i=1}^{N} (y_i - \bar{y})^2 = SS_{\text{reg}} + SS_{\text{res}}$$

Frequentist R^2:

$$R^2 = \frac{SS_{\text{reg}}}{SS_{\text{tot}}}$$

Added explained sum of squares for x_1:

$$SS_{\text{reg},x_1} = \sum_i (\hat{y}_{x_1,i} - \hat{y}_i)^2$$

Here $\hat{y}_{x_1,i}$ is the predicted value from a model without x_1.

Partial R^2:

$$R^2_{x_1} = \frac{SS_{x_1,\text{reg}}}{SS_{x_1,\text{reg}} + SS_{\text{res}}}$$

Partial η^2:

$$\eta^2_{x_1} = \frac{SS_{x_1,\text{reg}}}{SS_{\text{tot}}}$$

Cohen's f:

$$f = \frac{\sqrt{SS_{\text{reg},x_1}}}{\sqrt{SS_{\text{res}}}}$$

Figure

R^2,η^2, and f are calculated in terms of sums of squares

$$SS_{\text{res}} = \sum_i (y_i - \hat{y}_i)^2$$

$$SS_{reg} = \sum_i (\hat{y}_i - \bar{y})^2 \qquad R^2 = \frac{SS_{\text{reg}}}{SS_{\text{tot}}}$$

$$SS_{\text{tot}} = \sum_{i=1}^{N} (y_i - \bar{y}_i)^2$$

49

Figure 14: Slide 49 - R-squared, eta-squared, and f in terms of sums of squares

Patial R^2 is the added variance explained by a factor

- Normalized by the residual variance after adding all the other factors

$$SS_{\text{res}} = \sum_i (y_i - \hat{y}_i)^2$$

$$SS_{reg} = \sum_i (\hat{y}_i - \bar{y})^2 \qquad SS_{reg,x_1} = \sum_i (\hat{y}_{x_1,i} - \hat{y}_i)^2$$

$\hat{y}_{x_1,i}$ is the predicted value from a model without x_1

$$SS_{\text{tot}} = \sum_{i=1}^{N} (y_i - \bar{y}_i)^2 = SS_{\text{reg}} + SS_{\text{res}} \qquad R^2_{x_1} = \frac{SS_{x_1,\text{reg}}}{SS_{x_1,\text{reg}} + SS_{\text{res}}}$$

50

Figure 15: Slide 50 - Normalized by the residual variance after adding all the other factors

Patial η^2 is the added variance explained by a factor

- Normalized by the total variance for that factor plus noise

$$SS_{\text{res}} = \sum_i \left(y_i - \hat{y}_i\right)^2$$

$$SS_{reg} = \sum_i \left(\hat{y}_i - \bar{y}\right)^2$$

$$SS_{reg,x_1} = \sum_i \left(\hat{y}_{x_1,i} - \hat{y}_i\right)^2$$

$\hat{y}_{x_1,i}$ is the predicted value from a model without x_1

$$SS_{\text{tot}} = \sum_{i=1}^{N} \left(y_i - \bar{y}_i\right)^2 = SS_{\text{reg}} + SS_{\text{res}}$$

$$\eta_{x_1}^2 = \frac{SS_{x_1,\text{reg}}}{SS_{\text{tot}}}$$

51

Figure 16: Slide 51 - Normalized by the total variance for that factor plus noise

Cohen's f is the added variance explained by a factor

- Normalized by the residual standard deviation

$$SS_{\text{res}} = \sum_i \left(y_i - \hat{y}_i\right)^2$$

$$SS_{reg} = \sum_i \left(\hat{y}_i - \bar{y}\right)^2$$

$$SS_{reg,x_1} = \sum_i \left(\hat{y}_{x_1,i} - \hat{y}_i\right)^2$$

$\hat{y}_{x_1,i}$ is the predicted value from a model without x_1

$$SS_{\text{tot}} = \sum_{i=1}^{N} \left(y_i - \bar{y}_i\right)^2 = SS_{\text{reg}} + SS_{\text{res}}$$

$$f = \frac{\sqrt{SS_{reg,x_1}}}{\sqrt{SS_{res}}}$$

52

Figure 17: Slide 52 - Normalized by the residual standard deviation

Lecture 10 Review Handout

Statistics 367-1-4361

Lecture Overview

Lecture 10 introduces model comparison as a way to choose among plausible statistical models rather than only rejecting or accepting a null model. The lecture begins with overfitting and underfitting, then reframes model quality around scientific goals: theory building, hypothesis testing, and prediction. The main technical path is predictive model comparison: KL divergence motivates the score, information criteria approximate it, and cross-validation estimates it more directly. The lecture ends by placing model comparison inside the Bayesian workflow and returning to the birth-weight and smoking example.

Lecture Sections

10.1 — Overfitting and underfitting

This section starts from the problem that many models are possible for the same dataset. Measures such as R^2 reward fit to the observed data, but fit alone always improves as parameters are added. The goal is not maximal fit; it is useful complexity, balancing models that ignore structure against models that chase noise.

New Definitions and Terms

- **Overfitting** — fitting details of the observed data that do not generalize to new data
- **Underfitting** — using a model too simple to capture important structure in the data
- **Goodness of fit** — how closely a model fits the observed data
- **Occam's razor** — the preference for the simpler model when two models fit equally well
- **Chatton's anti-razor** — the reminder that added assumptions may be necessary when simpler assumptions do not explain the phenomena

Figure

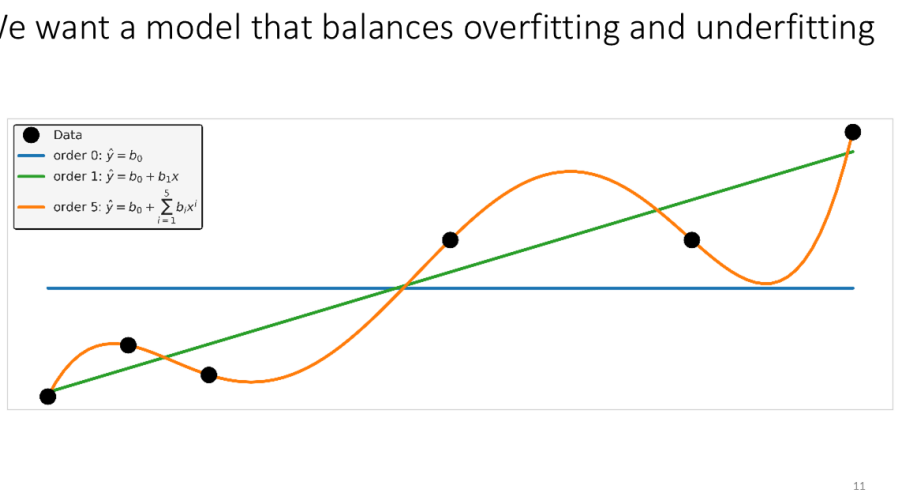


Figure 1: Slide 11 - A polynomial model balancing overfitting and underfitting

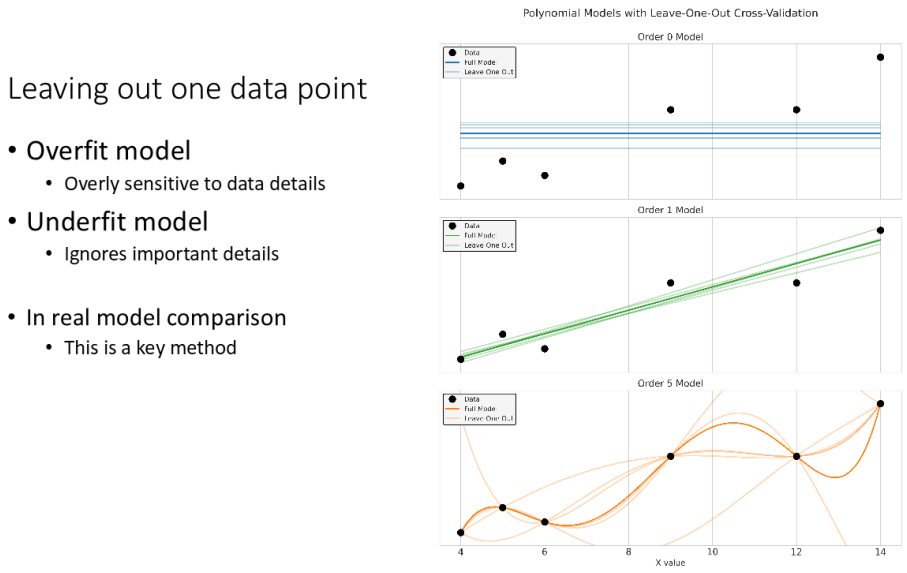


Figure 2: Slide 14 — Leaving out one data point

10.2 — What makes a good model

This section makes model evaluation explicitly goal-dependent. A model can be useful for theory building, hypothesis testing, or prediction, and each goal implies a different meaning of "good." The central shift is from asking whether a model is true to asking whether it is useful for the scientific goal.

New Definitions and Terms

- **Theory-building model** — a model evaluated by whether it is interpretable, meaningful, and generalizable
- **Predictive model** — a model evaluated by how well it predicts new data
- **Candidate model** — one of the models being compared for a particular scientific purpose

Figure

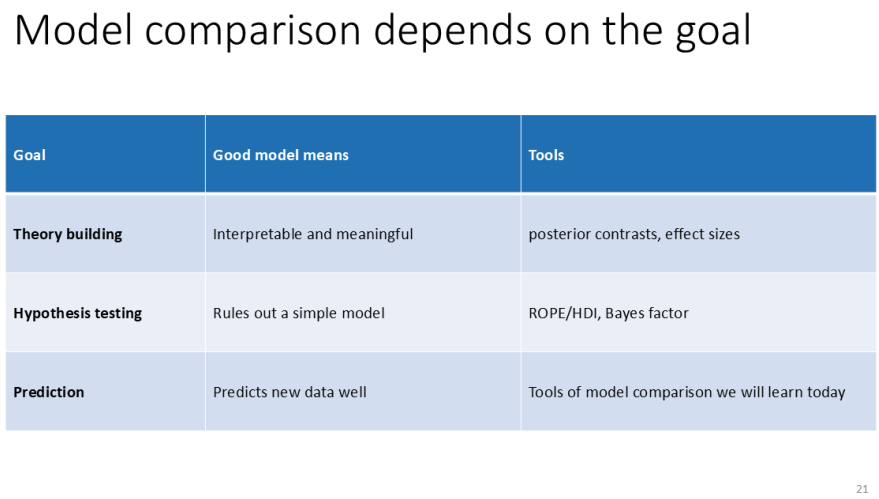


Figure 3: Slide 21 — Model comparison depends on the goal

10.3 — Kullback-Leibler divergence

This section motivates predictive comparison by imagining a distance between a candidate model and the true data-generating distribution. We do not know the true distribution, so we cannot calculate this distance directly. The key insight is that a model closer to the true data-generating distribution should assign higher probability to new data.

New Definitions and Terms

- **Surprise** — the information cost of an outcome; lower-probability outcomes are more surprising
- **Entropy** — the expected surprise under the true distribution
- **Cross-entropy** — the expected surprise when guesses are made using the wrong distribution
- **Kullback-Leibler divergence** — the extra expected surprise from using one distribution in place of another
- **Score** — the expected log predictive density of new data under a model
- **Deviance** — -2 times the score; the same predictive idea expressed on a different scale

Formulas

Entropy:

$$H(p) = -\int p(y) \log p(y)\, dy$$

Cross-entropy:

$$H(p, q) = -\int p(y) \log q(y)\, dy$$

KL divergence:

$$D_{\mathrm{KL}}(p \| q) = H(p, q) - H(p) = \int p(y) \log \frac{p(y)}{q(y)}\, dy$$

For model comparison, the wrong distribution is the model's posterior predictive distribution:

$$D_{\mathrm{KL}}\{p \| p(y \mid m)\} = \int p(y)\{\log p(y) - \log p(y \mid m)\}\, dy$$

The predictive score is the part that can differ between models:

$$\text{score}(m) \approx \sum_i \log p(\tilde{y}_i \mid m)$$

where

$$p(\tilde{y}_i \mid m) = \int p(\tilde{y}_i \mid \theta, m) p(\theta \mid m)\, d\theta$$

Deviance uses the same comparison on a different scale:

$$\text{deviance} = -2\,\text{score}$$

Figure

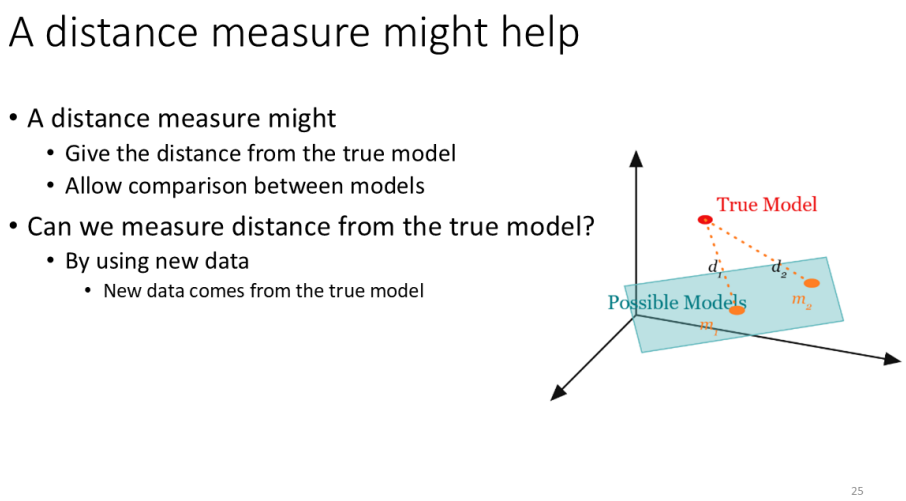


Figure 4: Slide 25 — A distance measure might help

KL divergence measures extra surprise

- The difference between the cross entropy and the entropy
- The added uncertainty of using the wrong model
- An (asymmetric) measure of distance between two distributions

$$D_{KL}\left(p\|q\right) = H\left(p,q\right) - H\left(p\right)$$

Insert definitions

$$= \left(-\int dp\, p\left(x\right)\log q\left(x\right)\right) - \left(-\int dp\, p\left(x\right)\log p\left(x\right)\right)$$

Rearrange terms

$$= \int dp\, p\left(x\right)\log p\left(x\right) - \int dp\, p\left(x\right)\log q\left(x\right)$$

Combine integrals

$$= \int dp\, p\left(x\right)\left(\log p\left(x\right) - \log q\left(x\right)\right)$$

The average (over the true distribution) of how wrong we are (in units of surprise)

30

Figure 5: Slide 30 — KL divergence measures extra surprise

Calculating the score

- To find likelihood of new data under the model
  - Average over the posterior distribution under the model

$$D_{KL}\left(p\|p\left(y\,|\,m\right)\right) \approx -\sum_i \log p\left(\tilde{y}_i\,|\,m\right) + H\left(p\right)$$

$$\sum_i \log p\left(\tilde{y}_i\,|\,m\right) = \sum_i \int_\theta d\theta \log p\left(\tilde{y}_i\,|\,\theta,m\right) p\left(\theta\,|\,m\right)$$

Often people calculate deviance instead of score
Different units, same idea

$$\text{deviance} = -2\cdot\text{Score} = -2\sum_i \log p\left(\tilde{y}_i\,|\,m\right)$$

Figure 6: Slide 33 — Calculating the score

10.4 — Information criteria

Information criteria estimate out-of-sample predictive performance using only the observed data. The problem is optimism: a model usually predicts the data it was fit on better than it predicts new data. Different information criteria estimate and subtract this optimism in different ways.

New Definitions and Terms

- **Information criterion** — an estimate of out-of-sample predictive performance based on in-sample fit plus an optimism correction
- **Optimism** — the excess predictive performance obtained by testing a model on the same data used to fit it
- **AIC** — Akaike information criterion; a frequentist information criterion using a point estimate and a parameter-count penalty
- **WAIC** — widely applicable information criterion; a Bayesian information criterion using the full posterior distribution
- **Effective number of parameters** — the posterior-based model-complexity estimate used by WAIC
- **Model weight** — a weight for combining model predictions according to estimated predictive performance

Formulas

AIC:

$$\text{AIC} = -2 \log p(y \mid \hat{\theta}, m) + 2k$$

Here k is the number of model parameters. Lower AIC means better estimated predictive performance.

WAIC:

$$\text{WAIC} = -2 \sum_i \log \int p(y_i \mid \theta, m) p(\theta \mid m) \, d\theta + 2 \sum_i \text{Var}_\theta \{\log p(y_i \mid \theta, m)\}$$

The first term measures posterior predictive fit to the observed data. The second term estimates optimism using the posterior variation in pointwise log likelihood.

Figure

Several criteria estimate predictive accuracy

Criterion	In sample log density	Optimism term
AIC	$-2\sum_i \log p(y_i \mid \hat{\theta}_{MLE}, m)$	$2k$ where k is the number of model parameters
DIC	$-2\sum_i \log p(y_i \mid \hat{\theta}, m)$	$2\left[\sum_i \log p(y_i \mid \hat{\theta}, m) - \mathbb{E}_{\theta \mid y} \sum_i \log p(y_i \mid \theta, m)\right]$
WAIC	$-2\sum_i \log \mathbb{E}_{\theta \mid y}[p(y_i \mid \theta, m)]$	$2\sum_i \mathrm{Var}_{\theta \mid y}[\log p(y_i \mid \theta, m)]$

39

Figure 7: Slide 39 — Several criteria estimate predictive accuracy

WAIC in an applied model comparison

- WAIC table gives estimate uncertainty
  - 'Best' model may not be much better
- WAIC model also shows test for influential data points
  - If one data point is overly influential, WAIC is biased estimate of score

```
az.compare(penguin_idata, ic="waic")
```

	rank	elpd_waic	p_waic	elpd_diff	weight	se	dse	warning	scale
species * sex	0	-2386.290200	6.967973	0.000000	0.862056	12.175076	0.000000	False	log
species + sex	1	-2392.875092	4.960643	6.584892	0.137944	12.005834	4.269239	False	log
species	2	-2517.111104	3.607536	130.820904	0.000000	11.230931	10.687399	False	log

- p_waic estimates the model complexity (the number of effective parameters)
- weight can be used to get improved predictions by using a weighted combination of models
- dse is the standard error of the difference
- Warning means that there are influential points

45

Figure 8: Slide 45 — WAIC in an applied model comparison

10.5 — Cross-validation

Cross-validation estimates predictive performance by repeatedly fitting a model on one part of the data and evaluating it on another part. True validation would use a new dataset, but that is often expensive or slow. Leave-one-out cross-validation is especially attractive in Bayesian modeling because it can often be approximated without refitting the model N times.

New Definitions and Terms

- **Validation** — testing a model on data not used to fit it
- **True validation** — validation on a newly collected dataset
- **Withheld data** — data held out from fitting and used for testing
- **Leakage** — information from the test data influencing fitting or model selection
- **Cross-validation** — repeated train-test splitting so each subset is used for evaluation
- **K-fold cross-validation** — cross-validation in which the data are divided into k chunks
- **Leave-one-out cross-validation** — cross-validation in which each observation is left out once
- **ELPD** — expected log predictive density; the expected log likelihood of new data
- **Importance sampling** — estimating an expectation under one distribution using samples from another distribution and weights
- **PSIS** — Pareto-smoothed importance sampling; a stabilized approximation used for LOO-CV
- **Pareto** k — a diagnostic for the reliability of the PSIS approximation

Formulas

Expected log predictive density for LOO-CV:

$$\mathrm{ELPD}_{\mathrm{LOO-CV}} = \sum_i \log \int p(y_i \mid \theta) p(\theta \mid y_{-i})\, d\theta$$

The importance-sampling approximation uses posterior draws from $p(\theta \mid y)$ instead of refitting with y_i left out:

$$\mathrm{ELPD}_{\mathrm{LOO-CV}} \approx \sum_i \log \frac{\frac{1}{S}\sum_s w_s p(y_i \mid \theta_s)}{\frac{1}{S}\sum_s w_s}$$

with

$$w_s = \frac{1}{p(y_i \mid \theta_s)}$$

The Pareto k diagnostic checks whether the importance weights are too extreme. If $k > 1$, the LOO-CV estimate may be unreliable and full cross-validation may be required.

Figure

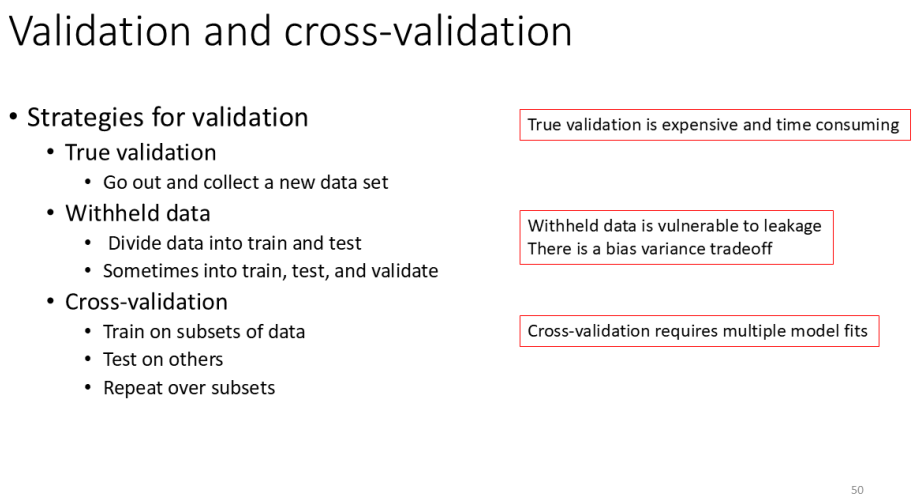


Figure 9: Slide 50 — Validation and cross-validation

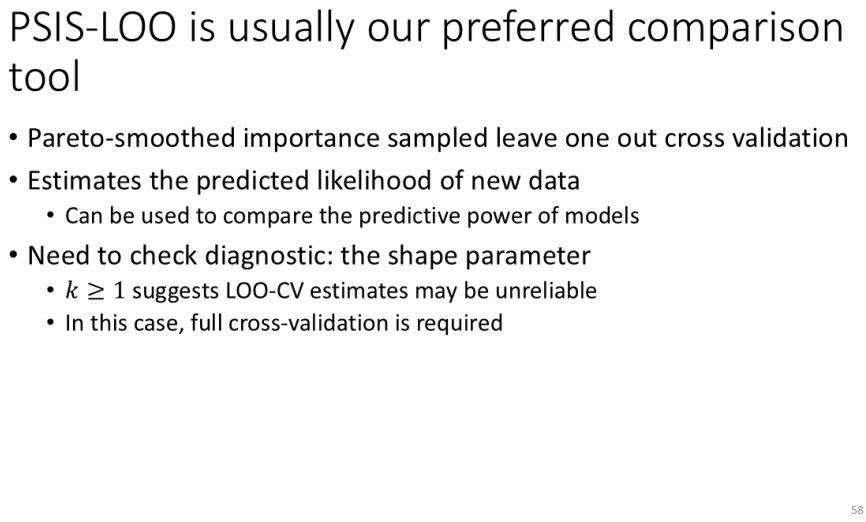


Figure 10: Slide 58 — PSIS-LOO is usually our preferred comparison tool

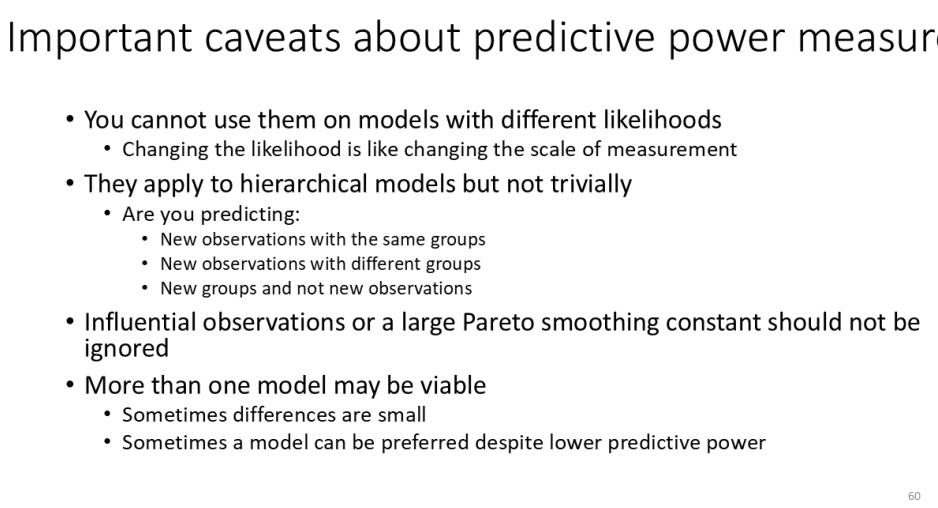


Figure 11: Slide 60 — Important caveats about predictive power measures

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10.6 — Updated Bayesian workflow

This section inserts model comparison into the full Bayesian workflow. Predictive comparison comes after scientific questions, candidate models, prior predictive checks, sampling, diagnostics, and posterior predictive checks. The ranking of models by predictive power is useful, but it does not replace interpretation of effects or scientific judgment.

The birth-weight example shows why this matters. The candidate models have similar overall effect sizes and similar predictive power, while the smoking parameter remains small but reasonably consistent across the leading models. The interaction model gives the largest smoking effect but also the largest uncertainty, so the analysis does not provide strong evidence for a smoking-by-gestation interaction.

New Definitions and Terms

- **Predictive ranking** — ordering models by estimated out-of-sample predictive performance
- **Model comparison workflow** — the use of predictive comparison after model checking and before final interpretation

Figure

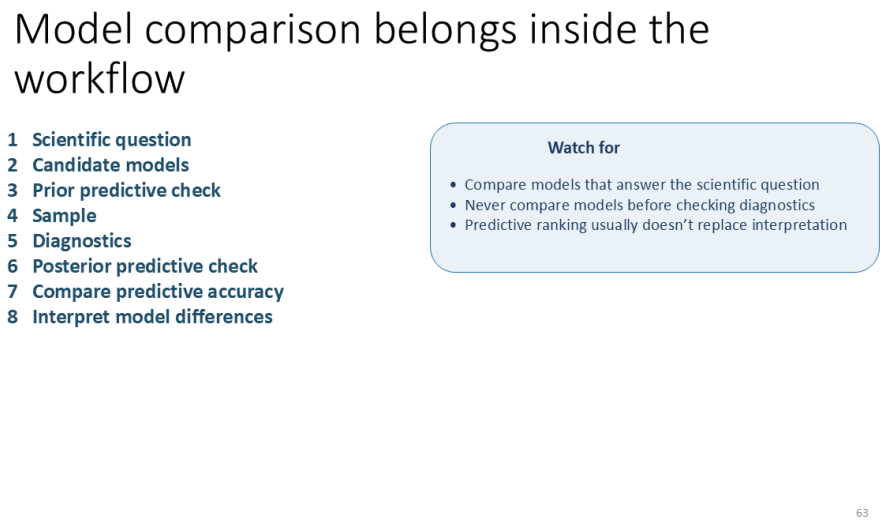


Figure 12: Slide 63 — Model comparison belongs inside the workflow

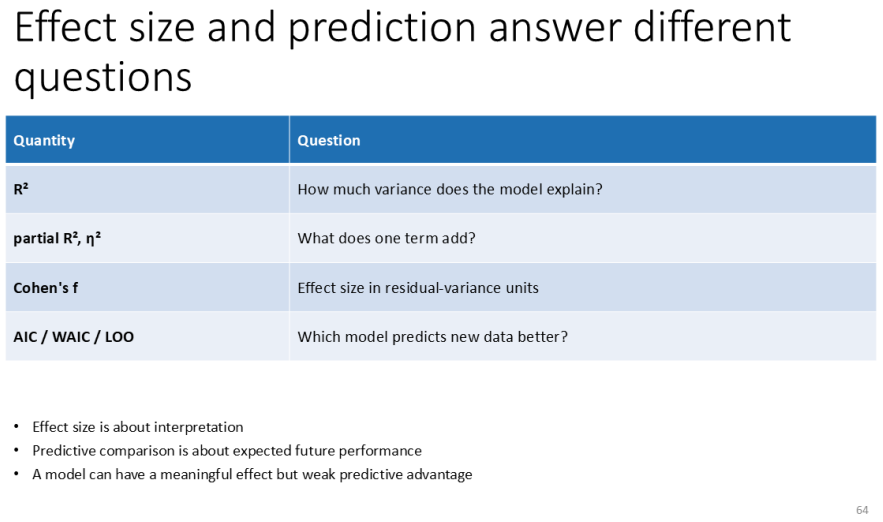


Figure 13: Slide 64 — Effect size and prediction answer different questions

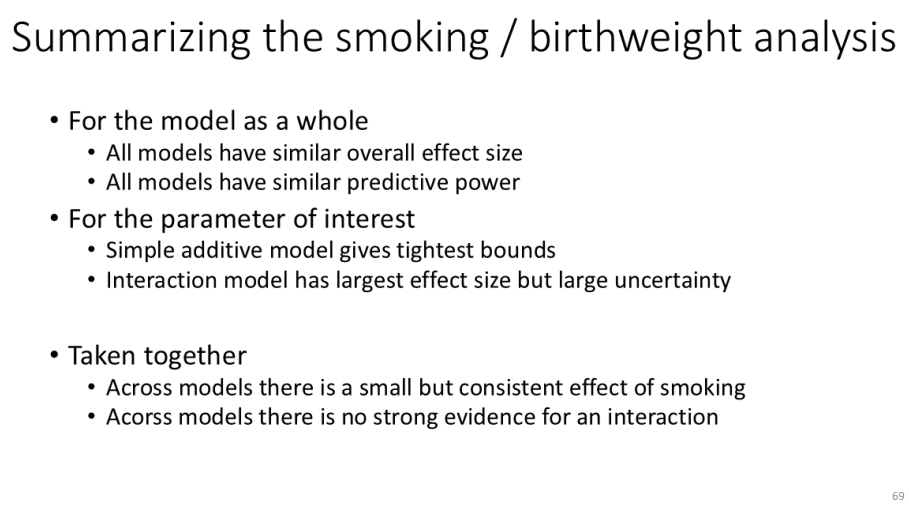


Figure 14: Slide 69 — Summarizing the smoking / birthweight analysis

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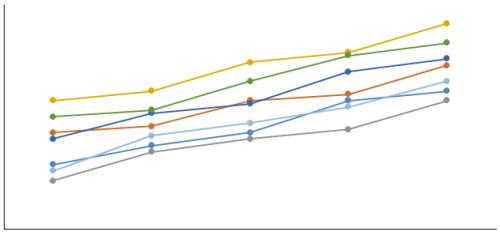
10.7 — Looking forward

The final slide previews the next step: combining regression with hierarchical models. The main takeaway from Lecture 10 is that model comparison is not an isolated calculation. It is one part of a larger workflow that starts with the scientific question and ends with interpreted model differences.

Figure

Next time: complex data one last time!

- Combining regression with hierarchical models!



71

Figure 15: Slide 71 - Next time: combining regression with hierarchical models

Lecture 11 Review Handout

Correlation and Repeated Measures | Statistics 367-1-4361

Lecture Overview

Lecture 11 moves from regression models, where one variable is treated as explanatory, to settings where variables or repeated measurements are jointly structured. The first half develops correlation as a population parameter inside a bivariate normal likelihood and compares correlations across two groups. The second half uses repeated wine ratings to separate latent wine quality from judge-specific generosity and discrimination. The lecture then translates the same modeling ideas into fixed/random effects, frequentist F-tests, and LOO-based model comparison.

11.1 — The Correlation Coefficient

Correlation is introduced as a symmetric, non-causal relationship between two noisy measurements. Unlike regression, neither variable has to be treated as the controlled or explanatory variable. The goal is to move from the sample correlation coefficient to a population correlation parameter that can appear inside a likelihood.

New Definitions and Terms

- **Correlation** — a symmetric association between two variables, without assigning one as the cause or predictor of the other.
- **Population correlation coefficient** — the parameter ρ describing the direction and strength of linear association in the population.
- **Sample correlation coefficient** — the estimate r, also called Pearson's r or the product-moment correlation.
- **Covariance matrix** — a matrix with variances on the diagonal and covariances on the off-diagonal.

- **Multivariate normal likelihood** — a likelihood for vector-valued observations, with means, variances, and covariances modeled jointly.

Formulas

$$\rho_{XY} = \frac{\operatorname{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

The population correlation is normalized covariance. Its sign gives direction and its magnitude gives strength.

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}$$

The sample correlation r estimates ρ. In the normal model it is the maximum-likelihood estimate.

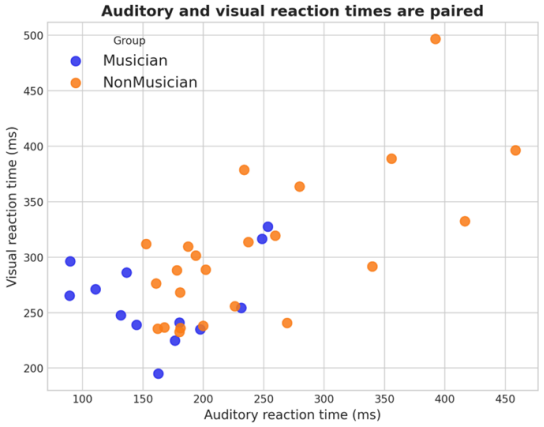
$$= \begin{pmatrix} \sigma_X^2 & \rho \sigma_X \sigma_Y \\ \rho \sigma_X \sigma_Y & \sigma_Y^2 \end{pmatrix}$$

The covariance matrix can be built from the two standard deviations and the correlation.

Figure

Correlation is symmetric

- Auditory and visual reaction times
  - Axes can be swapped
  - Both variables are measures with noise



4

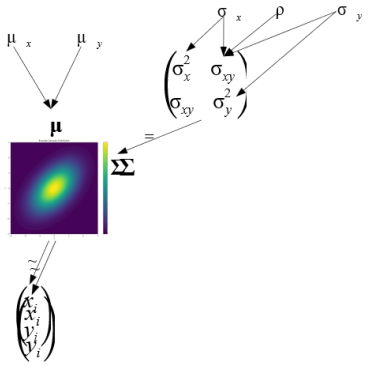
Figure 1: Slide 5 — Correlation is symmetric

How do we model correlations?

$$\begin{pmatrix} x \\ y \end{pmatrix} \sim \mathrm{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix} \qquad \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{xy} & \sigma_y^2 \end{pmatrix} \qquad \sigma_{xy} = \sigma_x \sigma_y \rho$$

- We want to estimate ρ.
  - For Bayesian inference, ρ must appear inside a likelihood.
  - We need a likelihood that describes a pair of outcomes jointly
- Each participant contributes a vector observation
  - (auditory RT, visual RT)
- The bivariate normal distribution gives us this
  - Our first multivariate likelihood



9

Figure 2: Slide 12 — How do we model correlations?

11.2 — Two-group Bivariate Normal Model

The audiovisual reaction-time data are modeled as two-dimensional observations: auditory and visual reaction time for each subject. The model estimates separate means, standard deviations, and correlations for musicians and non-musicians. The group difference in correlation is calculated inside the model as a deterministic parameter.

New Definitions and Terms

- **Vector observation** — one data point containing multiple measured outcomes, here (auditory RT, visual RT).
- **Group-specific correlation** — a separate ρ_g parameter for each group.
- **Scaled Beta prior for correlation** — a Beta prior on $\theta_\rho \in [0, 1]$, transformed to $\rho \in [-1, 1]$.
- **Deterministic contrast** — a parameter calculated from other model parameters, such as $\Delta\rho$.

Formulas

$$\theta_{\rho,g} \sim \text{Beta}(\alpha, \beta), \qquad \rho_g = 2\theta_{\rho,g} - 1$$

The Beta prior is convenient because it is bounded; the transformation maps it to the valid correlation range.

$$\mathbf{y}_i \sim \text{MvNormal}(\boldsymbol{\mu}_{g_i}, {}_{g_i})$$

Each subject contributes a two-variable response vector, and the group index selects that subject's mean vector and covariance matrix.

$$\Delta\rho = \rho_{\text{Musician}} - \rho_{\text{NonMusician}}$$

The group contrast is part of the posterior, not a separate post-hoc calculation.

Figure

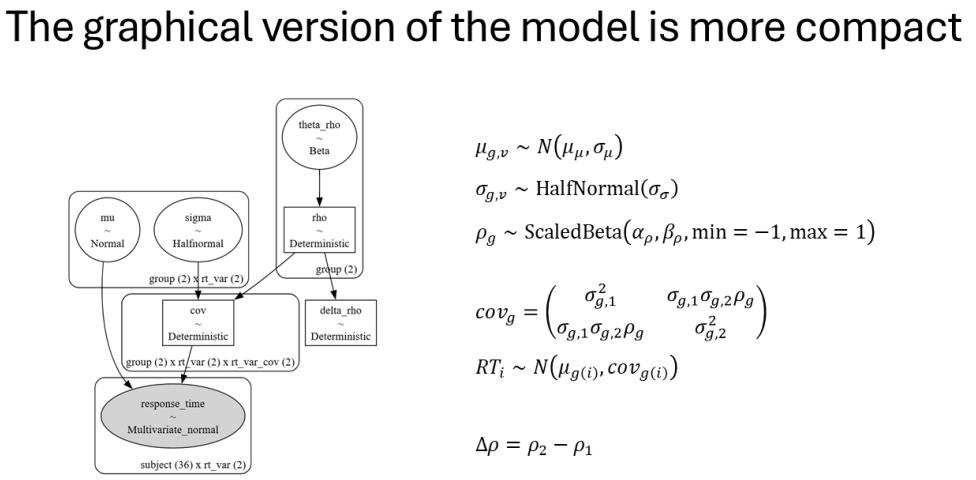


Figure 3: Slide 22 — The graphical version of the model is more compact

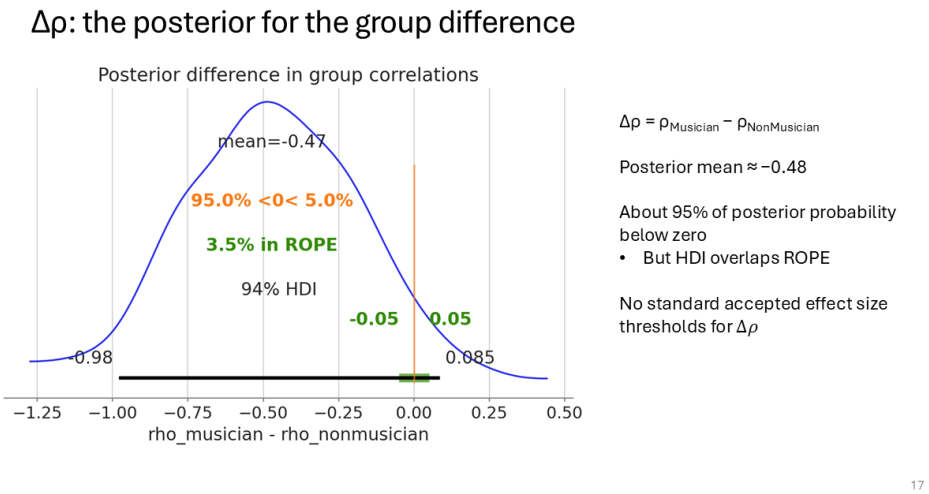


Figure 4: Slide 24 - posterior for the group difference in correlation

11.3 — Frequentist Translation

The frequentist translation uses Pearson's r as the point estimate and Fisher's z transform to construct confidence intervals and compare independent correlations. The Bayesian model estimates the same kind of population quantity, but represents uncertainty with posterior distributions rather than sampling intervals and null tests.

New Definitions and Terms

- **Fisher-z transform** — the transformation $\operatorname{arctanh}(r)$, used because transformed correlations are approximately normal.
- **Fisher-z confidence interval** — a confidence interval computed on the Fisher-z scale and then transformed back to the correlation scale.
- **Fisher-z difference test** — a frequentist test for whether two independent population correlations differ.

Formulas

$$z = \operatorname{arctanh}(r) = \frac{1}{2}\log\left(\frac{1+r}{1-r}\right)$$

Fisher's transform maps correlations from $[-1, 1]$ to the real line.

$$z \pm 1.96\sqrt{\frac{1}{n-3}}$$

This gives an approximate 95% confidence interval on the Fisher-z scale; applying $\tanh(\cdot)$ returns the interval to the correlation scale.

$$z_\Delta = \frac{z_1 - z_2}{\sqrt{\frac{1}{n_1-3} + \frac{1}{n_2-3}}}$$

This is the standard test statistic for comparing two independent correlations.

Figure

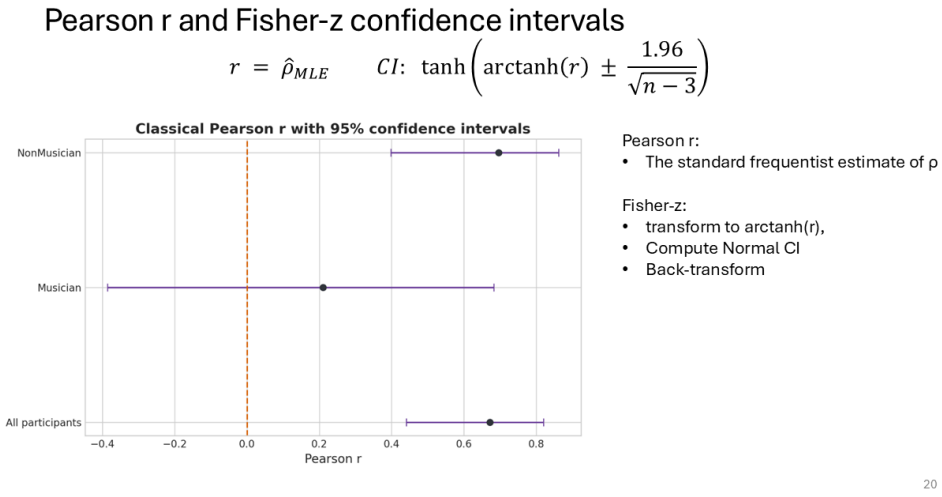


Figure 5: Slide 26 — Pearson r and Fisher-z confidence intervals

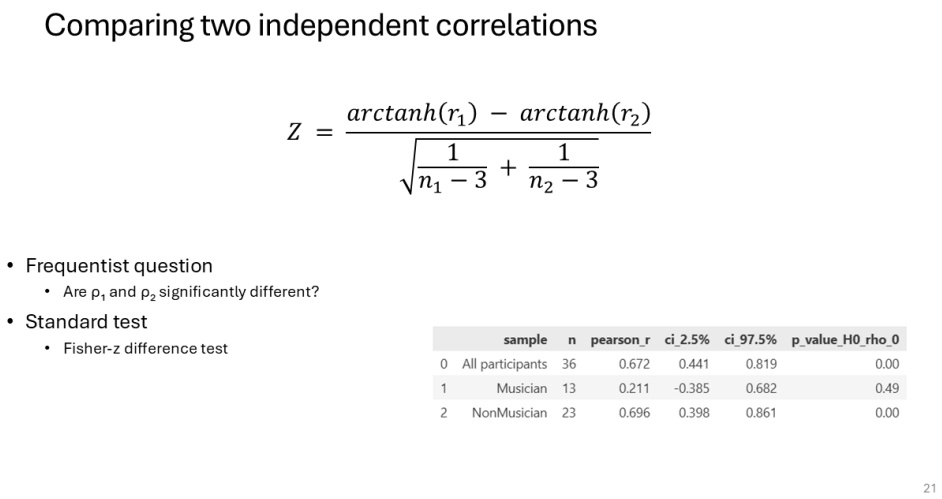


Figure 6: Slide 27 — Comparing two independent correlations

11.4 — Repeated Measurements: Wine Ratings

The wine ratings data introduce repeated measurements: every judge rates every wine. A first model ignores judge identity and lets scores vary only by wine. This creates a simple latent wine-quality model, with wine effects represented as deviations from the grand mean.

New Definitions and Terms

- **Repeated measurements** — multiple observations sharing a structured source, here many ratings from the same judges and many ratings of the same wines.
- **Zero-sum constraint** — a constraint that effects sum to zero, so each level is interpreted as a deviation from the grand mean.
- **ZeroSumNormal** — a PyMC distribution for normally distributed effects constrained to sum to zero.

Formulas

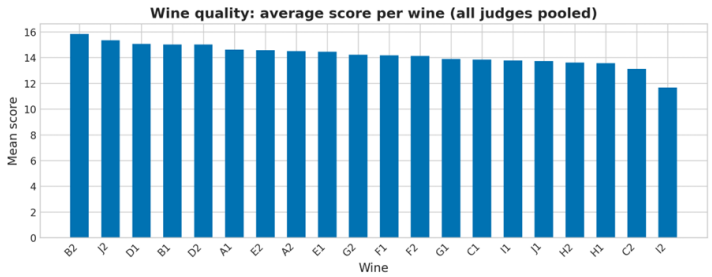
$$\mu_{ij} = \alpha + Q_i, \qquad \sum_i Q_i = 0$$

In the first wine model, the expected score depends only on wine identity. The wine-quality effects Q_i are deviations from the grand mean.

Figure

The Wines2012 data: wines and judges

9 judges, 20 wines, 180 ratings: every judge scores every wine

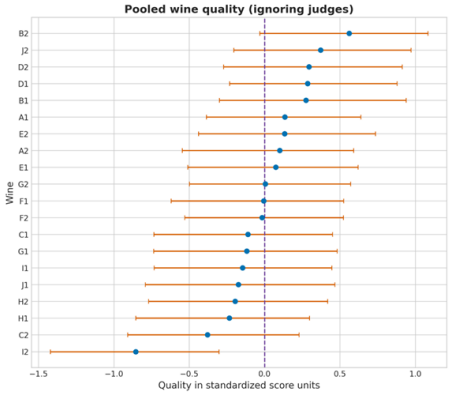


Trying to figure out which wines are better than which other wines

23

Figure 7: Slide 30 — The Wines2012 data: wines and judges

Pooled model quality estimates



Rankings visible; some wines clearly above or below average

25

Figure 8: Slide 33 — Pooled model quality estimates

11.5 — Wine Quality and Judge Generosity

The heatmap shows systematic row differences: some judges score higher or lower overall even when the wine ordering is fixed. The new model adds judge generosity as a hierarchical source of variation. Wine effects remain non-hierarchical because the goal is to estimate the specific wines, while judge effects share information through a population-level standard deviation.

New Definitions and Terms

- **Judge generosity** — a judge-specific level shift; generous judges give higher scores on average and harsh judges give lower scores on average.
- **Hierarchical judge effect** — judge effects are estimated with a shared variance parameter, allowing information sharing across judges.
- $\sigma_{\mathbf{judge}}$ — the population standard deviation of judge generosity.

Formulas

$$\mu_{ij} = \alpha + Q_i + G_j$$

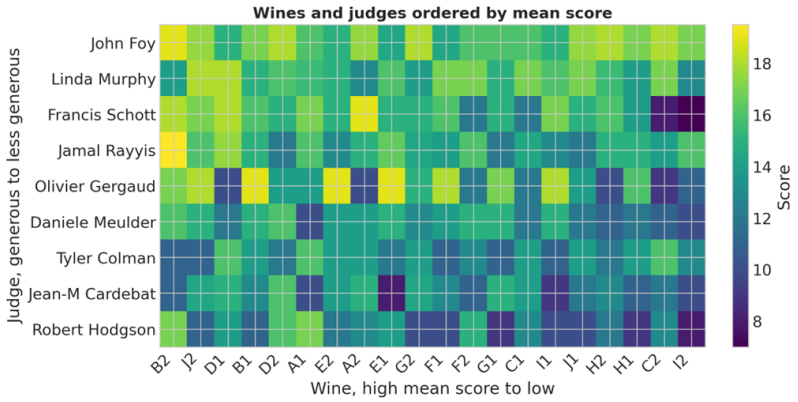
Wine score is modeled as the sum of a grand mean, a wine-quality effect, and a judge-generosity effect.

$$G_j \sim \text{ZeroSumNormal}(\sigma_{\text{judge}})$$

The spread of judge generosity is learned from the data. If the posterior for σ_{judge} is away from zero, judges differ meaningfully in their average scoring levels.

Figure

The heatmap reveals judge generosity

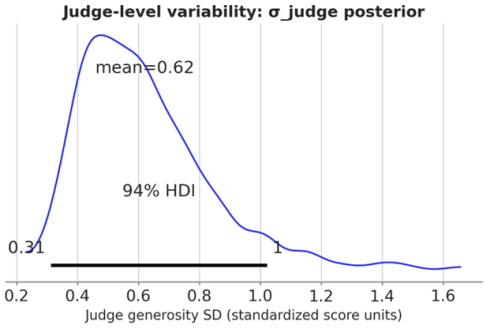


Rows differ systematically even after wines are ordered.

26

Figure 9: Slide 35 — The heatmap reveals judge generosity

Differences between judges could matter



$$G_j \sim Normal(0, \sigma_{judge})$$
$$\sigma_{judge} \sim HalfNormal(\dots)$$

σ_{judge}: population standard deviation of judge generosity

If posterior mass is away from zero → judges differ substantially

This matters if judges judge different wines

31

Figure 10: Slide 39 — Differences between judges could matter

11.6 — Fixed and Random Effects

Fixed and random effects are introduced as frequentist terminology that can be translated into Bayesian model structure. In the Bayesian translation, fixed effects are level-specific parameters with fixed priors, while random effects are level-specific parameters governed by a hierarchical variance. The wine model treats wines as fixed-effect-like because the levels matter, and judges as random-effect-like because the spread of judge effects is the main target.

New Definitions and Terms

- **Fixed effect** — an effect where the levels themselves matter and are estimated directly.
- **Random effect** — an effect where the levels are treated as replaceable, and the population-level spread is estimated.
- **Varying intercept** — a group-specific intercept, written in formula notation as (1 | group).
- **Wilkinson notation** — the formula grammar used by Bambi, lme4, nlme, and related modeling tools.

Formulas

```
score_z ~ 0 + wine + (1 | judge)
```

This formula means: one wine-specific term for each wine identity, plus a varying intercept for judge.

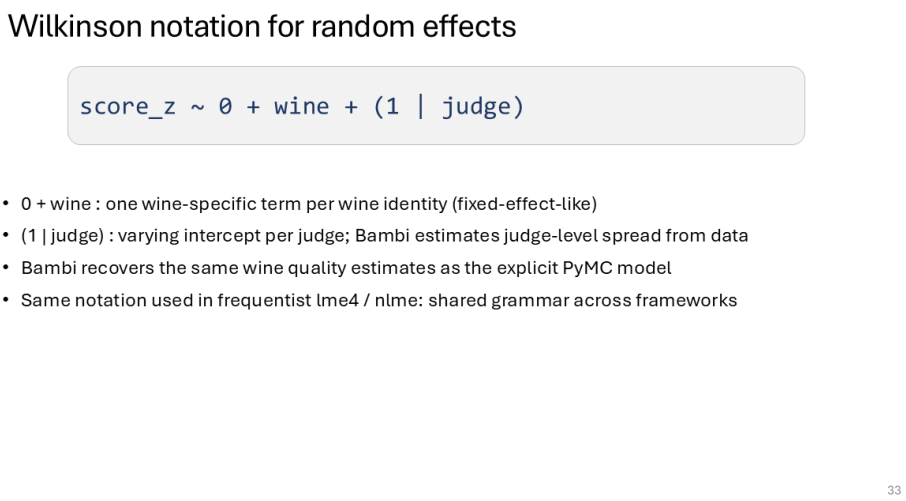


Figure 11: Slide 42 — Wilkinson notation for random effects

11.7 — Frequentist Multi-factor Models

The lecture then translates the wine structure into ordinary least squares and ANOVA language. A model with wine and judge factors is fit with OLS, and the ANOVA table partitions the fit into terms. The full-model F-test asks whether the model as a whole explains variation, while a partial F-test asks whether adding one factor improves fit relative to a reduced model.

New Definitions and Terms

- **Multi-factor model** — a regression model with more than one categorical or continuous term.
- **Dummy variable** — a coded regression variable representing one level or contrast of a categorical factor.
- **Full-model F-test** — a test of whether the full regression model improves fit relative to an intercept-only model.
- **Partial F-test** — a test of whether a factor improves fit relative to a reduced model without that factor.

Formulas

Let SS_{full} be the residual sum of squares for the full model and p the number of fitted regression coefficients.

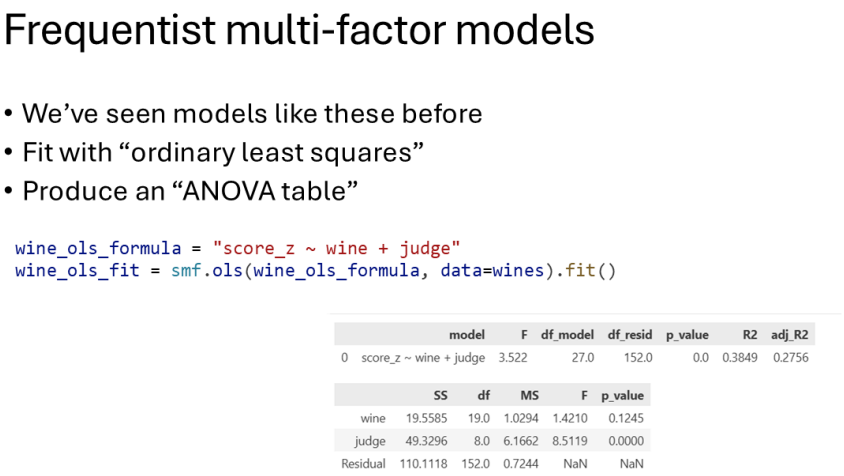
$$F_{\text{full}} = \frac{(SS_{\text{null}} - SS_{\text{full}})/(p-1)}{SS_{\text{full}}/(n-p)}$$

The numerator measures improvement over the null model per model degree of freedom. The denominator measures residual variance per residual degree of freedom.

$$F_{\text{factor}} = \frac{(SS_{\text{reduced}} - SS_{\text{full}})/(p_{\text{full}} - p_{\text{reduced}})}{SS_{\text{full}}/(n-p_{\text{full}})}$$

The partial F-test asks whether the additional factor reduces residual sum of squares enough to justify its degrees of freedom. In the wine example, 20 wines correspond to 19 dummy variables and 9 judges correspond to 8 dummy variables.

Figure



	model	F	df_model	df_resid	p_value	R2	adj_R2
0	score_z ~ wine + judge	3.522	27.0	152.0	0.0	0.3849	0.2756

	SS	df	MS	F	p_value
wine	19.5585	19.0	1.0294	1.4210	0.1245
judge	49.3296	8.0	6.1662	8.5119	0.0000
Residual	110.1118	152.0	0.7244	NaN	NaN

Figure 12: Slide 46 — Frequentist multi-factor models

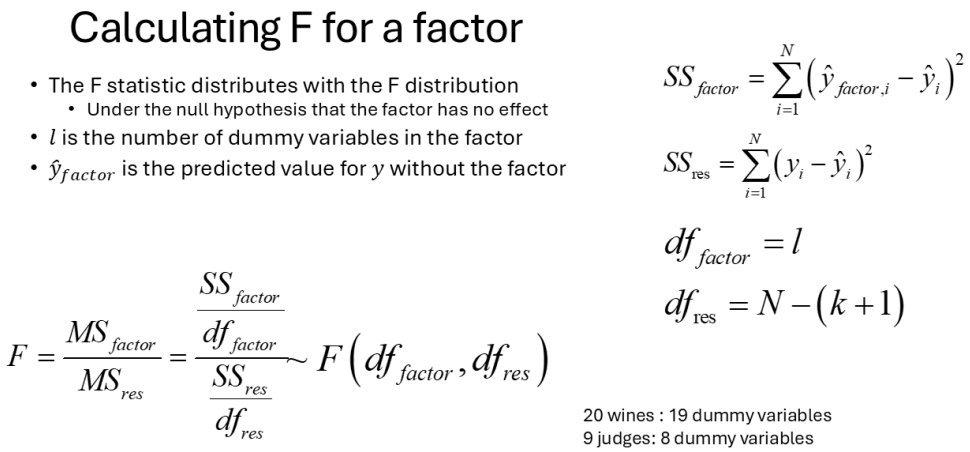


Figure 13: Slide 48 — Calculating F for a factor

11.8 — Discrimination: Motivation and Model

Judge generosity is only a level shift. The discrimination model adds the idea that judges may differ in how strongly their scores reflect latent wine quality. A judge with higher discrimination gives scores that vary more strongly with inferred wine quality, while a low-discrimination judge compresses quality differences.

New Definitions and Terms

- **Discrimination** — judge-specific sensitivity to latent wine quality; the strength of the quality-score relationship.
- **Latent wine quality** — wine quality inferred inside the model rather than treated as an observed predictor.
- **Log-discrimination** — a log-scale parameter used so discrimination remains positive after exponentiation.
- **Population ellipse** — the fitted population distribution of judge effects, distinct from individual judge posterior uncertainty.

Formulas

$$\mu_{ij} = G_j + D_j Q_i$$

The expected score is shifted by judge generosity G_j and scaled by judge discrimination D_j.

$$D_j = \exp(L_j), \qquad (G_j, L_j) \sim \text{MvNormal}(\boldsymbol{\mu}_{\text{judge}}, \text{judge})$$

Generosity and log-discrimination are modeled jointly, so their population correlation can be estimated.

Figure

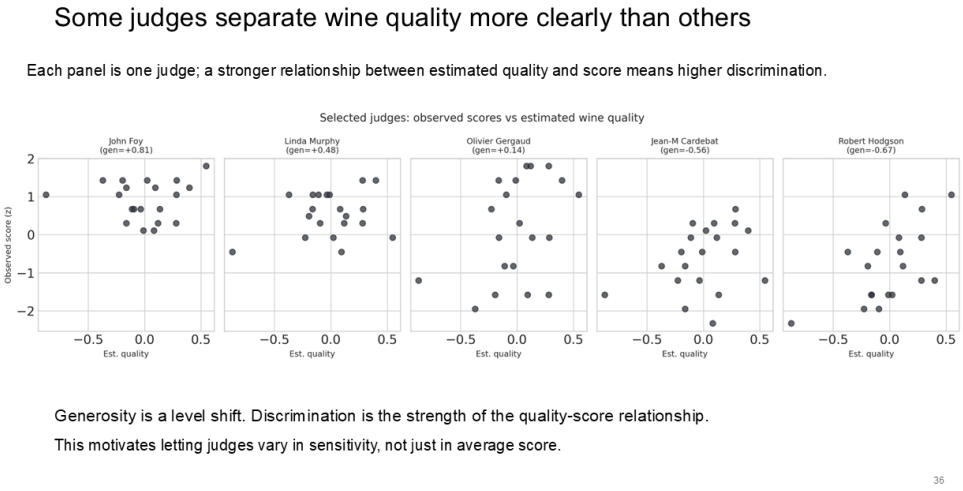


Figure 14: Slide 50 — Some judges separate wine quality more clearly than others

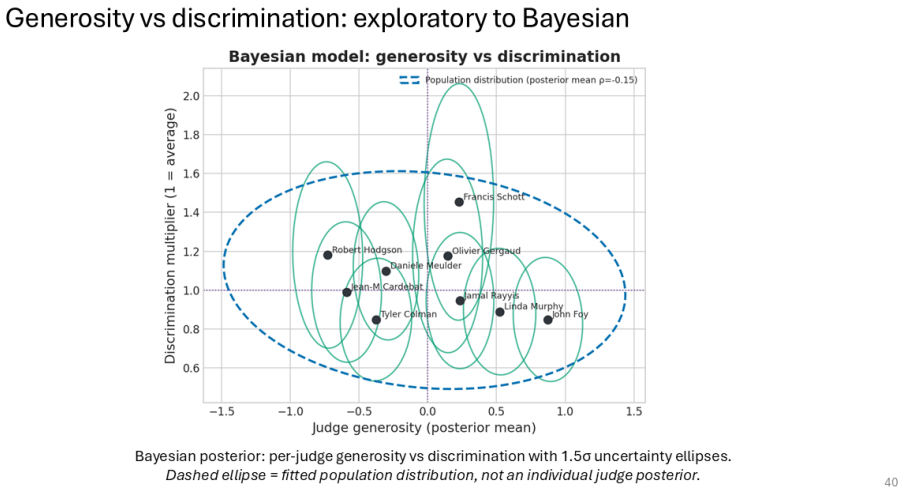


Figure 15: Slide 54 — Generosity vs discrimination: exploratory to Bayesian

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11.9 — Three-model Comparison

The final model comparison asks whether added complexity improves predictive performance. The pooled model allows scores to vary only by wine. The generosity model adds judge-specific intercepts, and the discrimination model adds judge-specific sensitivity to latent quality. LOO favors the generosity model over the pooled model, while the improvement from discrimination is small or uncertain.

New Definitions and Terms

- **Pooled wine model** — scores vary by wine only, with no judge structure.
- **Generosity model** — wine quality plus judge-specific average scoring level.
- **Discrimination model** — wine quality plus judge-specific average scoring level and judge-specific sensitivity to quality.

Figure

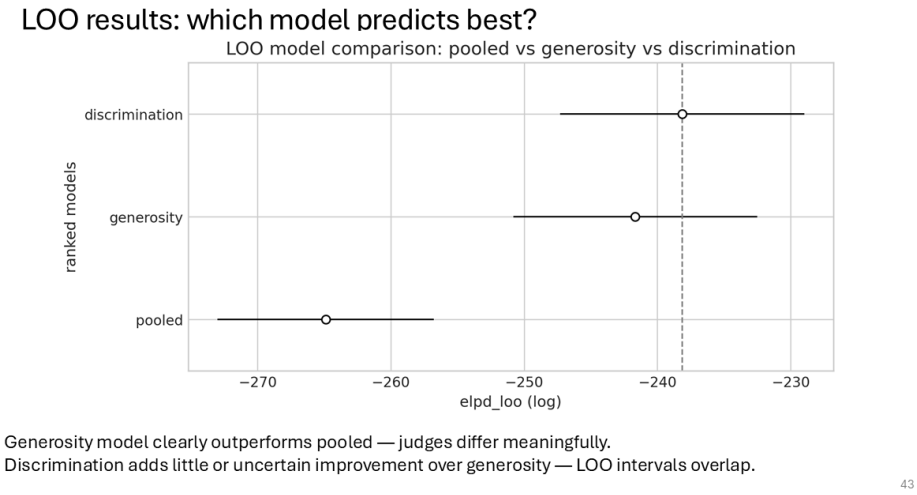


Figure 16: Slide 57 — LOO results: which model predicts best?

11.10 — Next Time

The lecture closes by returning to the Bayesian workflow. By this point, model specification, sampling, diagnostics, posterior predictive checks, interpretation, and model comparison have become a repeated pattern. The next lecture shifts attention upstream: before modeling begins, data can already mislead through measurement problems, sampling bias, missingness, bias, and confounding.

Figure

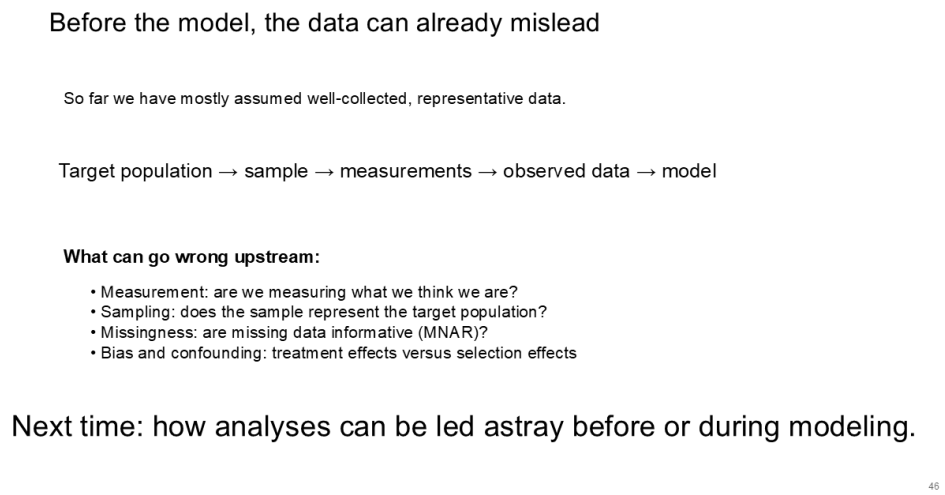


Figure 17: Slide 62 — Before the model, the data can already mislead

Lecture 12 Review Handout

Reproducibility | Statistics 367-1-4361

Lecture Overview

Lecture 12 steps back from individual statistical models and asks how scientific knowledge is produced, filtered, reported, and sometimes distorted. The lecture follows the scientific pipeline from study choice and design, through significance filtering, researcher degrees of freedom, questionable research practices, publication, incentives, and transparent reporting. The central message is that reproducible science requires more than correct code or a valid statistical model; it requires ownership of the full workflow. Bayesian workflow remains the statistical core, but this lecture extends the same logic of checking, transparency, and model criticism to the research process itself.

Lecture Sections

12.1 — Scientific knowledge

This section introduces reproducibility as a problem of scientific knowledge production, not only a problem of individual papers. The examples of chloroquine, MMR/autism, psychology, and cancer biology show that incorrect or fragile findings can enter the literature, influence behavior, and remain influential even after correction. The key idea is that scientific knowledge is the result of a long pipeline with many possible failure points.

New Definitions and Terms

- **Reproducibility crisis** — the recognition that many published findings are difficult to reproduce, replicate, or sustain in later work.
- **Retraction** — formal withdrawal of a paper from the scientific literature after serious problems are identified.
- **Replication** — a later study that tests whether an earlier empirical finding appears again under similar or related conditions.

Figure

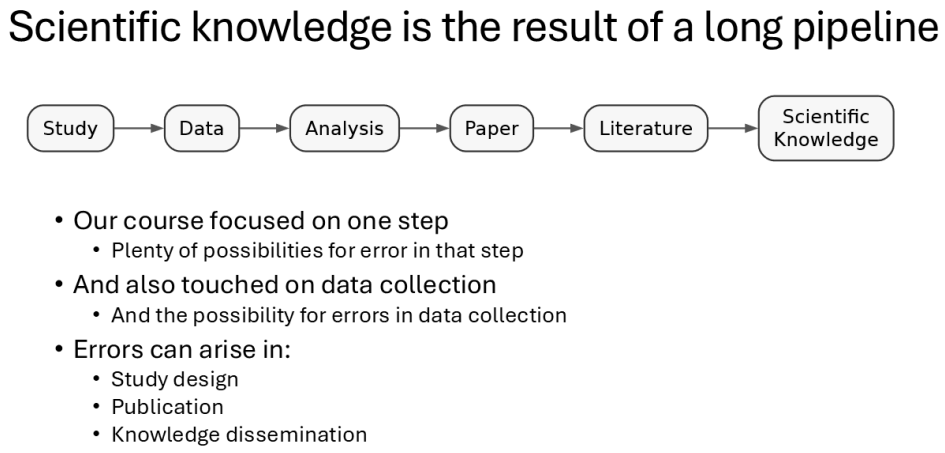


Figure 1: Slide 10 - scientific knowledge is the result of a long pipeline

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12.2 — Where do studies come from?

This section rejects the simple textbook picture in which studies arise only from a clean logical sequence of hypothesis, experiment, and conclusion. Actual study choice is shaped by missing knowledge, available tools, funding priorities, collaborations, previous commitments, and what a group knows how to do. A good study design turns those pressures into a central yes/no question with a clear primary measure and a plausible expected effect.

New Definitions and Terms

- **Pilot data** — preliminary data used to develop methods, test feasibility, clarify analysis, support grants, or motivate new studies.

Figure

Séralini *Environmental Science Europe* 2014, doi:10.1186/s12302-014-0014-5

Does genetically modified food cause cancer?

- 200 rats for 2 years
  - Male and female
  - 10 groups
    - With different amounts and types of GM corn in the diet
- Measured
  - 34 organs
  - Blood samples and urine samples at 11 time points
    - 20 blood parameters
    - 16 urine parameters
  - Tumor incidence
  - Mortality
- Found kidney and liver toxicity starting at 90 days
  - And escalating to kidney and liver relate deaths

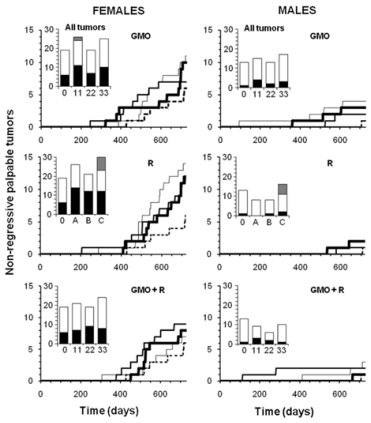


Figure 2: Slide 20 - criteria for good study design

12.3 — Fragile design

This section shows that problems can begin before analysis, when the design itself makes the result hard to interpret. The examples include low power combined with many measures, pseudo-replication, selection bias in observational studies, and confounding through experimenter expectations. The recurring lesson is that bad design is difficult to repair after the data have already been collected.

New Definitions and Terms

- **Pseudo-replication** — treating many measurements from the same experimental unit as if they were independent experimental units.
- **Selection bias** — systematic differences between groups caused by how participants or observations enter the study.
- **Confounding variable** — a variable that is mixed with the intended predictor or manipulation, making the causal interpretation unclear.

Figure

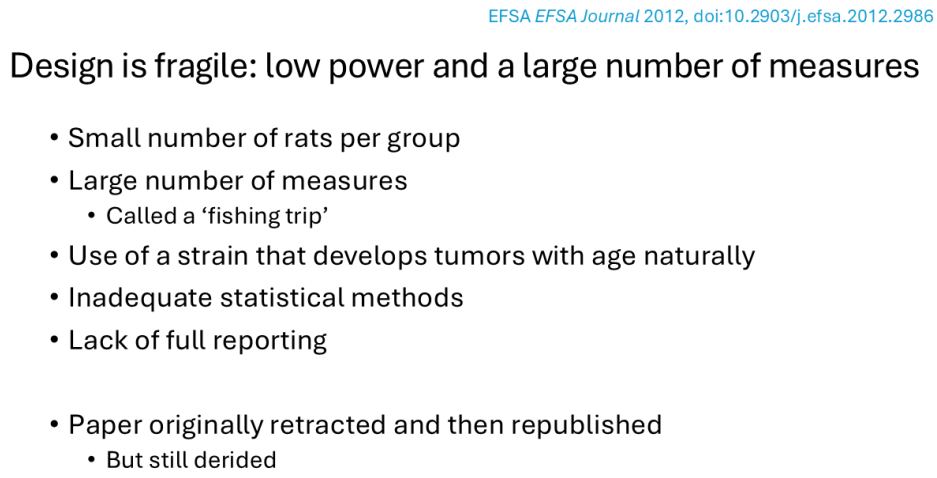


Figure 3: Slide 21 - design is fragile: low power and a large number of measures

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12.4 — The significance filter

This section connects earlier material on significance to bias in the scientific literature. A threshold such as $p < 0.05$ turns continuous evidence into a binary distinction between "real" and "null" findings. Because only some results pass that threshold and many unpublished or non-significant results disappear, the published literature can overrepresent large, noisy, or lucky estimates.

New Definitions and Terms

- **Significance filter** — the process by which results are more likely to become visible when they cross a significance threshold.
- **File-drawer problem** — the disappearance of studies or analyses that do not produce publishable or significant results.
- **Multiple comparison testing** — correction for the fact that many tests create many chances for false-positive findings.

Figure

Turner *NEJM* 2008, doi:10.1056/NEJMsa065779

Failed drug trials do not get published

- Examined 74 submissions to FDA
  - Almost all the approved studies were published
  - Very few that were not approvded were published

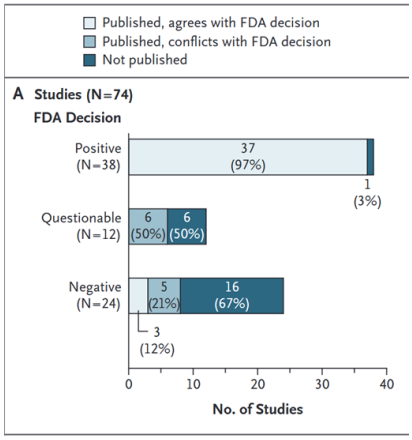
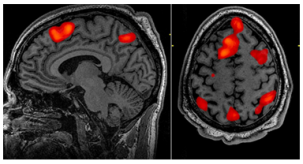


Figure 4: Slide 31 - selection biases which estimates get published

Eklund *PNAS* 2012, doi:10.1073/pnas.1602413113

Multiple tests multiply false positives

- An experiment can involve
  - Many voxels
  - Many electrodes
  - Many cells
  - Many hypotheses
- Correcting the significance is called multiple comparison testing
- Many methods
  - Bonferroni / Sidak
  - Random field theory
  - Permutation analysis
  - False discovery rate
- Different answers depending on the method chosen



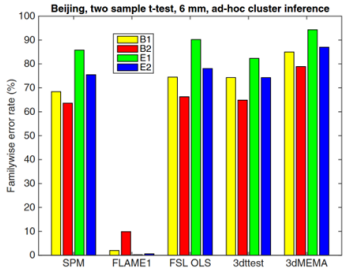


Figure 5: Slide 34 - multiple tests multiply false positives

12.5 — The garden of forking paths

This section moves from publication filtering to the many analysis choices researchers make before a result exists. Even without deliberate misconduct, different reasonable preprocessing choices, model choices, exclusion rules, and testing strategies can lead to different results. Specification curves and multiverse analyses make those unused paths visible rather than hiding them behind a single reported analysis.

New Definitions and Terms

- **Garden of forking paths** — the set of possible analysis decisions that could have been taken, often invisible in the final report.
- **Specification curve** — a display of results across many reasonable analytic specifications.
- **Multiverse analysis** — an analysis strategy that runs many plausible pipelines to examine how conclusions vary across choices.
- **Researcher degrees of freedom** — flexible decisions available to researchers that can change the result.

Figure

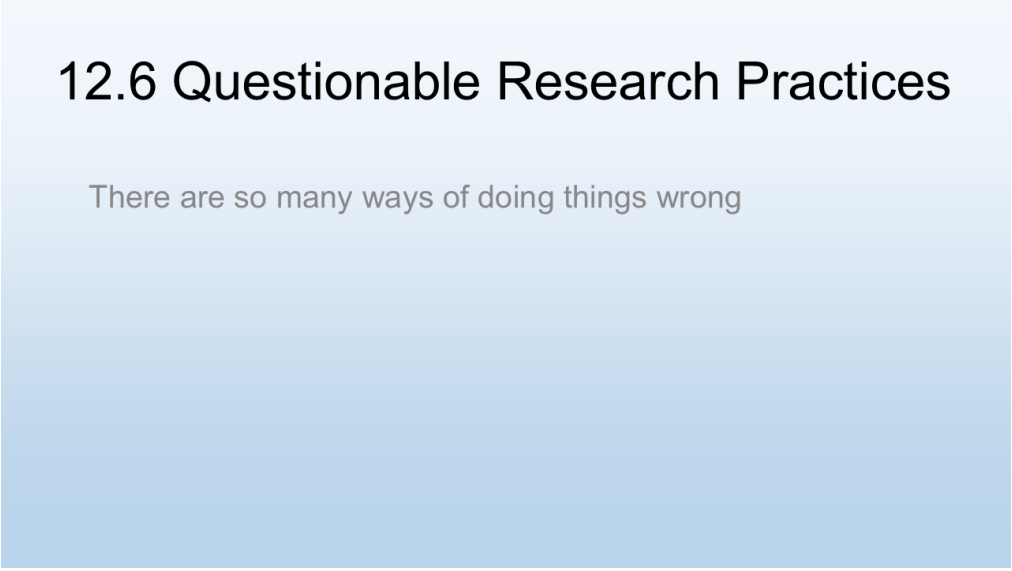


Figure 6: Slide 43 - specification curves can make forking paths visible

---

12.6 — Questionable Research Practices

This section groups several failures under the heading of questionable research practices. The emphasis is that bad research can be unintentional, but some practices violate scientific values because they make the final claim look more confirmatory, cleaner, or more certain than it really is. p-hacking is highlighted as widespread, hard to detect, easy to do, and damaging; HARKing is treated as more ambiguous because storytelling and dishonesty can look superficially similar.

New Definitions and Terms

- **Questionable research practice (QRP)** — a research behavior that undermines scientific reliability without necessarily being outright fabrication or falsification.
- **p-hacking** — repeatedly changing analyses, exclusions, outcomes, or stopping rules until a significant result appears.
- **HARKing** — hypothesizing after the results are known; presenting a post-hoc explanation as if it had been the original hypothesis.

Figure

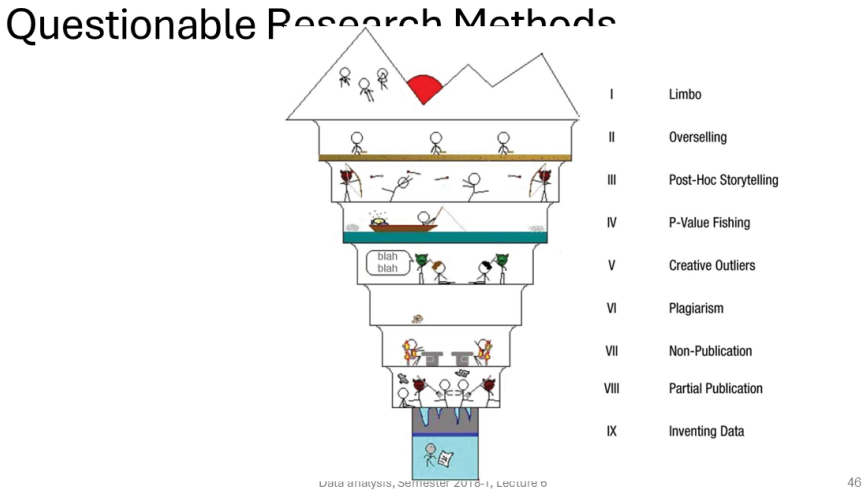


Figure 7: Slide 46 - central scientific values

Simmons *Psychological Science* 2011, doi:10.1177/0956797611417632

Simulation of p-hacking and significance

	Significance level		
Researcher degrees of freedom	***p* < .1**	***p* < .05**	***p* < .01**
Situation A: two dependent variables (*r* = .50)	17.8%	9.5%	2.2%
Situation B: addition of 10 more observations per cell	14.5%	7.7%	1.6%
Situation C: controlling for gender or interaction of gender with treatment	21.6%	11.7%	2.7%
Situation D: dropping (or not dropping) one of three conditions	23.2%	12.6%	2.8%
Combine Situations A and B	26.0%	14.4%	3.3%
Combine Situations A, B, and C	50.9%	30.9%	8.4%
Combine Situations A, B, C, and D	81.5%	60.7%	21.5%

Figure 8: Slide 49 - simulation of p-hacking and significance

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12.7 — Exploration and Confirmation

This section rehabilitates exploration as legitimate scientific work while distinguishing it from confirmation. Exploratory science is appropriate when expectations, methods, or analyses are still being developed; confirmatory science is appropriate when the question, method, and story are already public and stable. Preregistration, Registered Reports, and transparent deviations are presented as tools for making the timing of decisions visible.

New Definitions and Terms

- **Preregistration** — recording hypotheses, methods, and analysis plans before observing the results.

Figure

Exploration, confirmation and harking

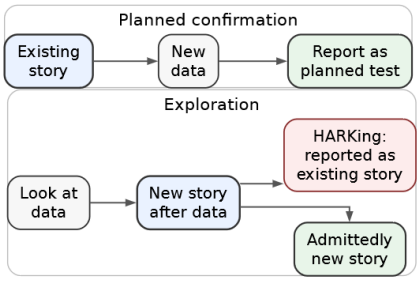


Figure 9: Slide 54 - exploration, confirmation and harking

Preregistration is a way of creating a timestamp

- Stage 1 review
  - Review science without knowing results
- Stage 2 review
  - Check that methods have been followed



Data analysis, 2020-2, Lecture 10 56/ 86

Figure 10: Slide 56 - preregistration is a way of creating a timestamp

12.8 — Creating the literature

This section asks how results become part of what other scientists read and use. Completion of a study is not the end of the process: results move through writing, submission, peer review, revision, and eventual acceptance or rejection. The literature includes primary papers, reviews, and meta-analyses, each with different roles in organizing, persuading, aggregating, or narrating scientific evidence.

New Definitions and Terms

- **Peer review** — evaluation of a submitted paper by external readers before publication.
- **Review paper** — a paper that organizes, interprets, argues about, or summarizes an area of research.
- **Meta-analysis** — a quantitative review that aggregates results, effect sizes, or significance patterns across studies.

Figure

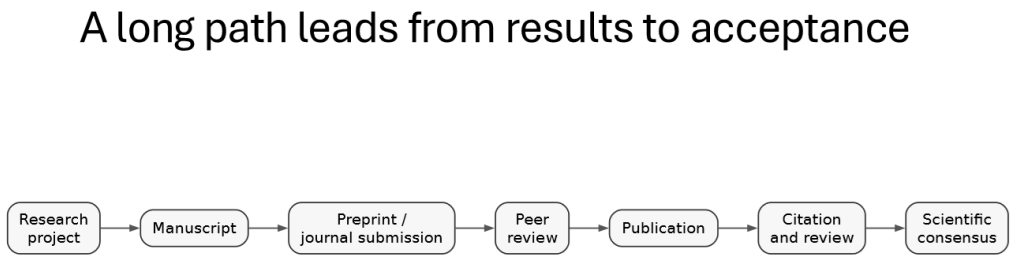


Figure 11: Slide 59 - a long path leads from results to acceptance

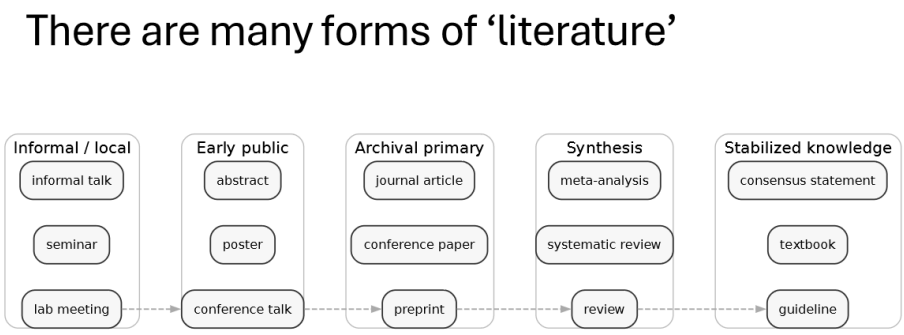


Figure 12: Slide 60 - there are many forms of literature

12.9 — Incentives and publication bias

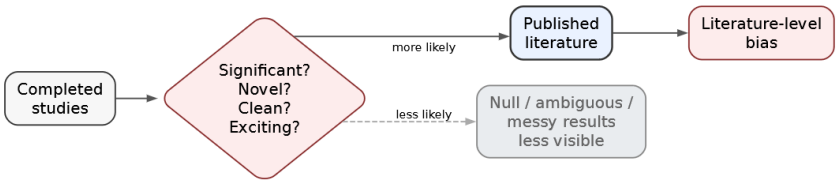
This section focuses on structural incentives rather than individual statistical choices. Positive results are easier to publish because they seem more interesting, fit narrative expectations, and are easier to interpret than failures to find an effect. The incentive system rewards publications, grants, hiring, promotion, and institutional visibility, so the literature is shaped by what the system rewards.

New Definitions and Terms

- **Publication bias** — systematic distortion of the literature because some results are more likely to be published than others.

Figure

It is always easier to publish positive results



- The literature needs to be interesting to scientists
- Our understanding needs a story
- Negative results can be poor technique
  - Harder to prove the lack of an effect

Figure 13: Slide 64 - it is always easier to publish positive results

Incentives for the scientist

- Everything depends on publications
  - Getting hired
  - Getting promoted
  - Getting funded
  - Getting students
    - Who can do experiments so that there can be more papers
- Institutions are also incentivized
  - Budgets
  - Special projects
  - Hiring quality faculty

Figure 14: Slide 66 - incentives for the scientist

12.10 — Transparent reporting

This section turns from diagnosing problems to reporting in a way that makes the literature better. Statistical reporting is needed in scientific papers, grant applications, preregistered research, ethical approval, medical reporting, engineering, and patents. The recommended structure separates what belongs in the main document, what belongs in an appendix, and what belongs in a repository.

New Definitions and Terms

- **Repository** — a public or shared location for data, code, additional analyses, and reproducible computational materials.

Figure

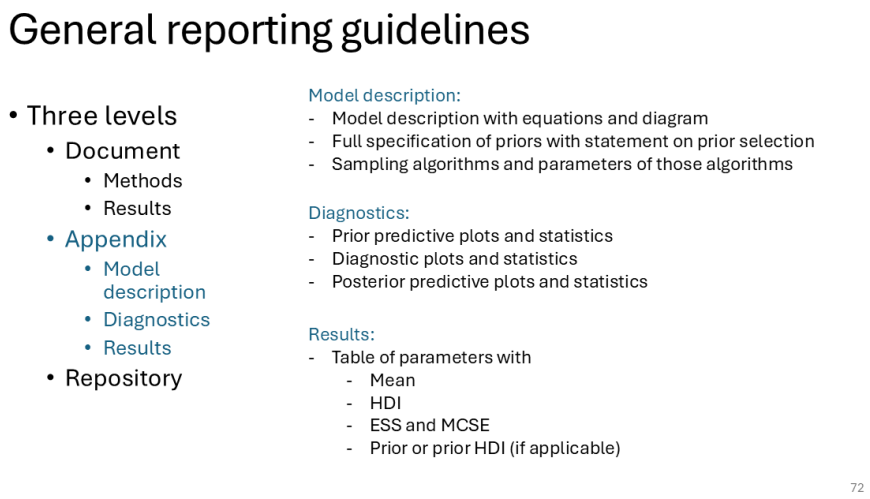


Figure 15: Slide 72 - Transparent reporting guidelines for document, appendix, and repository

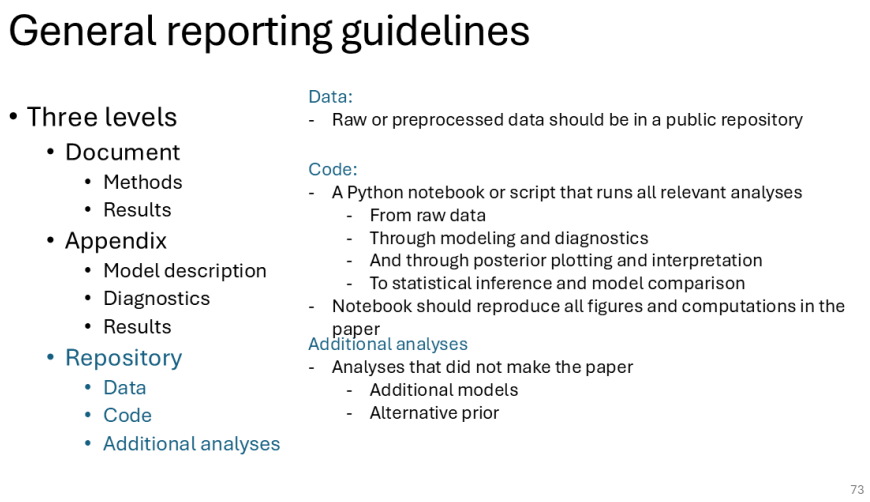


Figure 16: Slide 73 - Repository materials for data, code, and additional analyses

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12.11 — Ownership of the workflow

The final section returns to the whole process. Understanding reproducibility requires taking responsibility not only for a final statistical model but also for study design, data collection, analysis choices, reporting, dissemination, and later interpretation. The complete workflow is therefore broader than the Bayesian workflow alone, but it preserves the same principle: make assumptions explicit, check each step, and keep the process transparent.

New Definitions and Terms

Figure

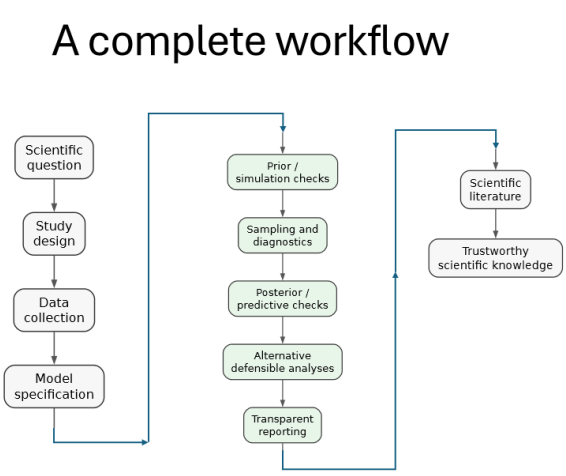


Figure 17: Slide 76 - a complete workflow